Regression Analysis

Lecture notes, Chapters 3–16

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Chapter 3. Simple Linear Regression

3.1 Regression and the straight-line model

Regression analysis studies how a **response** (dependent) variable Y changes with one or more **explanatory** (independent) variables. The name goes back to Galton, who observed that sons of unusually tall or short fathers tend to be closer to the average height: their heights "regress" towards the mean. In Pearson's father–son data the fitted line has slope about 0.5, so a father 6 inches above average has sons only about 3 inches above average on average.

With one explanatory variable and a linear relationship we speak of **simple linear regression**.

Definition (Simple linear regression model)

For data $(x_1, y_1), \ldots, (x_n, y_n)$,

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2), \qquad i = 1, \ldots, n,$$

where the x_i are treated as fixed (nonrandom) constants and $\beta_0, \beta_1, \sigma^2$ are unknown.

The model packs three assumptions.

(1) **Linearity**: $E(Y \mid X = x) = \beta_0 + \beta_1 x$.

(2) **Normality and equal variance**: given $X = x$, $Y \sim N(\beta_0 + \beta_1 x, \sigma^2)$, with σ^2 not depending on x (**homoscedasticity**); otherwise the errors are **heteroscedastic** (Chapter 12).

(3) **Independence**: the errors are uncorrelated, $\mathrm{Cov}(\epsilon_i, \epsilon_j) = 0$ for $i \neq j$ (independent under normality).

The model is linear *in the parameters*; a model such as $\beta_0 + \beta_1 x + \beta_2 x^2$ is still a linear model. If X is itself random, the same analysis applies conditionally on the observed x_i; this is the **regression of Y on X**, which differs from the regression of X on Y.

It is often convenient to centre the explanatory variable:

$$y_i = \beta_0^* + \beta_1 (x_i - \bar{x}) + \epsilon_i, \qquad \beta_0^* = \beta_0 + \beta_1 \bar{x}. \tag{3.1}$$

3.2 Least squares

Motivation. We want the line that best describes how y changes with x. "Best" needs a criterion; least squares chooses the line minimizing the sum of squared vertical deviations, which gives closed-form estimates and, under normal errors, coincides with maximum likelihood.

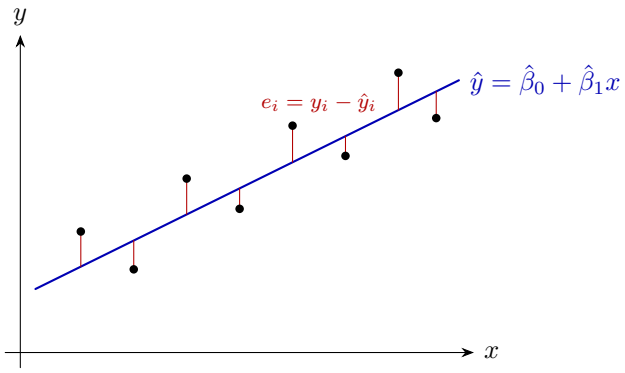


Figure. *Least squares chooses the line minimizing the sum of the squared vertical deviations e_i (red), not perpendicular distances.*

Definition (Least squares estimator)

The **least squares estimators** (LSE) $\hat{\beta}_0, \hat{\beta}_1$ minimize

$$S(\beta_0, \beta_1) = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2.$$

Write

$$S_{xx} = \sum (x_i - \bar{x})^2, \qquad S_{yy} = \sum (y_i - \bar{y})^2, \qquad S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y}).$$

Theorem 3.1 (Normal equations)

If $S_{xx} > 0$, the LSE is the unique solution of the **normal equations**

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum x_i = \sum y_i, \qquad \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2 = \sum x_i y_i,$$

namely

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.$$

Proof. Setting the partial derivatives to zero,

$$\frac{\partial S}{\partial \beta_0} = -2 \sum (y_i - \beta_0 - \beta_1 x_i) = 0, \qquad \frac{\partial S}{\partial \beta_1} = -2 \sum x_i (y_i - \beta_0 - \beta_1 x_i) = 0,$$

gives the normal equations. The first gives $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$; substituting into the second, $\hat{\beta}_1 (\sum x_i^2 - n\bar{x}^2) = \sum x_i y_i - n\bar{x}\bar{y}$, i.e. $\hat{\beta}_1 S_{xx} = S_{xy}$. The Hessian

$$J = 2 \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$$

has $n > 0$ and determinant $4nS_{xx} > 0$, so it is positive definite. S is a convex quadratic, and the stationary point is its unique minimum. ∎

In the centred form (3.1) the normal equations decouple: $\hat{\beta}_0^* = \bar{y}$ and $\hat{\beta}_1 = S_{xy}/S_{xx}$, so the fitted line is

$$\hat{y} = \bar{y} + \hat{\beta}_1 (x - \bar{x}).$$

Example 3.1

Advertising spending x and sales y for six stores:

x	2	3	5	6	8	10
y	5	7	8	11	13	16

Here $\bar{x} = 34/6 = 5.667$, $\bar{y} = 10$, $S_{xx} = 238 - 34^2/6 = 45.33$ and $S_{xy} = 401 - 34 \cdot 60/6 = 61$. So $\hat{\beta}_1 = 61/45.33 = 1.346$, $\hat{\beta}_0 = 10 - 1.346 \cdot 5.667 = 2.37$, and $\hat{y} = 2.37 + 1.346x$.

Maximum likelihood. Under normal errors the log-likelihood is

$$l(\beta_0, \beta_1, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}S(\beta_0, \beta_1).$$

For each σ^2 it is maximized by minimizing S, so the MLEs of β_0, β_1 are the LSEs, and $\hat{\sigma}^2_{\text{MLE}} = S(\hat{\beta}_0, \hat{\beta}_1)/n$.

3.3 Properties of the fitted line

The **fitted values** are $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$ and the **residuals** are $e_i = y_i - \hat{y}_i$.

Theorem 3.2

(a) $\sum e_i = 0$; equivalently $\sum y_i = \sum \hat{y}_i$.

(b) $\sum x_i e_i = 0$.

(c) $\sum \hat{y}_i e_i = 0$.

(d) The fitted line passes through $(\bar{x}, \bar{y})$.

(e) $\sum e_i^2$ is the minimum of $S(\beta_0, \beta_1)$.

Proof. (a) and (b) are the two normal equations, since $\sum e_i = \sum(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$ and $\sum x_i e_i = \sum x_i(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)$.

(c) $\sum \hat{y}_i e_i = \hat{\beta}_0 \sum e_i + \hat{\beta}_1 \sum x_i e_i = 0$.

(d) $\hat{\beta}_0 + \hat{\beta}_1 \bar{x} = \bar{y}$.

(e) is the definition of the LSE. ■

Geometrically, (a)–(c) say that the residual vector is orthogonal to $\mathbf{1}$, to $\mathbf{x}$, and hence to the fitted vector. This is why the residual plot of e_i against $\hat{y}_i$ (or x_i) should show no linear trend.

3.4 Decomposition of variation

Motivation. How much of the variation in y does the line explain? The total sum of squares splits exactly into a regression part and a residual part, and their ratio R^2 is the standard summary of fit.

Theorem 3.3 (Sum of squares decomposition)

$$\underbrace{\sum(y_i - \bar{y})^2}_{SST} = \underbrace{\sum(y_i - \hat{y}_i)^2}_{SSE} + \underbrace{\sum(\hat{y}_i - \bar{y})^2}_{SSR},$$

and $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2/S_{xx}$.

Proof. Write $y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})$ and square. The cross term is $2\sum e_i(\hat{y}_i - \bar{y}) = 2\sum \hat{y}_i e_i - 2\bar{y}\sum e_i = 0$ by Theorem 3.2. Since $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$, $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2/S_{xx}$. ■

SST is the **total** sum of squares, SSE the **residual** (error) sum of squares, and SSR the **regression** sum of squares. Their degrees of freedom are $n-1$, $n-2$ and 1.

Definition (Coefficient of determination)

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST} \in [0,1].$$

R^2 is the proportion of the variation in y explained by the regression. $R^2 = 1$ means a perfect fit; $R^2 = 0$ means $\hat{\beta}_1 = 0$, i.e. the fitted line is horizontal. The standard error of estimate is $s = \sqrt{MSE}$ with $MSE = SSE/(n-2)$; we will see that MSE is unbiased for σ^2 (Chapter 4).

Example 3.2

In Example 3.1, $\sum y_i^2 = 684$, so $SST = S_{yy} = 684 - 600 = 84$. Then $SSR = 61^2/45.33 = 82.08$, $SSE = 84 - 82.08 = 1.92$ and $R^2 = 0.977$: advertising explains about 98% of the variation in sales.

3.5 Correlation

For a random pair (X, Y) the **population correlation** is $\rho_{XY} = \mathrm{Cov}(X,Y)/\sqrt{\mathrm{Var}\,X\,\mathrm{Var}\,Y}$, and the **sample correlation** is

$$r_{xy} = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} \in [-1,1].$$

Theorem 3.4

(a) $R^2 = r_{xy}^2$ in simple linear regression.

(b) $r_{xy} = \hat{\beta}_1 \, s_x/s_y$, where $s_x^2 = S_{xx}/(n-1)$ and $s_y^2 = S_{yy}/(n-1)$; hence r_{xy} has the sign of $\hat{\beta}_1$.

Proof. (a) By Theorem 3.3, $R^2 = \frac{SSR}{SST} = \frac{S_{xy}^2}{S_{xx}S_{yy}} = r_{xy}^2$.

(b) $r_{xy} = \frac{S_{xy}}{S_{xx}}\sqrt{\frac{S_{xx}}{S_{yy}}} = \hat{\beta}_1 \frac{s_x}{s_y}$. ■

r_{xy} measures only *linear* association. For $x = 0,1,2,3,4$ and $y = x^2 - 4x + 6 = (6,3,2,3,6)$, $S_{xy} = 0$ and $r_{xy} = 0$, although y is a deterministic function of x.

Remark

The slope and the correlation answer different questions: $\hat{\beta}_1$ is the expected change in y per unit of x, while r_{xy} is unit-free and symmetric in x and y. The regression of x on y has slope S_{xy}/S_{yy}, and the product of the two slopes is $r_{xy}^2 \le 1$; the two lines coincide only if $|r_{xy}| = 1$.

3.6 Selected exercises

Motivation. These problems test the least squares formulas on special designs: a line forced through the origin, a design with only two distinct x values, and a description of the slope as an average of pairwise slopes.

Exercise 3.1 (Regression through the origin)

Consider $y_i = \beta x_i + \epsilon_i$, $i = 1, \dots, n$, with $\sum x_i^2 > 0$ and no intercept.
(a) Show that the least squares estimator is $\hat{\beta} = \sum x_i y_i / \sum x_i^2$ and that the residuals $e_i = y_i - \hat{\beta} x_i$ satisfy $\sum x_i e_i = 0$, but in general $\sum e_i \neq 0$.
(b) Show that $\sum y_i^2 = \sum \hat{y}_i^2 + \sum e_i^2$, and give an example in which the centred decomposition $SST = SSR + SSE$ of Theorem 3.3 fails.

Solution [Supplement].

(a) $S(\beta) = \sum (y_i - \beta x_i)^2$ has $S'(\beta) = -2 \sum x_i (y_i - \beta x_i)$ and $S''(\beta) = 2 \sum x_i^2 > 0$. So S is a strictly convex quadratic, and its unique minimizer solves $\sum x_i (y_i - \beta x_i) = 0$, which gives $\hat{\beta}$. This equation is exactly $\sum x_i e_i = 0$. There is no equation forcing $\sum e_i = 0$: for $x = (1, 2)$ and $y = (1, 1)$, $\hat{\beta} = 3/5$ and $e = (0.4, -0.2)$, so $\sum e_i = 0.2$.

(b) Since $\hat{y}_i = \hat{\beta} x_i$,

$$\sum y_i^2 = \sum (\hat{y}_i + e_i)^2 = \sum \hat{y}_i^2 + 2\hat{\beta} \sum x_i e_i + \sum e_i^2 = \sum \hat{y}_i^2 + \sum e_i^2.$$

In the example of (a), $\bar{y} = 1$, so $SST = 0$, while $SSE = 0.16 + 0.04 = 0.2 > 0$. Hence $SST \neq SSR + SSE$ (the right side is positive), and $R^2 = 1 - SSE/SST$ is meaningless. For models without intercept one uses the uncorrected sums $\sum y_i^2$ and $\sum \hat{y}_i^2$. ■

Exercise 3.2 (Two design points)

Suppose the x_i take only two distinct values $a < b$, with $n_a \geq 1$ observations at a and $n_b \geq 1$ at b ($n = n_a + n_b$). Let $\bar{y}_a$ and $\bar{y}_b$ be the means of the responses at a and at b. Show that

$$S_{xx} = \frac{n_a n_b (b - a)^2}{n}, \qquad \hat{\beta}_1 = \frac{\bar{y}_b - \bar{y}_a}{b - a},$$

and that the fitted line passes through $(a, \bar{y}_a)$ and $(b, \bar{y}_b)$.

Solution [Supplement].

Step 1. $\bar{x} = (n_a a + n_b b)/n$, so $a - \bar{x} = -n_b (b - a)/n$ and $b - \bar{x} = n_a (b - a)/n$.

Step 2. Hence

$$S_{xx} = n_a \frac{n_b^2 (b - a)^2}{n^2} + n_b \frac{n_a^2 (b - a)^2}{n^2} = \frac{n_a n_b (b - a)^2 (n_a + n_b)}{n^2} = \frac{n_a n_b (b - a)^2}{n}.$$

Step 3. Using $S_{xy} = \sum (x_i - \bar{x}) y_i$ and grouping the observations by their x value,

$$S_{xy} = -\frac{n_b (b - a)}{n} n_a \bar{y}_a + \frac{n_a (b - a)}{n} n_b \bar{y}_b = \frac{n_a n_b (b - a)(\bar{y}_b - \bar{y}_a)}{n},$$

and $\hat{\beta}_1 = S_{xy}/S_{xx} = (\bar{y}_b - \bar{y}_a)/(b - a)$.

Step 4. The fitted line is $\hat{y}(x) = \bar{y} + \hat{\beta}_1 (x - \bar{x})$ with $\bar{y} = (n_a \bar{y}_a + n_b \bar{y}_b)/n$. At $x = a$, by Step 1,

$$\hat{y}(a) = \frac{n_a \bar{y}_a + n_b \bar{y}_b}{n} - \frac{n_b (\bar{y}_b - \bar{y}_a)}{n} = \frac{(n_a + n_b) \bar{y}_a}{n} = \bar{y}_a,$$

and in the same way $\hat{y}(b) = \bar{y}_b$. So the line joins the two group means, and all the variation within the groups goes into SSE. ■

Exercise 3.3 (The slope as an average of pairwise slopes)

Assume $S_{xx} > 0$. For $i < j$ with $x_i \neq x_j$, let $s_{ij} = (y_j - y_i)/(x_j - x_i)$ be the slope of the line through observations i and j. Show that

$$\hat{\beta}_1 = \sum_{i<j,\ x_i \neq x_j} w_{ij} s_{ij}, \qquad w_{ij} = \frac{(x_j - x_i)^2}{\sum_{k<l}(x_l - x_k)^2},$$

a weighted average of the pairwise slopes with weights proportional to the squared distance between the x values.

Solution [Supplement].

Step 1. For any numbers u_i, v_i, expanding the double sum,

$$\sum_{i<j}(u_j - u_i)(v_j - v_i) = \frac{1}{2}\sum_{i,j}(u_j - u_i)(v_j - v_i) = n\sum_i u_i v_i - \Big(\sum_i u_i\Big)\Big(\sum_i v_i\Big) = n\sum_i (u_i - \bar{u})(v_i - \bar{v}).$$

Step 2. With $u = v = x$ this gives $\sum_{k<l}(x_l - x_k)^2 = nS_{xx} > 0$, so the weights are well defined, nonnegative and sum to 1 (pairs with $x_i = x_j$ have zero weight). With $u = x$, $v = y$ it gives $\sum_{i<j}(x_j - x_i)(y_j - y_i) = nS_{xy}$.

Step 3. For a pair with $x_i \neq x_j$, $(x_j - x_i)(y_j - y_i) = (x_j - x_i)^2 s_{ij}$, and pairs with $x_i = x_j$ contribute 0 to the sum in Step 2. Therefore

$$\sum_{i<j,\ x_i \neq x_j} w_{ij} s_{ij} = \frac{\sum_{i<j}(x_j - x_i)(y_j - y_i)}{nS_{xx}} = \frac{nS_{xy}}{nS_{xx}} = \hat{\beta}_1.$$

Pairs of points far apart in x determine their slope more reliably, and least squares gives them more weight. ■

Chapter 4. Inference in Simple Regression

Throughout, $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with $\epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2)$ and fixed x_i with $S_{xx} > 0$.

4.1 Sampling distributions of the estimators

Motivation. The fitted slope would change with a new sample. To judge whether an observed slope reflects a real relationship we need its sampling distribution; under normal errors $\hat\beta_1$ is normal and the studentized slope has a t distribution.

Both estimators are linear in y:

$$\hat\beta_1 = \frac{S_{xy}}{S_{xx}} = \sum_i k_i y_i, \qquad k_i = \frac{x_i - \bar x}{S_{xx}}, \qquad \sum k_i = 0, \quad \sum k_i x_i = 1, \quad \sum k_i^2 = \frac{1}{S_{xx}},$$

(using $S_{xy} = \sum (x_i - \bar x) y_i$), and $\hat\beta_0 = \bar y - \hat\beta_1 \bar x$.

Theorem 4.1

(a) $\hat\beta_1 \sim N(\beta_1, \sigma^2/S_{xx})$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$.

(c) $\hat\beta_0 \sim N\Big(\beta_0,\ \sigma^2\big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\big)\Big)$ and $\operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\dfrac{\bar x \sigma^2}{S_{xx}}$.

Proof. (a) $\hat\beta_1$ is a linear combination of independent normals, with mean $\sum k_i(\beta_0 + \beta_1 x_i) = \beta_1$ and variance $\sigma^2 \sum k_i^2 = \sigma^2 / S_{xx}$.

(b) $\operatorname{Cov}(\bar y, \hat\beta_1) = \sum_i \frac{1}{n} k_i \sigma^2 = 0$.

(c) $E\hat\beta_0 = (\beta_0 + \beta_1 \bar x) - \bar x \beta_1 = \beta_0$ and, by (b),

$$\operatorname{Var} \hat\beta_0 = \operatorname{Var} \bar y + \bar x^2 \operatorname{Var} \hat\beta_1 = \sigma^2\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(\hat\beta_0, \hat\beta_1) = -\bar x \operatorname{Var} \hat\beta_1.$$

■

Theorem 4.2

Let $SSE = \sum e_i^2$ and $MSE = SSE/(n-2)$. Then $SSE/\sigma^2 \sim \chi^2(n-2)$, independent of $(\hat\beta_0, \hat\beta_1)$. In particular $E(MSE) = \sigma^2$.

This is the case $p = 1$ of Theorem 6.4; a direct proof uses an orthogonal transformation whose first two rows span $\mathbf{1}$ and $\mathbf{x} - \bar x \mathbf{1}$, exactly as for the normal sample variance. Consequently

$$\frac{\hat\beta_1 - \beta_1}{\sqrt{MSE/S_{xx}}} \sim t(n-2), \qquad \frac{\hat\beta_0 - \beta_0}{\sqrt{MSE(1/n + \bar x^2/S_{xx})}} \sim t(n-2).$$

Corollary 4.3 (Intervals and tests for the coefficients)

$100(1-\alpha)\%$ confidence intervals are

$$\beta_1 :\ \hat\beta_1 \pm t_{\alpha/2}(n-2)\sqrt{\frac{MSE}{S_{xx}}}, \qquad \beta_0 :\ \hat\beta_0 \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(\frac{1}{n} + \frac{\bar x^2}{S_{xx}}\Big)}.$$

The test of $H_0 : \beta_1 = 0$ rejects when $|t_0| > t_{\alpha/2}(n-2)$, $t_0 = \hat{\beta}_1/\sqrt{MSE/S_{xx}}$. If σ^2 is known, replace MSE by σ^2 and t by z.

Theorem 4.4 (ANOVA F test)

$E(SSR) = \sigma^2 + \beta_1^2 S_{xx}$ and $E(SSE) = (n-2)\sigma^2$. The statistic $F_0 = MSR/MSE$, $MSR = SSR/1$, satisfies $F_0 = t_0^2$, so the F test with $F_\alpha(1, n-2) = t_{\alpha/2}(n-2)^2$ is the two-sided t test of $\beta_1 = 0$.

Proof. $E(SSR) = S_{xx}E\hat{\beta}_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. Also $SSR/MSE = \hat{\beta}_1^2 S_{xx}/MSE = t_0^2$. ■

So $E(MSR)/E(MSE) = 1 + \beta_1^2 S_{xx}/\sigma^2$, which is 1 exactly when $\beta_1 = 0$; large F_0 is evidence against H_0.

4.2 Estimating the mean response and predicting

Motivation. Estimating the average response at x_0 and predicting a single new response there are different problems: the prediction interval must also account for the new observation's own error, so it is always wider.

At a point x, the mean response $\mu_{y\cdot x} = \beta_0 + \beta_1 x$ is estimated by $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x = \bar{y} + \hat{\beta}_1(x - \bar{x})$.

Theorem 4.5

(a) $\hat{y} \sim N\Big(\mu_{y\cdot x},\ \sigma^2\Big(\dfrac{1}{n} + \dfrac{(x-\bar{x})^2}{S_{xx}}\Big)\Big)$, and a $100(1-\alpha)\%$ confidence interval for $\mu_{y\cdot x}$ is

$$\hat{y} \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(\frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}\Big)}.$$

(b) For a new observation $y_s = \beta_0 + \beta_1 x + \epsilon_s$, independent of the sample, $y_s - \hat{y} \sim N\big(0, \sigma^2(1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}})\big)$, and a $100(1-\alpha)\%$ prediction interval is

$$\hat{y} \pm t_{\alpha/2}(n-2)\sqrt{MSE\Big(1 + \frac{1}{n} + \frac{(x-\bar{x})^2}{S_{xx}}\Big)}.$$

Proof. (a) By Theorem 4.1(b), $\operatorname{Var}\hat{y} = \operatorname{Var}\bar{y} + (x-\bar{x})^2 \operatorname{Var}\hat{\beta}_1$. Studentize with MSE, which is independent of $\hat{y}$.

(b) y_s is independent of $\hat{y}$, so the variances add. ■

Both intervals are narrowest at $x = \bar{x}$ and widen quadratically away from it; extrapolation is costly. The prediction interval does not shrink to zero as $n \to \infty$, because of the irreducible variance σ^2 of a new observation.

Example 4.1

For the advertising data of Example 3.1 ($n = 6$, $\bar{x} = 5.667$, $S_{xx} = 45.33$, $MSE = 1.92/4 = 0.48$, $\hat{y} = 2.37 + 1.346x$), at $x = 7$: $\hat{y} = 11.79$, $\frac{1}{6} + \frac{(1.333)^2}{45.33} = 0.206$, and $t_{0.025}(4) = 2.776$. The 95% confidence interval for the mean is $11.79 \pm 2.776\sqrt{0.48 \cdot 0.206} = 11.79 \pm 0.87$, and the prediction interval is $11.79 \pm 2.776\sqrt{0.48 \cdot 1.206} = 11.79 \pm 2.11$.

4.3 Inference on the correlation

When (X, Y) is bivariate normal with correlation ρ, the conditional mean is linear with slope $\beta_1 = \rho\sigma_Y/\sigma_X$, so $\rho = 0 \iff \beta_1 = 0$.

Theorem 4.6

(a) The t statistic for $\beta_1 = 0$ equals $t_0 = \dfrac{r\sqrt{n-2}}{\sqrt{1-r^2}}$, which is $t(n-2)$ under $\rho = 0$.

(b) (Fisher) $Z = \frac{1}{2}\log\frac{1+r}{1-r}$ is approximately $N\big(\frac{1}{2}\log\frac{1+\rho}{1-\rho}, \frac{1}{n-3}\big)$ for moderate n (say $n \geq 25$).

Proof. (a) $SSE = SST(1-r^2)$ and $\hat\beta_1^2 S_{xx} = SSR = r^2 SST$, so

$$t_0^2 = \frac{\hat\beta_1^2 S_{xx}}{SSE/(n-2)} = \frac{(n-2)r^2}{1-r^2}.$$

The null distribution is conditionally $t(n-2)$ given the x_i, hence unconditionally as well.

(b) is the variance-stabilizing transformation of the asymptotic law $\sqrt{n}(r-\rho) \to N(0, (1-\rho^2)^2)$; the correction $n-3$ improves the approximation. ■

Part (b) gives tests of $H_0 : \rho = \rho_0 \neq 0$ and confidence intervals: $\tanh\big(Z \pm z_{\alpha/2}/\sqrt{n-3}\big)$ covers ρ with probability about $1-\alpha$.

4.4 Residuals

Motivation. All inference rests on model assumptions. Residuals are our window on the unobservable errors, and plotting them is how we check linearity, constant variance and normality.

Residuals are our window on the errors. In vector form $\mathbf{e} = (I - H)\mathbf{y}$ with H the hat matrix (Chapter 6), and:

Theorem 4.7

$Ee_i = 0$ and

$$\operatorname{Var}(e_i) = \sigma^2\Big(1 - \frac{1}{n} - \frac{(x_i-\bar x)^2}{S_{xx}}\Big), \qquad \operatorname{Cov}(e_i, e_j) = -\sigma^2\Big(\frac{1}{n} + \frac{(x_i-\bar x)(x_j-\bar x)}{S_{xx}}\Big) \ (i \neq j).$$

Proof. $e_i = y_i - \bar y - \hat\beta_1(x_i - \bar x)$. Using $\operatorname{Var}\bar y = \sigma^2/n$, $\operatorname{Var}\hat\beta_1 = \sigma^2/S_{xx}$, $\operatorname{Cov}(y_i, \bar y) = \sigma^2/n$, $\operatorname{Cov}(y_i, \hat\beta_1) = \sigma^2 k_i$ and $\operatorname{Cov}(\bar y, \hat\beta_1) = 0$,

$$\operatorname{Var} e_i = \sigma^2\Big(1 + \frac{1}{n} + \frac{(x_i-\bar x)^2}{S_{xx}} - \frac{2}{n} - 2\frac{(x_i-\bar x)^2}{S_{xx}}\Big),$$

which simplifies to the claim; the covariance is computed in the same way. ■

Thus residuals are neither independent nor equally variable; points far from $\bar x$ have smaller residual variance (Chapter 10). Since $\sum e_i = 0$ and $\sum x_i e_i = 0$, residual plots against x_i or $\hat y_i$ have no linear trend by construction; what we look for is *non-random* structure:

- a curved band suggests a missing quadratic term or a transformation;
- a funnel suggests nonconstant variance (weighted least squares or a variance-stabilizing transformation of y);
- a trend in time order suggests correlated errors (Chapter 12).

4.5 Selected exercises

Motivation. The problems below add what the sampling theory above leaves implicit: when the two coefficient estimators are independent, why a one-sided t test has the right size over a composite null, how $E(SSE)$ follows from the residual variances alone, and how the design controls $\operatorname{Var}\hat{\beta}_1$.

Exercise 4.1 (Independence of the coefficient estimators)

Show that $\bar{y}$ and $\hat{\beta}_1$ are always independent, and that $\hat{\beta}_0$ and $\hat{\beta}_1$ are independent if and only if $\bar{x} = 0$.

Solution.

Step 1. (Joint normality.) Each of $\bar{y}$, $\hat{\beta}_0$, $\hat{\beta}_1$ is a fixed linear combination of $\mathbf{y} = (y_1, \dots, y_n)'$, so $(\hat{\beta}_0, \hat{\beta}_1, \bar{y})' = A\mathbf{y}$ for a fixed $3 \times n$ matrix A. Since $\mathbf{y} \sim N_n(X\beta, \sigma^2 I)$, $A\mathbf{y}$ is multivariate normal (Section 5.4).

Step 2. (Uncorrelated implies independent.) For a jointly normal pair (U, W) with $\operatorname{Cov}(U, W) = 0$, the mgf is $\exp\{t_1 EU + t_2 EW + \frac{1}{2}(t_1^2 \operatorname{Var} U + t_2^2 \operatorname{Var} W)\}$, the product of the two marginal mgfs; hence U and W are independent. Conversely, independence implies zero covariance.

Step 3. By Theorem 4.1(b), $\operatorname{Cov}(\bar{y}, \hat{\beta}_1) = 0$, so $\bar{y}$ and $\hat{\beta}_1$ are independent for every design.

Step 4. By Theorem 4.1(c), $\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$, which is zero if and only if $\bar{x} = 0$. By Step 2, $\hat{\beta}_0$ and $\hat{\beta}_1$ are independent if and only if $\bar{x} = 0$. (If $\bar{x} = 0$, then $\hat{\beta}_0 = \bar{y}$, in line with Step 3.) ■

Exercise 4.2 (A one-sided test with a composite null)

Let $0 < \alpha < 1$ and test $H_0 : \beta_1 \le 0$ against $H_1 : \beta_1 > 0$. Show that the test rejecting H_0 when

$$t_0 = \frac{\hat{\beta}_1}{\sqrt{MSE/S_{xx}}} > t_\alpha(n-2)$$

satisfies

$$\sup_{\beta_0 \in \mathbb{R},\ \beta_1 \le 0,\ \sigma^2 > 0} P_{\beta_0, \beta_1, \sigma^2}(\text{the test rejects } H_0) = \alpha,$$

and that its rejection probability exceeds α whenever $\beta_1 > 0$.

Solution.

Step 1. For every parameter value, $T = (\hat{\beta}_1 - \beta_1)/\sqrt{MSE/S_{xx}} \sim t(n-2)$ (Section 4.1). Since $MSE > 0$ with probability one,

$$t_0 = T + \frac{\beta_1}{\sqrt{MSE/S_{xx}}}.$$

Step 2. If $\beta_1 \le 0$, the last term is ≤ 0, so $t_0 \le T$ and $\{t_0 > t_\alpha(n-2)\} \subseteq \{T > t_\alpha(n-2)\}$. Hence the rejection probability is at most $P(T > t_\alpha(n-2)) = \alpha$.

Step 3. If $\beta_1 = 0$, then $t_0 = T$ and the rejection probability is exactly α, for every β_0 and σ^2. With Step 2, the supremum over Θ_0 equals α and is attained on the boundary $\beta_1 = 0$.

Step 4. (Power.) $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$ is independent of MSE (Theorem 4.2). Write $c = t_\alpha(n-2)$ and $\tau = \sigma/\sqrt{S_{xx}}$. Conditioning on MSE,

$$P(t_0 > c) = E\Big[P\big(\hat{\beta}_1 > c\sqrt{MSE/S_{xx}} \mid MSE\big)\Big] = E\Big[1 - \Phi\Big(\frac{c\sqrt{MSE/S_{xx}} - \beta_1}{\tau}\Big)\Big].$$

The integrand is strictly increasing in β_1, so the rejection probability is a strictly increasing function of β_1. It equals α at $\beta_1 = 0$ by Step 3, hence exceeds α for every $\beta_1 > 0$: the test is unbiased. (This also reproves Step 2.) ■

Exercise 4.3 (Expected residual sum of squares without chi-square theory)

(a) Using only Theorem 4.7, show that $E(SSE) = (n-2)\sigma^2$.
(b) Show that $\operatorname{Var}\hat{y}_i + \operatorname{Var} e_i = \sigma^2$ for each i, where $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$, and that $\sum_i \operatorname{Var}\hat{y}_i = 2\sigma^2$.

Solution [Supplement].

(a) Since $Ee_i = 0$, $Ee_i^2 = \operatorname{Var} e_i$. Summing the formula of Theorem 4.7 and using $\sum(x_i - \bar{x})^2 = S_{xx}$,

$$E(SSE) = \sum_i Ee_i^2 = \sigma^2\Big(n - 1 - \frac{S_{xx}}{S_{xx}}\Big) = (n-2)\sigma^2.$$

No normality is needed: only $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I$ were used.

(b) By Theorem 4.5(a) at $x = x_i$, $\operatorname{Var}\hat{y}_i = \sigma^2\{1/n + (x_i - \bar{x})^2/S_{xx}\}$; adding $\operatorname{Var} e_i$ from Theorem 4.7 gives σ^2. Summing over i,

$$\sum_i \operatorname{Var}\hat{y}_i = \sigma^2\Big(1 + \frac{S_{xx}}{S_{xx}}\Big) = 2\sigma^2.$$

So the total variance $n\sigma^2$ of the y_i splits into $2\sigma^2$ in the fitted values (two parameters) and $(n-2)\sigma^2$ in the residuals. ■

Exercise 4.4 (The best design for the slope)

Let n be even, and suppose the experimenter may choose each x_i in a fixed interval $[a, b]$. Show that

$$\operatorname{Var}\hat{\beta}_1 \geq \frac{4\sigma^2}{n(b-a)^2},$$

with equality if and only if half of the x_i equal a and the other half equal b.

Solution [Supplement].

Step 1. For any constant c, $\sum(x_i - c)^2 = S_{xx} + n(\bar{x} - c)^2 \geq S_{xx}$ (expand $x_i - c = (x_i - \bar{x}) + (\bar{x} - c)$; the cross term vanishes). Take $c = m = (a+b)/2$.

Step 2. Since $x_i \in [a, b]$, $|x_i - m| \leq (b-a)/2$. Hence

$$S_{xx} \leq \sum(x_i - m)^2 \leq \frac{n(b-a)^2}{4}, \qquad \operatorname{Var}\hat{\beta}_1 = \frac{\sigma^2}{S_{xx}} \geq \frac{4\sigma^2}{n(b-a)^2}.$$

Step 3. Equality holds in both inequalities of Step 2 if and only if $\bar{x} = m$ and $|x_i - m| = (b-a)/2$ for all i, i.e. every $x_i \in \{a, b\}$ and $\bar{x} = m$. If n_a points are at a and $n - n_a$ at b, then $\bar{x} = m$ means $n_a a + (n - n_a)b = n(a+b)/2$, i.e. $n_a = n/2$. Conversely the half-and-half design gives $S_{xx} = n(b-a)^2/4$ (Exercise 3.2).

Such a design cannot reveal curvature, since $E(y)$ is observed at two points only; in practice some points are placed in the interior to check linearity. ■

Chapter 5. Transformations, Comparisons and Quadratic Forms

5.1 Transformations and the Box–Cox family

When the scatter plot or the residual plot shows curvature or a funnel shape, a transformation of y (or x) can restore linearity, constant variance and normality at once. Common choices are $\log y$, $\sqrt{y}$ and $1/y$. Box and Cox embedded them in one family:

$$y^{(\lambda)} = \begin{cases} \dfrac{y^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log y, & \lambda = 0, \end{cases} \qquad y > 0,$$

which is continuous in λ since $(y^\lambda - 1)/\lambda \to \log y$. $\lambda = 1$ is (essentially) no transformation, $\lambda = \frac{1}{2}$ the square root and $\lambda = -1$ the reciprocal. The model is

$$y_i^{(\lambda)} = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

with λ an additional parameter.

Choosing λ. Comparing SSE across λ directly is meaningless, because the scale of $y^{(\lambda)}$ changes with λ. Divide by the Jacobian: with the geometric mean $\dot{y} = (\prod y_i)^{1/n}$, set

$$z_i^{(\lambda)} = \begin{cases} \dfrac{y_i^\lambda - 1}{\lambda \dot{y}^{\lambda-1}}, & \lambda \neq 0, \\ \dot{y} \log y_i, & \lambda = 0. \end{cases}$$

Then fit $z^{(\lambda)} = \beta_0 + \beta_1 x + \epsilon$ for a grid of λ and choose the λ with the smallest $SSE(\lambda)$.

Proposition 5.1

The profile log-likelihood of λ in the Box–Cox model is $-\frac{n}{2}\log SSE(\lambda) + \text{const}$, where $SSE(\lambda)$ is the residual sum of squares of the regression of $z^{(\lambda)}$. Hence minimizing $SSE(\lambda)$ is maximum likelihood estimation of λ.

Proof. The density of y_i is the normal density of $y_i^{(\lambda)}$ times the Jacobian $y_i^{\lambda-1}$. Maximizing over β and σ^2 for fixed λ gives

$$l_{\max}(\lambda) = -\frac{n}{2}\log\frac{SSE_y(\lambda)}{n} - \frac{n}{2} + (\lambda - 1)\sum \log y_i,$$

where $SSE_y(\lambda)$ is the residual sum of squares for $y^{(\lambda)}$. Since $z^{(\lambda)} = y^{(\lambda)}/\dot{y}^{\lambda-1}$, $SSE(\lambda) = SSE_y(\lambda)\dot{y}^{-2(\lambda-1)}$, and $(\lambda-1)\sum \log y_i = n(\lambda-1)\log \dot{y}$. Substituting, $l_{\max}(\lambda) = -\frac{n}{2}\log SSE(\lambda) + \text{const}$. ∎

An approximate 95% confidence set for λ is $\{\lambda : 2\{l_{\max}(\hat{\lambda}) - l_{\max}(\lambda)\} \le \chi^2_{0.05}(1)\}$; in practice one picks a simple value $(0, \frac{1}{2}, 1)$ inside it.

5.2 Measurement error in the explanatory variable

Suppose the true model is $y_i = \beta_0 + \beta_1 x_i^* + \epsilon_i$, but we observe $x_i = x_i^* + \delta_i$, where δ_i has mean 0 and variance σ_δ^2 and is independent of x_i^* and ϵ_i. Treat the x_i^* as random with variance σ_*^2. Then $y_i = \beta_0 + \beta_1 x_i + (\epsilon_i - \beta_1\delta_i)$, and the new error is correlated with the regressor x_i.

Proposition 5.2 (Attenuation)

The least squares slope from regressing y on the observed x satisfies

$$\hat{\beta}_1 \xrightarrow{p} \frac{\mathrm{Cov}(x,y)}{\mathrm{Var}(x)} = \beta_1 \frac{\sigma_*^2}{\sigma_*^2+\sigma_\delta^2} = \beta_1\Big(1-\frac{\sigma_\delta^2}{\mathrm{Var}\,x}\Big).$$

Proof. $\hat{\beta}_1 = \frac{\frac{1}{n}\sum(x_i-\bar{x})(y_i-\bar{y})}{\frac{1}{n}\sum(x_i-\bar{x})^2}$, a ratio of sample moments that converges to $\mathrm{Cov}(x,y)/\mathrm{Var}(x)$. Here $\mathrm{Cov}(x,y) = \mathrm{Cov}(x^*+\delta, \beta_1 x^* + \epsilon) = \beta_1\sigma_*^2$ and $\mathrm{Var}\,x = \sigma_*^2+\sigma_\delta^2$. ■

So the slope is biased towards zero (**attenuation** or regression dilution). The bias is small when σ_δ^2 is small relative to the spread of the x's. If the experimenter *sets* the observed value x and the actual value fluctuates around it (Berkson's model, $x^* = x + \delta$), the error is independent of x and least squares remains unbiased.

5.3 Comparing two regression lines

Two groups, $i = 1, 2$, give

$$y_{ij} = \beta_{0i} + \beta_{1i}x_{ij} + \epsilon_{ij}, \qquad \epsilon_{ij} \overset{\text{iid}}{\sim} N(0,\sigma^2), \qquad j = 1,\dots,n_i,$$

with a *common* error variance. This is the **full model**.

Are the lines identical? Test $H_0 : \beta_{01} = \beta_{02},\ \beta_{11} = \beta_{12}$. Under H_0 we have the **reduced model** $y_{ij} = \beta_0 + \beta_1 x_{ij} + \epsilon_{ij}$ for the pooled $n_1 + n_2$ observations.

(1) Fit each group separately: $SSE(F) = SSE_1 + SSE_2$, with $df_F = n_1 + n_2 - 4$.

(2) Fit one line to the pooled data: $SSE(R)$, with $df_R = n_1 + n_2 - 2$.

(3) Compute

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/(df_R - df_F)}{SSE(F)/df_F} = \frac{\{SSE(R) - SSE(F)\}/2}{SSE(F)/(n_1+n_2-4)}.$$

(4) Reject H_0 when $F_0 > F_\alpha(2, n_1+n_2-4)$.

That $F_0 \sim F(2, n_1+n_2-4)$ under H_0 follows from the general theory of Chapters 7 and 8: $SSE(F)/\sigma^2 \sim \chi^2(df_F)$, and $\{SSE(R) - SSE(F)\}/\sigma^2 \sim \chi^2(2)$ independently (a general linear hypothesis with two constraints).

Are the lines parallel? For $H_0 : \beta_{11} = \beta_{12}$, the separate slope estimates are independent, so

$$\mathrm{Var}(\hat{\beta}_{11} - \hat{\beta}_{12}) = \sigma^2\Big(\frac{1}{S_{xx,1}} + \frac{1}{S_{xx,2}}\Big), \qquad t_0 = \frac{\hat{\beta}_{11} - \hat{\beta}_{12}}{\sqrt{MSE(F)\big(\frac{1}{S_{xx,1}} + \frac{1}{S_{xx,2}}\big)}} \sim t(n_1+n_2-4)$$

under H_0, with $MSE(F) = SSE(F)/(n_1+n_2-4)$. Reject when $|t_0| > t_{\alpha/2}(n_1+n_2-4)$. The same comparisons can be done in a single regression with indicator variables (Chapter 13).

5.4 Multivariate normal distribution

The distribution theory of regression rests on the multivariate normal distribution and quadratic forms.

Definition (Multivariate normal)

$\mathbf{X} \sim N_n(\mu, \Sigma)$, with Σ symmetric nonnegative definite (possibly singular), if any of the following equivalent conditions holds:

(1) $\mathbf{X} \overset{d}{=} A\mathbf{Z} + \mu$ for some $n \times k$ matrix A with $AA' = \Sigma$ and $\mathbf{Z} \sim N_k(\mathbf{0}, I)$;

(2) $\mathbf{X} \overset{d}{=} \Sigma^{1/2}\mathbf{Z} + \mu$ with $\mathbf{Z} \sim N_n(\mathbf{0}, I)$;

(3) $Ee^{\mathbf{t}'\mathbf{X}} = \exp(\mu'\mathbf{t} + \frac{1}{2}\mathbf{t}'\Sigma\mathbf{t})$ for all $\mathbf{t} \in \mathbb{R}^n$;

(4) $\mathbf{a}'\mathbf{X} \sim N(\mathbf{a}'\mu, \mathbf{a}'\Sigma\mathbf{a})$ for every $\mathbf{a}$.

The equivalences follow from $Ee^{\mathbf{s}'\mathbf{Z}} = e^{\mathbf{s}'\mathbf{s}/2}$: for instance, (1) gives $Ee^{\mathbf{t}'(A\mathbf{Z}+\mu)} = e^{\mathbf{t}'\mu}e^{\frac{1}{2}\mathbf{t}'AA'\mathbf{t}}$, which is (3). Linear maps preserve normality: $B\mathbf{X} + \mathbf{c} \sim N(B\mu + \mathbf{c}, B\Sigma B')$, and jointly normal vectors are independent iff they are uncorrelated.

5.5 Quadratic forms

Noncentral chi-square: if $X_i \sim N(\mu_i, 1)$ are independent, $\sum_{i=1}^r X_i^2 \sim \chi^2(r, \lambda)$ with noncentrality $\lambda = \frac{1}{2}\sum \mu_i^2$. (Some books use $\sum \mu_i^2$ instead of half of it; we follow the half convention of the lecture.)

Theorem 5.3

If $\mathbf{Z} \sim N(\mathbf{0}, I)$ and A is symmetric with $A^2 = A$, then $\mathbf{Z}'A\mathbf{Z} \sim \chi^2(r)$ with $r = \mathrm{tr}A$. Conversely, if $\mathbf{Z}'A\mathbf{Z}$ is chi-square, then $A^2 = A$.

Proof. Diagonalize $A = P\,\mathrm{diag}(\lambda_i)P'$ with P orthogonal. $A^2 = A$ forces $\lambda_i \in \{0, 1\}$, and the number of ones is $\mathrm{tr}A = r$. With $\mathbf{W} = P'\mathbf{Z} \sim N(\mathbf{0}, I)$, $\mathbf{Z}'A\mathbf{Z} = \sum_i \lambda_i W_i^2$, a sum of r squared independent standard normals. For the converse, compare the mgfs $\prod_i (1 - 2\lambda_i t)^{-1/2}$ and $(1 - 2t)^{-r/2}$: equality of the polynomials $\prod(1 - 2\lambda_i t)$ and $(1 - 2t)^r$ forces each $\lambda_i \in \{0, 1\}$. ∎

Theorem 5.4 (Moments of quadratic forms)

Let $\mathbf{y} \sim N(\mu, V)$ and A symmetric. Then

(a) $E(\mathbf{y}'A\mathbf{y}) = \mathrm{tr}(AV) + \mu'A\mu$ (this needs only the mean and covariance);

(b) $\mathrm{Cov}(\mathbf{y}, \mathbf{y}'A\mathbf{y}) = 2VA\mu$;

(c) $\mathrm{Var}(\mathbf{y}'A\mathbf{y}) = 2\mathrm{tr}\{(AV)^2\} + 4\mu'AVA\mu$.

Proof. (a) $\mathbf{y}'A\mathbf{y} = \mathrm{tr}(A\mathbf{y}\mathbf{y}')$ and $E\mathbf{y}\mathbf{y}' = V + \mu\mu'$. (b), (c) Write $\mathbf{y} = \mu + \mathbf{w}$ with $\mathbf{w} \sim N(\mathbf{0}, V)$: $\mathbf{y}'A\mathbf{y} = \mathbf{w}'A\mathbf{w} + 2\mu'A\mathbf{w} + \mu'A\mu$. Odd moments of $\mathbf{w}$ vanish, so $\mathrm{Cov}(\mathbf{w}, \mathbf{w}'A\mathbf{w}) = 0$ and $\mathrm{Cov}(\mathbf{y}, \mathbf{y}'A\mathbf{y}) = \mathrm{Cov}(\mathbf{w}, 2\mu'A\mathbf{w}) = 2VA\mu$. Likewise $\mathrm{Var}(\mathbf{y}'A\mathbf{y}) = \mathrm{Var}(\mathbf{w}'A\mathbf{w}) + 4\mu'AVA\mu$. Write $\mathbf{w} = V^{1/2}\mathbf{z}$ and diagonalize $V^{1/2}AV^{1/2}$ with eigenvalues λ_i: then $\mathrm{Var}(\mathbf{w}'A\mathbf{w}) = 2\sum \lambda_i^2 = 2\mathrm{tr}\{(AV)^2\}$. ∎

Theorem 5.5

Let $\mathbf{y} \sim N(\mu, V)$ with V nonsingular and A symmetric.

(a) $\mathbf{y}'A\mathbf{y} \sim \chi^2(r(A), \frac{1}{2}\mu'A\mu)$ if and only if AV is idempotent.

(b) In particular, if $\mathbf{y} \sim N(\mu, \sigma^2 I)$, then $\mathbf{y}'A\mathbf{y}/\sigma^2 \sim \chi^2(r(A), \frac{1}{2\sigma^2}\mu'A\mu)$ iff A is idempotent; and $\mathbf{y}'\mathbf{y}/\sigma^2 \sim \chi^2(n, \mu'\mu/(2\sigma^2))$.

Proof of sufficiency in (a). Let $\mathbf{z} = V^{-1/2}\mathbf{y} \sim N(V^{-1/2}\mu, I)$ and $B = V^{1/2}AV^{1/2}$. Then $\mathbf{y}'A\mathbf{y} = \mathbf{z}'B\mathbf{z}$, and $B^2 = V^{1/2}AVAV^{1/2} = V^{1/2}AV^{1/2} = B$ when $AVA = A$, which follows from $(AV)^2 = AV$. Diagonalize $B = P\,\mathrm{diag}(I_r, 0)P'$; with $\mathbf{w} = P'\mathbf{z} \sim N(P'V^{-1/2}\mu, I)$, $\mathbf{z}'B\mathbf{z} = \sum_{i\le r} w_i^2$, a noncentral chi-square with $r = \mathrm{tr}B = r(A)$ degrees of freedom and noncentrality $\frac{1}{2}\sum_{i\le r}(Ew_i)^2 = \frac{1}{2}\mu'V^{-1/2}BV^{-1/2}\mu = \frac{1}{2}\mu'A\mu$. Necessity follows by comparing mgfs as in Theorem 5.3. ■

Theorem 5.6 (Independence)

Let $\mathbf{y} \sim N(\mu, V)$.

(a) $\mathbf{y}'A\mathbf{y}$ and $B\mathbf{y}$ are independent if $BVA = 0$.

(b) $\mathbf{y}'A\mathbf{y}$ and $\mathbf{y}'B\mathbf{y}$ (with A, B symmetric) are independent if $AVB = 0$.

(The conditions are also necessary.)

Proof. We give the proof when A and B are nonnegative definite (for instance idempotent), the case used in regression.

(a) Write $A = LL'$ with L of full column rank. $L'\mathbf{y}$ and $B\mathbf{y}$ are jointly normal with $\mathrm{Cov}(B\mathbf{y}, L'\mathbf{y}) = BVL$. From $BVA = BVLL' = 0$, multiplying by $L(L'L)^{-1}$ on the right gives $BVL = 0$, so $L'\mathbf{y}$ and $B\mathbf{y}$ are independent, and $\mathbf{y}'A\mathbf{y} = \|L'\mathbf{y}\|^2$ is a function of $L'\mathbf{y}$.

(b) The same argument applied to both forms: $\mathrm{Cov}(L_A'\mathbf{y}, L_B'\mathbf{y}) = L_A'VL_B = 0$ when $AVB = 0$. ■

Theorem 5.7 (Cochran)

Let $\mathbf{y} \sim N(\mathbf{0}, I_n)$ and let $A_1, \dots, A_p$ be symmetric with $\sum A_j = I_n$ and $r(A_j) = k_j$. Then the forms $\mathbf{y}'A_j\mathbf{y}$ are independent with $\mathbf{y}'A_j\mathbf{y} \sim \chi^2(k_j)$ if and only if $\sum k_j = n$.

Proof of sufficiency. Since $\sum A_j = I$ and $\sum r(A_j) = n$, the column spaces of the A_j together span $\mathbb{R}^n$ with no overlap: $\mathbb{R}^n = \bigoplus_j \mathrm{col}(A_j)$. For $\mathbf{v} \in \mathrm{col}(A_k)$, $\mathbf{v} = \sum_j A_j\mathbf{v}$ with $A_j\mathbf{v} \in \mathrm{col}(A_j)$; uniqueness of the decomposition gives $A_k\mathbf{v} = \mathbf{v}$ and $A_j\mathbf{v} = \mathbf{0}$ for $j \neq k$. Hence $A_k^2 = A_k$ and $A_jA_k = 0$ $(j \neq k)$. Theorems 5.3 and 5.6 finish the proof. ■

5.6 Expected mean squares in simple regression

Motivation. The F-test compares the regression and error mean squares. Their expectations show why: both estimate σ^2 when $\beta_1 = 0$, while the regression mean square is inflated by $\beta_1^2 S_{xx}$ otherwise.

In the centred model $y_i = \beta_0^* + \beta_1(x_i - \bar{x}) + \epsilon_i$:

Proposition 5.8

$$E(SST) = \beta_1^2 S_{xx} + (n-1)\sigma^2, \qquad E(SSR) = \sigma^2 + \beta_1^2 S_{xx}, \qquad E(SSE) = (n-2)\sigma^2.$$

Proof. $y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (\epsilon_i - \bar{\epsilon})$, so $E(SST) = \beta_1^2 S_{xx} + E\sum(\epsilon_i - \bar{\epsilon})^2 = \beta_1^2 S_{xx} + (n-1)\sigma^2$ (the cross term has mean 0). $E(SSR) = S_{xx}E\hat{\beta}_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. Subtract. ■

In matrix form, with $J_n = \mathbf{1}\mathbf{1}'$, $SST = \mathbf{y}'(I - J_n/n)\mathbf{y}$; $I - J_n/n$ is idempotent of rank $n-1$, and Theorem 5.4(a) reproduces $E(SST)$.

5.7 Selected exercises

Motivation. The quadratic-form theorems of Section 5.5 are used here to find exact distributions and variances, and the measurement-error discussion is completed by showing why Berkson-type errors do no harm.

Exercise 5.1 (Quadratic forms in a projection)

Let $\mathbf{y} \sim N_n(\mathbf{0}, I_n)$, let X be a fixed $n \times k$ matrix of rank $k < n$, and $H = X(X'X)^{-1}X'$.
(a) Find the distribution of $\mathbf{y}'H\mathbf{y}$.
(b) Find the distribution of $\mathbf{y}'(I-H)\mathbf{y}$.
(c) Find the distribution of $T = \mathbf{y}'H\mathbf{y}/\mathbf{y}'(I-H)\mathbf{y}$.

Solution.

(a) H is symmetric, and $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$. Its rank equals its trace, $\operatorname{tr}\{(X'X)^{-1}X'X\} = \operatorname{tr} I_k = k$. By Theorem 5.3, $\mathbf{y}'H\mathbf{y} \sim \chi^2(k)$.

(b) $I - H$ is symmetric with $(I-H)^2 = I - 2H + H^2 = I - H$ and $\operatorname{tr}(I-H) = n - k$. By Theorem 5.3, $\mathbf{y}'(I-H)\mathbf{y} \sim \chi^2(n-k)$.

(c) Since $H(I-H) = H - H^2 = 0$, Theorem 5.6(b) (with $V = I$) shows that the two forms are independent. The denominator is a $\chi^2(n-k)$ variable, positive with probability one, so T is well defined almost surely, and

$$\frac{n-k}{k}\,T = \frac{\mathbf{y}'H\mathbf{y}/k}{\mathbf{y}'(I-H)\mathbf{y}/(n-k)} \sim F(k, n-k).$$

Thus T is distributed as $\frac{k}{n-k}F$ with $F \sim F(k, n-k)$. ■

Exercise 5.2 (Variance of the residual mean square)

(a) Let $y_1, \dots, y_n$ be i.i.d. $N(\mu, \sigma^2)$ and $S^2 = \sum(y_i - \bar{y})^2/(n-1)$. Use Theorem 5.4 to show $\operatorname{Var} S^2 = 2\sigma^4/(n-1)$.
(b) In the regression model $\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$ with X of size $n \times (p+1)$ and full column rank, show $\operatorname{Var}(MSE) = 2\sigma^4/(n-p-1)$.

Solution [Supplement].

(a) $S^2 = \mathbf{y}'A\mathbf{y}$ with $A = (I - J_n/n)/(n-1)$, $\mathbf{y} \sim N(\mu\mathbf{1}, \sigma^2 I)$. Since $(I - J_n/n)\mathbf{1} = \mathbf{0}$, $A\mu\mathbf{1} = \mathbf{0}$, so the second term of Theorem 5.4(c) vanishes. Using that $I - J_n/n$ is idempotent with trace $n-1$,

$$\operatorname{Var} S^2 = 2\operatorname{tr}\{(\sigma^2 A)^2\} = \frac{2\sigma^4}{(n-1)^2}\operatorname{tr}\Big(I - \frac{J_n}{n}\Big) = \frac{2\sigma^4}{n-1}.$$

(b) $MSE = \mathbf{y}'B\mathbf{y}$ with $B = (I-H)/(n-p-1)$. Since $(I-H)X = 0$, $BX\beta = \mathbf{0}$, and since $I - H$ is idempotent with trace $n-p-1$,

$$\operatorname{Var}(MSE) = 2\sigma^4 \operatorname{tr}(B^2) = \frac{2\sigma^4}{(n-p-1)^2}\operatorname{tr}(I-H) = \frac{2\sigma^4}{n-p-1}.$$

(The same follows from $SSE/\sigma^2 \sim \chi^2(n-p-1)$, whose variance is $2(n-p-1)$.) The relative standard error of MSE is $\sqrt{2/(n-p-1)}$, which is large when the residual degrees of freedom are few. ■

Exercise 5.3 (Berkson errors)

The experimenter sets the nominal values $x_1, \dots, x_n$ (fixed, with $S_{xx} > 0$), but the actual values are $x_i^* = x_i + \delta_i$, and $y_i = \beta_0 + \beta_1 x_i^* + \epsilon_i$. Assume the $2n$ variables $\epsilon_1, \dots, \epsilon_n, \delta_1, \dots, \delta_n$ are independent, with $E\epsilon_i = E\delta_i = 0$, $\operatorname{Var}\epsilon_i = \sigma^2$ and $\operatorname{Var}\delta_i = \sigma_\delta^2$. Show that the least squares slope from regressing y on the nominal x is unbiased for β_1, with variance $(\sigma^2 + \beta_1^2\sigma_\delta^2)/S_{xx}$.

Solution [Supplement].

Step 1. Substituting x_i^*, $y_i = \beta_0 + \beta_1 x_i + u_i$ with $u_i = \epsilon_i + \beta_1\delta_i$. The u_i are independent (functions of disjoint sets of independent variables), with $Eu_i = 0$ and $\operatorname{Var} u_i = \sigma^2 + \beta_1^2\sigma_\delta^2$. So the regression of y on the *fixed* x_i is an ordinary simple linear regression with error variance $\sigma^2 + \beta_1^2\sigma_\delta^2$.

Step 2. With $k_i = (x_i - \bar{x})/S_{xx}$ as in Section 4.1, $\hat{\beta}_1 = \sum k_i y_i$, so $E\hat{\beta}_1 = \sum k_i(\beta_0 + \beta_1 x_i) = \beta_1$ and $\operatorname{Var}\hat{\beta}_1 = (\sigma^2 + \beta_1^2\sigma_\delta^2)\sum k_i^2 = (\sigma^2 + \beta_1^2\sigma_\delta^2)/S_{xx}$.

The contrast with Proposition 5.2 is in Step 1: there the error δ_i sits in the *observed* regressor, so the composite error is correlated with it; here the regressor is fixed and the error only adds noise. If ϵ_i and δ_i are normal, u_i is normal and all the inference of Chapter 4 applies with the inflated variance. ■

Chapter 6. Multiple Linear Regression

6.1 The model in matrix form

With p explanatory variables,

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i, \qquad i = 1, \ldots, n,$$

or, stacking the observations,

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n),$$

where $\mathbf{y}$ is $n \times 1$, X is the $n \times (p+1)$ **design matrix** whose i-th row is $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$, and $\beta = (\beta_0, \ldots, \beta_p)'$. We assume $n > p+1$ and $\operatorname{rank} X = p+1$. Simple regression is the case $p = 1$, with rows $(1, x_i)$.

6.2 Least squares

Motivation. With several predictors, matrix notation turns all the simple-regression formulas into one: $\hat{\boldsymbol{\beta}} = (X^\top X)^{-1} X^\top \mathbf{y}$, the projection of $\mathbf{y}$ onto the column space of X.

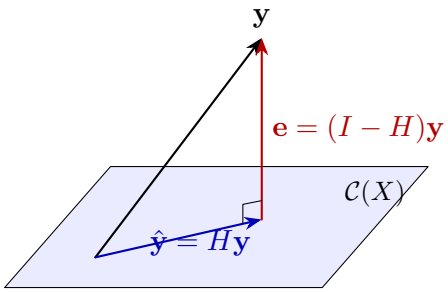


Figure. *The geometry of least squares: $\hat{\mathbf{y}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space $\mathcal{C}(X)$, and the residual vector $\mathbf{e}$ is perpendicular to every column of X, which is exactly the normal equations $X^\top \mathbf{e} = \mathbf{0}$.*

Theorem 6.1

The least squares estimator minimizing $S(\beta) = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta)$ is the solution of the normal equations $X'X\hat{\beta} = X'\mathbf{y}$:

$$\hat{\beta} = (X'X)^{-1} X'\mathbf{y}.$$

Proof. $S(\beta) = \mathbf{y}'\mathbf{y} - 2\beta' X'\mathbf{y} + \beta' X'X\beta$ has gradient $-2X'\mathbf{y} + 2X'X\beta$ and Hessian $2X'X$, which is positive definite because X has full column rank ($\mathbf{a}'X'X\mathbf{a} = \|X\mathbf{a}\|^2 > 0$ for $\mathbf{a} \neq \mathbf{0}$). So the unique stationary point is the global minimum. Alternatively, for any β,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat{\beta}\|^2 + \|X(\hat{\beta} - \beta)\|^2,$$

because the cross term $2(\hat{\beta} - \beta)' X'(\mathbf{y} - X\hat{\beta})$ vanishes by the normal equations. ∎

For $p = 1$,

$$X'X = \begin{pmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}, \qquad (X'X)^{-1} = \frac{1}{n S_{xx}} \begin{pmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{pmatrix},$$

which reproduces the formulas of Chapter 3.

Example 6.1

For $n = 6$ observations of (x_1, x_2, y):

x_1	1	2	3	4	5	6
x_2	2	1	4	3	6	5
y	3	6	7	10	12	13

Centring gives $\bar{x}_1 = \bar{x}_2 = 3.5$, $\bar{y} = 8.5$, and the centred cross products $S_{11} = S_{22} = 17.5$, $S_{12} = 14.5$, $S_{1y} = 35.5$, $S_{2y} = 28.5$. The slopes solve

$$\begin{pmatrix} 17.5 & 14.5 \\ 14.5 & 17.5 \end{pmatrix} \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} 35.5 \\ 28.5 \end{pmatrix},$$

so, with determinant $17.5^2 - 14.5^2 = 96$,

$$\hat{\beta}_1 = \frac{17.5 \cdot 35.5 - 14.5 \cdot 28.5}{96} = 2.167, \qquad \hat{\beta}_2 = \frac{17.5 \cdot 28.5 - 14.5 \cdot 35.5}{96} = -0.167,$$

and $\hat{\beta}_0 = 8.5 - 3.5(2.167 - 0.167) = 1.5$. Note that y is strongly positively correlated with x_2 on its own ($S_{2y} > 0$), yet the coefficient of x_2 is slightly negative once x_1 is in the model: x_2 carries little information beyond x_1 (Section 6.6).

The **fitted values** are $\hat{\mathbf{y}} = X\hat{\beta} = H\mathbf{y}$ with the **hat matrix** $H = X(X'X)^{-1}X'$, and the residuals are $\mathbf{e} = (I - H)\mathbf{y}$. H is symmetric and idempotent with $HX = X$ and $\mathrm{tr}H = p + 1$: it is the orthogonal projection onto the column space of X. Hence $X'\mathbf{e} = \mathbf{0}$: the residuals sum to zero (the intercept column) and are orthogonal to every regressor and to $\hat{\mathbf{y}}$.

6.3 Analysis of variance

With $J_n = \mathbf{1}\mathbf{1}'$, $n\bar{y}^2 = \mathbf{y}'(J_n/n)\mathbf{y}$ is the **correction term**. The sums of squares are quadratic forms:

$$SST = \mathbf{y}'\Big(I - \frac{J_n}{n}\Big)\mathbf{y},$$

$$SSE = \mathbf{y}'(I - H)\mathbf{y} = \mathbf{y}'\mathbf{y} - \hat{\beta}'X'\mathbf{y}, \qquad SSR = \mathbf{y}'\Big(H - \frac{J_n}{n}\Big)\mathbf{y} = \hat{\beta}'X'\mathbf{y} - n\bar{y}^2,$$

with $SST = SSR + SSE$. The degrees of freedom are the ranks of the matrices: $n - 1$, p and $n - p - 1$. (SST has $n - 1$ degrees of freedom because the n deviations $y_i - \bar{y}$ satisfy one linear constraint.)

source	SS	df	MS	F_0
regression	SSR	p	$MSR = SSR/p$	MSR/MSE
residual	SSE	$n - p - 1$	$MSE = SSE/(n - p - 1)$	
total	SST	$n - 1$		

Theorem 6.2 (Expected mean squares)

$$E(MSE) = \sigma^2, \qquad E(MSR) = \sigma^2 + \frac{1}{p}\beta'X'\Big(I - \frac{J_n}{n}\Big)X\beta.$$

The second term is ≥ 0, with equality iff $\beta_1 = \cdots = \beta_p = 0$.

Proof. By Theorem 5.4(a) with $E\mathbf{y} = X\beta$ and $\mathrm{Var}\,\mathbf{y} = \sigma^2 I$: for $B = I - H$, $\mathrm{tr}B = n - (p+1)$ and $X'BX = 0$, so $E(SSE) = (n - p - 1)\sigma^2$. For $A = H - J_n/n$, $\mathrm{tr}A = (p + 1) - 1 = p$ and $X'AX =$

$X'X - X'J_nX/n = X'(I - J_n/n)X$, since $HX = X$. Finally $\beta'X'(I - J_n/n)X\beta = \|(I - J_n/n)X\beta\|^2$ is the squared length of the centred mean vector, which is zero iff $X\beta$ is constant, i.e. iff all slopes vanish (the centred columns of X are linearly independent). ■

For $p = 1$, $\beta'X'(I - J_n/n)X\beta = \beta_1^2 S_{xx}$, recovering Chapter 5. The **coefficient of determination** is $R^2 = SSR/SST = 1 - SSE/SST$, and $\hat{\sigma}^2 = MSE$ is the **residual mean square**.

6.4 Properties of the estimators

Motivation. Writing $\hat{\boldsymbol{\beta}}$ as a linear function of $\mathbf{y}$ makes its mean and covariance immediate, and the hat matrix H summarizes how fitted values depend on observations.

Theorem 6.3

(a) $E\hat{\beta} = \beta$ and $\operatorname{Var}\hat{\beta} = \sigma^2(X'X)^{-1}$; writing $(X'X)^{-1} = (c_{ij})$, $\operatorname{Var}\hat{\beta}_j = c_{jj}\sigma^2$ and $\operatorname{Cov}(\hat{\beta}_i, \hat{\beta}_j) = c_{ij}\sigma^2$.

(b) At a point $\mathbf{x} = (1, x_1, \ldots, x_p)'$, $\hat{y} = \mathbf{x}'\hat{\beta}$ estimates $E(y \mid \mathbf{x})$ with $\operatorname{Var}\hat{y} = \mathbf{x}'(X'X)^{-1}\mathbf{x}\,\sigma^2$.

(c) A new observation y_s at $\mathbf{x}$ has $\operatorname{Var}(y_s - \hat{y}) = \{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}\sigma^2$.

Proof. (a) $E\hat{\beta} = (X'X)^{-1}X'X\beta = \beta$ and $\operatorname{Var}\hat{\beta} = (X'X)^{-1}X'(\sigma^2 I)X(X'X)^{-1} = \sigma^2(X'X)^{-1}$.

(b) $\operatorname{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}'\operatorname{Var}(\hat{\beta})\mathbf{x}$.

(c) y_s is independent of $\hat{y}$. ■

For $p = 1$ these reduce to $\operatorname{Var}\hat{\beta}_0 = \sigma^2(1/n + \bar{x}^2/S_{xx})$, $\operatorname{Var}\hat{\beta}_1 = \sigma^2/S_{xx}$, $\operatorname{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\bar{x}\sigma^2/S_{xx}$ and $\operatorname{Var}\hat{y} = \sigma^2\{1/n + (x - \bar{x})^2/S_{xx}\}$, so $\hat{\beta}_0$ and $\hat{\beta}_1$ are uncorrelated iff $\bar{x} = 0$. The precision of $\hat{\beta}$ depends on the design through $(X'X)^{-1}$; choosing the design well (spreading the x's, avoiding near-collinear columns) reduces the variances.

6.5 Distribution theory

Motivation. Exact tests in multiple regression need the joint distribution of $\hat{\boldsymbol{\beta}}$ and the residual sum of squares. Under normal errors they are independent, and SSE/σ^2 is χ^2, which yields t and F tests.

Theorem 6.4

Under $\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$:

(a) $SSE/\sigma^2 \sim \chi^2(n - p - 1)$;

(b) $SSR/\sigma^2 \sim \chi^2(p, \lambda)$ with $\lambda = \beta'X'(I - J_n/n)X\beta/(2\sigma^2)$;

(c) $SST/\sigma^2 \sim \chi^2(n - 1, \lambda)$ with the same λ;

(d) SSR and SSE are independent, and $\hat{\beta}$ is independent of SSE;

(e) $F_0 = MSR/MSE \sim F(p, n{-}p{-}1, \lambda)$; under $H_0 : \beta_1 = \cdots = \beta_p = 0$, $F_0 \sim F(p, n{-}p{-}1)$.

Proof. All three matrices $I - H$, $H - J_n/n$ and $I - J_n/n$ are symmetric idempotent: for $H - J_n/n$ use $HJ_n = J_nH = J_n$ (since $H\mathbf{1} = \mathbf{1}$, the intercept column) and $J_n^2 = nJ_n$:

$$\left(H - \frac{J_n}{n}\right)^2 = H - \frac{J_n}{n} - \frac{J_n}{n} + \frac{J_n}{n} = H - \frac{J_n}{n}.$$

Their ranks (traces) are $n-p-1$, p and $n-1$. Theorem 5.5(b) gives (a)–(c), the noncentralities being $\frac{1}{2\sigma^2}(X\beta)'A(X\beta)$; for $I-H$ it is 0 because $(I-H)X=0$.

(d) $(H-J_n/n)(I-H)=0$, so Theorem 5.6(b) gives independence of SSR and SSE. Also $\hat\beta=(X'X)^{-1}X'\mathbf{y}$ and $(X'X)^{-1}X'(I-H)=0$, so Theorem 5.6(a) applies.

(e) follows from (a), (b), (d). ■

Consequently, with c_{jj} the diagonal of $(X'X)^{-1}$,

$$\frac{\hat\beta_j-\beta_j}{\sqrt{c_{jj}MSE}}\sim t(n-p-1).$$

Example 6.2

In Example 6.1, $SST=\sum(y_i-8.5)^2=73.5$ and $SSR=\hat\beta_1 S_{1y}+\hat\beta_2 S_{2y}=2.167\cdot 35.5-0.167\cdot 28.5=72.17$, so $SSE=1.33$ with 3 degrees of freedom and $R^2=0.982$. Then $F_0=(72.17/2)/(1.33/3)=81.2>F_{0.05}(2,3)=9.55$: the regression is highly significant.

6.6 Partial regression: the Frisch–Waugh–Lovell theorem

Partition $X=(X_1,X_2)$, with X_1 $(n\times(q+1))$ containing the intercept and X_2 $(n\times(p-q))$ the remaining columns, and write $\mathbf{y}=X_1\beta_1+X_2\beta_2+\epsilon$. Let $H_1=X_1(X_1'X_1)^{-1}X_1'$.

Theorem 6.5 (Frisch–Waugh–Lovell)

Let $\mathbf{e}=(I-H_1)\mathbf{y}$ be the residuals of regressing $\mathbf{y}$ on X_1, and $X_{2\cdot1}=(I-H_1)X_2$ the residuals of regressing each column of X_2 on X_1. The least squares coefficient from regressing $\mathbf{e}$ on $X_{2\cdot1}$,

$$\tilde\beta_2=(X_{2\cdot1}'X_{2\cdot1})^{-1}X_{2\cdot1}'\mathbf{e},$$

equals the $\hat\beta_2$ part of the full least squares estimator. Moreover the residuals of the two regressions coincide.

Proof [Supplement]. The normal equations of the full model are

$$\begin{pmatrix}X_2'X_2 & X_2'X_1\\ X_1'X_2 & X_1'X_1\end{pmatrix}\begin{pmatrix}\hat\beta_2\\ \hat\beta_1\end{pmatrix}=\begin{pmatrix}X_2'\mathbf{y}\\ X_1'\mathbf{y}\end{pmatrix}.$$

Multiply on the left by $\begin{pmatrix}I & -X_2'X_1(X_1'X_1)^{-1}\\ 0 & I\end{pmatrix}$. The first block row becomes

$$X_2'(I-H_1)X_2\hat\beta_2+\{X_2'X_1-X_2'H_1X_1\}\hat\beta_1=X_2'(I-H_1)\mathbf{y},$$

and the bracket is 0 since $H_1X_1=X_1$. Hence $\hat\beta_2=\{X_2'(I-H_1)X_2\}^{-1}X_2'(I-H_1)\mathbf{y}$. Because $I-H_1$ is symmetric idempotent, this is $(X_{2\cdot1}'X_{2\cdot1})^{-1}X_{2\cdot1}'\mathbf{e}=\tilde\beta_2$. For the residuals: the full fit is $X_1\hat\beta_1+X_2\hat\beta_2$; applying $I-H_1$ to $\mathbf{y}-X_1\hat\beta_1-X_2\hat\beta_2$ (which is orthogonal to X_1) gives $\mathbf{e}-X_{2\cdot1}\hat\beta_2$, the residual of the second regression. ■

Interpretation

A multiple regression coefficient $\hat\beta_j$ measures the effect of x_j on y *after both have been adjusted for the other regressors*: it is the slope of the residuals of y against the residuals of x_j. It is

generally different from the simple regression slope of y on x_j, and the two agree when x_j is orthogonal to the other (centred) regressors.

6.7 Selected exercises

Motivation. Three facts about the matrix form of least squares: the estimator and the residual vector are independent, the multiple correlation is the largest correlation attainable by a linear combination of the regressors, and the fit does not depend on how the column space is parametrized.

Exercise 6.1 (Joint distribution of $\hat\beta$ and e)

Under $\mathbf{y} \sim N_n(X\beta, \sigma^2 I_n)$, find the joint distribution of $(\hat\beta', \mathbf{e}')'$, with its mean vector and covariance matrix, and deduce that $\hat\beta$ and $\mathbf{e}$ are independent.

Solution.

Step 1. $\begin{pmatrix}\hat\beta \\ \mathbf{e}\end{pmatrix} = M\mathbf{y}$ with $M = \begin{pmatrix}(X'X)^{-1}X' \\ I - H\end{pmatrix}$, a fixed $(p+1+n)\times n$ matrix, so the vector is multivariate normal (Section 5.4).

Step 2. (Mean.) $E\hat\beta = \beta$ and $E\mathbf{e} = (I-H)X\beta = \mathbf{0}$, since $HX = X$.

Step 3. (Covariance.) $\operatorname{Var}(M\mathbf{y}) = \sigma^2 MM'$. The diagonal blocks are $(X'X)^{-1}X'X(X'X)^{-1} = (X'X)^{-1}$ and $(I-H)(I-H)' = I-H$; the off-diagonal block is $(X'X)^{-1}X'(I-H) = (X'X)^{-1}(X'-X'H) = 0$, because $X'H = (HX)' = X'$. Hence

$$\begin{pmatrix}\hat\beta \\ \mathbf{e}\end{pmatrix} \sim N_{p+1+n}\left(\begin{pmatrix}\beta \\ \mathbf{0}\end{pmatrix},\ \sigma^2\begin{pmatrix}(X'X)^{-1} & 0 \\ 0 & I-H\end{pmatrix}\right).$$

The distribution is singular, since $I-H$ has rank $n-p-1 < n$.

Step 4. (Independence.) By the mgf characterization (3) of Section 5.4, for $\mathbf{t} = (\mathbf{t}_1', \mathbf{t}_2')'$ the joint mgf is

$$\exp\{\mathbf{t}_1'\beta + \tfrac{\sigma^2}{2}\mathbf{t}_1'(X'X)^{-1}\mathbf{t}_1\}\cdot\exp\{\tfrac{\sigma^2}{2}\mathbf{t}_2'(I-H)\mathbf{t}_2\},$$

the product of the mgfs of $\hat\beta$ and of $\mathbf{e}$. So $\hat\beta$ and $\mathbf{e}$ are independent. In particular $\hat\beta$ is independent of every function of $\mathbf{e}$, such as $SSE = \mathbf{e}'\mathbf{e}$ and the residual plots, which strengthens Theorem 6.4(d). ■

Exercise 6.2 (Multiple correlation as a maximal correlation)

Consider $Y_i = \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$ with $\sum_i Y_i = 0$ and $\sum_i x_{ij} = 0$ for every j (centred data, no intercept), $X = (x_{ij})$ of size $n \times p$ with full rank and $n > p$. Let $\hat\beta$ be the least squares estimator, $\hat Y_i = \sum_j \hat\beta_j x_{ij}$, $SSR = \sum \hat Y_i^2$ and $SST = \sum Y_i^2$. Write $\langle \mathbf{a}, \mathbf{b}\rangle = \sum a_i b_i$ and $\|\mathbf{a}\|^2 = \langle \mathbf{a}, \mathbf{a}\rangle$.

(a) Express SSR and SST through X and $\mathbf{Y}$.

(b) Assuming $SST \neq 0$, prove that

$$\max_{\beta\in\mathbb{R}^p,\ \beta\neq\mathbf{0}} \frac{\langle X\beta, \mathbf{Y}\rangle}{\|X\beta\|\,\|\mathbf{Y}\|} = \sqrt{\frac{SSR}{SST}}.$$

(c) Which β attain the maximum in (b)?

Solution.

(a) $\hat{\mathbf{Y}} = H\mathbf{Y}$ with $H = X(X'X)^{-1}X'$ symmetric idempotent, so

$$SSR = \mathbf{Y}'H'H\mathbf{Y} = \mathbf{Y}'X(X'X)^{-1}X'\mathbf{Y}, \qquad SST = \mathbf{Y}'\mathbf{Y}.$$

(b) For $\beta \neq \mathbf{0}$, $X\beta \neq \mathbf{0}$ by full rank, so the ratio is defined. Since $X\beta$ lies in the column space, $HX\beta = X\beta$, and by symmetry of H and the Cauchy–Schwarz inequality,

$$\langle X\beta, \mathbf{Y}\rangle = \langle HX\beta, \mathbf{Y}\rangle = \langle X\beta, H\mathbf{Y}\rangle \leq \|X\beta\|\,\|H\mathbf{Y}\| = \|X\beta\|\sqrt{SSR}.$$

Dividing by $\|X\beta\|\|\mathbf{Y}\| = \|X\beta\|\sqrt{SST}$ shows that the ratio is at most $\sqrt{SSR/SST}$. If $SSR > 0$, the choice $\beta = \hat{\beta}$ gives $\langle H\mathbf{Y}, \mathbf{Y}\rangle/(\|H\mathbf{Y}\|\|\mathbf{Y}\|) = SSR/\sqrt{SSR \cdot SST} = \sqrt{SSR/SST}$, so the bound is attained. If $SSR = 0$, then $H\mathbf{Y} = \mathbf{0}$ and the ratio is 0 for every β, which again equals $\sqrt{SSR/SST}$.

(c) If $SSR > 0$, equality in the Cauchy–Schwarz step with a positive left side holds iff $X\beta = cH\mathbf{Y} = cX\hat{\beta}$ for some $c > 0$, i.e. (full rank) iff $\beta = c\hat{\beta}$ with $c > 0$. If $SSR = 0$, every $\beta \neq \mathbf{0}$ attains the maximum.

Because the data are centred, the ratio in (b) is the sample correlation between $\mathbf{Y}$ and the linear combination $X\beta$. So the multiple correlation $R = \sqrt{R^2}$ is the largest correlation between y and any linear combination of the regressors, attained by the fitted values. ■

Exercise 6.3 (Reparametrization does not change the fit)

Let A be a nonsingular $(p+1) \times (p+1)$ matrix and $Z = XA$.
(a) Show that the least squares coefficient of $\mathbf{y}$ on Z is $\hat{\gamma} = A^{-1}\hat{\beta}$, with $\operatorname{Var}\hat{\gamma} = \sigma^2 A^{-1}(X'X)^{-1}(A^{-1})'$, and that the hat matrix, the fitted values, the residuals, SSE, MSE and R^2 are the same for X and Z.
(b) Replace each regressor x_j by $z_j = (x_j - c_j)/d_j$ with constants c_j and $d_j \neq 0$, keeping the intercept. Show that the new slopes are $d_j\hat{\beta}_j$ and that the t statistics of the slopes change at most in sign.

Solution [Supplement].

(a) Z has full column rank, because $Z\mathbf{a} = \mathbf{0}$ gives $XA\mathbf{a} = \mathbf{0}$, so $A\mathbf{a} = \mathbf{0}$ and $\mathbf{a} = \mathbf{0}$. Then

$$\hat{\gamma} = (A'X'XA)^{-1}A'X'\mathbf{y} = A^{-1}(X'X)^{-1}(A')^{-1}A'X'\mathbf{y} = A^{-1}\hat{\beta},$$

and $\operatorname{Var}\hat{\gamma} = \sigma^2(Z'Z)^{-1} = \sigma^2 A^{-1}(X'X)^{-1}(A^{-1})'$. The hat matrix is

$$Z(Z'Z)^{-1}Z' = XAA^{-1}(X'X)^{-1}(A')^{-1}A'X' = H,$$

so the fitted values $H\mathbf{y}$, the residuals $(I - H)\mathbf{y}$ and SSE coincide; MSE has the same degrees of freedom $n - p - 1$, and SST does not involve the design, so R^2 coincides too.

(b) The new design has columns $\mathbf{1}$ and $\mathbf{z}_j = (\mathbf{x}_j - c_j\mathbf{1})/d_j$, so $Z = XA$ with A upper triangular with diagonal $(1, 1/d_1, \ldots, 1/d_p)$, hence nonsingular. By (a) the fitted values agree:

$$\hat{\beta}_0\mathbf{1} + \sum_j \hat{\beta}_j\mathbf{x}_j = \hat{\gamma}_0\mathbf{1} + \sum_j \hat{\gamma}_j\frac{\mathbf{x}_j - c_j\mathbf{1}}{d_j}.$$

Since the columns of X are linearly independent, the coefficients of $\mathbf{x}_j$ match: $\hat{\gamma}_j = d_j\hat{\beta}_j$, and $\hat{\gamma}_0 = \hat{\beta}_0 + \sum_j \hat{\beta}_j c_j$. The same linear identity holds for the random vectors, so $\operatorname{Var}\hat{\gamma}_j = d_j^2 \operatorname{Var}\hat{\beta}_j$, i.e. c_{jj} is multiplied by d_j^2. With the same MSE,

$$\frac{\hat{\gamma}_j}{\sqrt{d_j^2 c_{jj} MSE}} = \frac{d_j}{|d_j|} \cdot \frac{\hat{\beta}_j}{\sqrt{c_{jj} MSE}}.$$

So centring and rescaling the regressors (as in Chapter 15) change the units of the slopes but no test or fitted value. ■

Chapter 7. Inference in Multiple Regression I

Throughout, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times (p+1)$ and full column rank. We distinguish two sets of assumptions:

- **(A)** $\epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n)$, needed for exact t and F inference;
- **(B)** $E\epsilon = \mathbf{0}$ and $\operatorname{Var}\epsilon = \sigma^2 I_n$ only, enough for the optimality of least squares.

7.1 The Gauss–Markov theorem

Motivation. Why least squares? Among all linear unbiased estimators, it has the smallest variance for every linear combination of coefficients, under uncorrelated errors with equal variances and without normality.

Theorem 7.1 (Gauss–Markov)

Under (B), among all linear unbiased estimators $\tilde{\beta} = A\mathbf{y}$ of β, the least squares estimator $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ has the smallest variance: $\operatorname{Var}\tilde{\beta} - \operatorname{Var}\hat{\beta}$ is nonnegative definite. Equivalently, for every $\mathbf{c}$, $\mathbf{c}'\hat{\beta}$ has the smallest variance among linear unbiased estimators of $\mathbf{c}'\beta$.

Proof [Supplement]. $E(A\mathbf{y}) = AX\beta = \beta$ for all β forces $AX = I$. Write $A = (X'X)^{-1}X' + D$; then $DX = AX - I = 0$. Hence

$$\operatorname{Var}(A\mathbf{y}) = \sigma^2 AA' = \sigma^2\{(X'X)^{-1} + (X'X)^{-1}X'D' + DX(X'X)^{-1} + DD'\} = \sigma^2(X'X)^{-1} + \sigma^2 DD',$$

and DD' is nonnegative definite, with $\mathbf{c}'DD'\mathbf{c} = \|D'\mathbf{c}\|^2$. Equality for all $\mathbf{c}$ holds iff $D = 0$. ■

So $\hat{\beta}$ is the **best linear unbiased estimator** (BLUE). The theorem needs neither normality nor independence, only uncorrelated errors with a common variance. If $\operatorname{Var}\epsilon = \sigma^2 V$ with $V \neq I$, the BLUE is the generalized least squares estimator (Chapter 12).

7.2 Confidence intervals

Motivation. Individual coefficient intervals follow from the t distribution, but each one describes the effect of a predictor holding the others fixed, which is why coefficients change when variables are added.

Under (A), with $(X'X)^{-1} = (c_{ij})$ and $MSE = SSE/(n-p-1)$ independent of $\hat{\beta}$ (Theorem 6.4):

Theorem 7.2

$100(1-\alpha)\%$ confidence intervals are:

(a) for a coefficient: $\hat{\beta}_j \pm t_{\alpha/2}(n-p-1)\sqrt{c_{jj}MSE}$;

(b) for a linear combination $\mathbf{q}'\beta$: $\mathbf{q}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{q}'(X'X)^{-1}\mathbf{q}\,MSE}$;

(c) for the mean response at $\mathbf{x}$: $\mathbf{x}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\mathbf{x}'(X'X)^{-1}\mathbf{x}\,MSE}$;

(d) a prediction interval for a new observation at $\mathbf{x}$: $\mathbf{x}'\hat{\beta} \pm t_{\alpha/2}(n-p-1)\sqrt{\{1 + \mathbf{x}'(X'X)^{-1}\mathbf{x}\}MSE}$.

With σ^2 known, replace MSE by σ^2 and $t_{\alpha/2}$ by $z_{\alpha/2}$.

Proof. (b) contains (a) and (c) as special cases. $\mathbf{q}'\hat\beta \sim N(\mathbf{q}'\beta, \sigma^2\mathbf{q}'(X'X)^{-1}\mathbf{q})$ is independent of $(n-p-1)MSE/\sigma^2 \sim \chi^2(n-p-1)$, so the studentized ratio is $t(n-p-1)$.

(d) adds the variance of the new error. ■

7.3 The general linear hypothesis

Many hypotheses are linear restrictions on β:

$$H_0 : C\beta = \mathbf{m} \quad \text{vs} \quad H_1 : C\beta \neq \mathbf{m},$$

with C a known $k\times(p+1)$ matrix of rank k and $\mathbf{m} \in \mathbb{R}^k$. Examples, for $E(y) = \beta_0+\beta_1x_1+\cdots+\beta_4x_4$:

- $\beta_1 = \beta_2$ and $\beta_3 = 2\beta_4$: $C = \begin{pmatrix} 0 & 1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 & -2 \end{pmatrix}$, $\mathbf{m} = \mathbf{0}$;
- $\beta_1 = \beta_2 = \beta_3 = \beta_4$: three rows $(0,1,-1,0,0)$, $(0,0,1,-1,0)$, $(0,0,0,1,-1)$;
- $\beta_j = 0$: $C = \mathbf{e}_j'$, the j-th unit row.

There are two equivalent routes: minimize SSE under the restriction, or reparametrize to a reduced model.

7.3.1 Method I: constrained least squares

Theorem 7.3 (Restricted least squares)

The minimizer of $\|\mathbf{y} - X\beta\|^2$ subject to $C\beta = \mathbf{m}$ is

$$\tilde\beta = \hat\beta - (X'X)^{-1}C'\{C(X'X)^{-1}C'\}^{-1}(C\hat\beta - \mathbf{m}),$$

and its residual sum of squares is

$$SSE(\tilde\beta) = SSE + Q, \qquad Q = (C\hat\beta - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat\beta - \mathbf{m}).$$

Proof [Supplement]. With a Lagrange multiplier vector $\lambda \in \mathbb{R}^k$, minimize $L = (\mathbf{y} - X\beta)'(\mathbf{y} - X\beta) + 2\lambda'(C\beta - \mathbf{m})$. Setting derivatives to zero:

$$X'X\tilde\beta + C'\lambda = X'\mathbf{y}, \qquad C\tilde\beta = \mathbf{m}.$$

The first gives $\tilde\beta = \hat\beta - (X'X)^{-1}C'\lambda$, and substituting into the second, $C\hat\beta - C(X'X)^{-1}C'\lambda = \mathbf{m}$, so $\lambda = \{C(X'X)^{-1}C'\}^{-1}(C\hat\beta - \mathbf{m})$. ($C(X'X)^{-1}C'$ is invertible since C has full row rank.) That this stationary point is the constrained minimum follows from the decomposition below. For any β,

$$\|\mathbf{y} - X\beta\|^2 = \|\mathbf{y} - X\hat\beta\|^2 + (\hat\beta - \beta)'X'X(\hat\beta - \beta),$$

since $X'(\mathbf{y} - X\hat\beta) = \mathbf{0}$. With $\beta = \tilde\beta$ and $\hat\beta - \tilde\beta = (X'X)^{-1}C'\lambda$,

$$(\hat\beta - \tilde\beta)'X'X(\hat\beta - \tilde\beta) = \lambda'C(X'X)^{-1}C'\lambda = (C\hat\beta - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\hat\beta - \mathbf{m}) = Q.$$

For any feasible β, $(\hat\beta - \beta)'X'X(\hat\beta - \beta) \geq Q$ (it is the squared distance, in the $X'X$ metric, from $\hat\beta$ to the affine set $\{C\beta = \mathbf{m}\}$, attained at $\tilde\beta$), so $\tilde\beta$ is optimal. ■

Theorem 7.4 (F test of the general linear hypothesis)

Under (A),

$$\frac{Q}{\sigma^2} \sim \chi^2(k, \lambda), \qquad \lambda = \frac{(C\beta - \mathbf{m})'\{C(X'X)^{-1}C'\}^{-1}(C\beta - \mathbf{m})}{2\sigma^2},$$

Q is independent of SSE, and

$$F_0 = \frac{Q/k}{SSE/(n-p-1)} = \frac{Q}{k \cdot MSE} \sim F(k, n-p-1, \lambda).$$

Under H_0, $\lambda = 0$, and the test rejects when $F_0 > F_\alpha(k, n-p-1)$.

Proof. $C\hat{\beta} - \mathbf{m} \sim N(C\beta - \mathbf{m}, \sigma^2 C(X'X)^{-1}C')$, and Q is the corresponding quadratic form with matrix $W = \{C(X'X)^{-1}C'\}^{-1}$. Since $W \cdot \sigma^2 C(X'X)^{-1}C'/\sigma^2 = I_k$ is idempotent, Theorem 5.5 gives the noncentral chi-square with k degrees of freedom. Q is a function of $\hat{\beta}$, which is independent of SSE (Theorem 6.4). ■

7.3.2 Method II: full and reduced models

Substitute the restriction into the model to get a **reduced model** with $p + 1 - k$ free parameters; fit it and compute

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/k}{SSE(F)/(n-p-1)},$$

where $SSE(F) = SSE$ is from the full model, with $df_F = n - p - 1$ and $df_R = n - p - 1 + k$. If the restriction involves a nonzero $\mathbf{m}$, move the known part to the left-hand side (an offset). Since $SSE(R)$ is exactly the constrained minimum $SSE(\tilde{\beta})$, Theorem 7.3 gives $SSE(R) - SSE(F) = Q$: *the two methods give the same statistic.*

For a model without intercept, $\mathbf{y} = X\beta + \epsilon$ with X of size $n \times p$, the same holds with $n - p$ in place of $n - p - 1$.

Example 7.1

A no-intercept model $y = \beta_1 x_1 + \beta_2 x_2 + \epsilon$ is fitted to

x_1	1	2	0	1	3
x_2	0	1	1	2	1
y	3	7	2	5	10

Test $H_0 : \beta_1 = 2\beta_2$. Here

$$X'X = \begin{pmatrix} 15 & 7 \\ 7 & 7 \end{pmatrix}, \qquad X'\mathbf{y} = \begin{pmatrix} 52 \\ 29 \end{pmatrix}, \qquad \hat{\beta} = \frac{1}{56}\begin{pmatrix} 7 & -7 \\ -7 & 15 \end{pmatrix}\begin{pmatrix} 52 \\ 29 \end{pmatrix} = \begin{pmatrix} 2.875 \\ 1.268 \end{pmatrix},$$

$\mathbf{y}'\mathbf{y} = 187$, and $SSE = 187 - \hat{\beta}'X'\mathbf{y} = 0.732$ with 3 degrees of freedom, so $MSE = 0.244$. *Method I*: $C = (1, -2)$, $C\hat{\beta} = 0.339$, $C(X'X)^{-1}C' = \frac{7+28+60}{56} = 1.696$, so $Q = 0.339^2/1.696 = 0.068$ and $F_0 = 0.068/0.244 = 0.28$. *Method II*: under H_0, $y = \beta_2(2x_1 + x_2) + \epsilon$; with $z = 2x_1 + x_2 = (2, 5, 1, 4, 7)$, $\sum z^2 = 95$ and $\sum zy = 133$, so $SSE(R) = 187 - 133^2/95 = 0.800$ and $\{SSE(R) - SSE\}/MSE = 0.068/0.244 = 0.28$. Both give $F_0 = 0.28 < F_{0.05}(1, 3) = 10.13$: H_0 is not rejected.

7.4 Properties of residuals

Motivation. Residuals are not the errors: they are correlated and have unequal variances $\sigma^2(1-h_{ii})$. This motivates the standardized and studentized residuals of Chapter 10.

Theorem 7.5

For the fitted model with intercept:

(a) $\sum_i e_i = 0$, $\sum_i x_{ij} e_i = 0$ for each j, and $\sum_i \hat{y}_i e_i = 0$;

(b) under (A), $\mathbf{e} = (I-H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I-H))$; in particular $\operatorname{Var} e_i = \sigma^2(1-h_{ii})$ and $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$.

Proof. (a) is $X'\mathbf{e} = \mathbf{0}$ together with $\hat{\mathbf{y}}'\mathbf{e} = \hat{\beta}' X'\mathbf{e}$.

(b) $E\mathbf{e} = (I-H)X\beta = \mathbf{0}$ and $\operatorname{Var}\mathbf{e} = \sigma^2(I-H)(I-H)' = \sigma^2(I-H)$. ■

So residuals are correlated, with correlation $-h_{ij}/\sqrt{(1-h_{ii})(1-h_{jj})}$, and have unequal variances depending on the design. Plots of residuals against $\hat{y}_i$, against each x_{ij}, and against time order are the basic checks:

- a horizontal band: no problem;
- a funnel: nonconstant variance; use weighted least squares or transform y;
- a linear trend against x_j: a computational error, since (a) forces zero correlation;
- a curve: add a quadratic term (or other function) of x_j, or transform.

7.5 Selected exercises

Motivation. The general linear hypothesis is applied to a model whose mean is itself constrained and to weights that must sum to one; a last exercise shows what imposing a restriction does to the bias and the variance of the estimator.

Exercise 7.1 (An F statistic under a side condition)

Let $\mathbf{y} = \theta + \epsilon$ with $\mathbf{y}, \theta, \epsilon \in \mathbb{R}^4$ and $\epsilon \sim N_4(\mathbf{0}, \sigma^2 I_4)$, where θ is known to satisfy $\theta_1+\theta_2+\theta_3+\theta_4 = 0$. Show that the F statistic for $H_0 : \theta_1 = \theta_3$ is

$$F_0 = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2},$$

and state its null distribution.

Solution.

Step 1. (The model as a regression.) The full model says $\theta \in \Omega = \{\theta : \mathbf{1}'\theta = 0\}$, a 3-dimensional subspace. With the basis $(1,-1,0,0)'$, $(0,1,-1,0)'$, $(0,0,1,-1)'$ of Ω as columns of a 4×3 matrix X of full rank, the model is $\mathbf{y} = X\beta + \epsilon$ without intercept, and H_0 is one linear restriction on β. So Method II of Section 7.3 applies: $F_0 = \{SSE(R) - SSE(F)\}/\{SSE(F)/(4-3)\}$, with 1 numerator degree of freedom. For a model whose mean ranges over a subspace L, $SSE = \|\mathbf{y} - P_L\mathbf{y}\|^2$ with P_L the orthogonal projection onto L (Theorem 6.1).

Step 2. (Full model.) Ω is the orthogonal complement of $\mathbf{1}$, so $P_\Omega \mathbf{y} = \mathbf{y} - \bar{y}\mathbf{1}$ and

$$SSE(F) = \|\bar{y}\mathbf{1}\|^2 = 4\bar{y}^2 = \frac{(y_1 + y_2 + y_3 + y_4)^2}{4}.$$

Step 3. (Reduced model.) Let $\mathbf{c} = (1, 0, -1, 0)'$. Then $\mathbf{c} \in \Omega$, and the reduced mean space is $\omega = \{\theta \in \Omega : \mathbf{c}'\theta = 0\}$, the orthogonal complement of $\mathbf{c}$ inside Ω. Hence $\Omega = \omega \oplus \operatorname{span}(\mathbf{c})$ orthogonally and $P_\Omega = P_\omega + P_{\mathbf{c}}$, with $P_{\mathbf{c}}\mathbf{y} = (\mathbf{c}'\mathbf{y}/\|\mathbf{c}\|^2)\mathbf{c}$. Since $\mathbf{y} - P_\Omega\mathbf{y} \perp \Omega \ni P_{\mathbf{c}}\mathbf{y}$, Pythagoras gives

$$SSE(R) = \|\mathbf{y} - P_\Omega\mathbf{y} + P_{\mathbf{c}}\mathbf{y}\|^2 = SSE(F) + \frac{(\mathbf{c}'\mathbf{y})^2}{\|\mathbf{c}\|^2} = SSE(F) + \frac{(y_1 - y_3)^2}{2}.$$

Step 4. Therefore

$$F_0 = \frac{(y_1 - y_3)^2/2}{(y_1 + y_2 + y_3 + y_4)^2/4} = \frac{2(y_1 - y_3)^2}{(y_1 + y_2 + y_3 + y_4)^2},$$

and under H_0, $F_0 \sim F(1, 1)$ by Theorem 7.4 (with $n - p$ in place of $n - p - 1$ for a model without intercept, here $4 - 3 = 1$). ■

Exercise 7.2 (Weights that sum to one)

In a survey, n respondents report scores $X_{i1}, \ldots, X_{ip}$ on p items and an overall score Y_i. The model is

$$Y_i = \sum_{j=1}^{p} \omega_j X_{ij} + \epsilon_i, \qquad \sum_{j=1}^{p} \omega_j = 1, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 I_n),$$

with $X = (X_{ij})$ of size $n \times p$ and full rank, $n > p$ (negative weights allowed). Derive the least squares estimator of $\omega = (\omega_1, \ldots, \omega_p)'$ and an unbiased estimator of σ^2, and give their distributions.

Solution. Write $G = (X'X)^{-1}$, $\hat{\omega} = GX'\mathbf{Y}$ (the unconstrained estimator), $\mathbf{g} = G\mathbf{1}$ and $s = \mathbf{1}'G\mathbf{1} > 0$.

Step 1. (Estimator.) Theorem 7.3 with $C = \mathbf{1}'$ ($k = 1$) and $\mathbf{m} = 1$ gives

$$\tilde{\omega} = \hat{\omega} - \frac{\mathbf{1}'\hat{\omega} - 1}{s}\mathbf{g} = \Big(I - \frac{\mathbf{g}\mathbf{1}'}{s}\Big)\hat{\omega} + \frac{\mathbf{g}}{s},$$

and indeed $\mathbf{1}'\tilde{\omega} = \mathbf{1}'\hat{\omega} - (\mathbf{1}'\hat{\omega} - 1) = 1$.

Step 2. (Reduced model.) Substituting $\omega_p = 1 - \sum_{j<p} \omega_j$,

$$Y_i - X_{ip} = \sum_{j<p} \omega_j (X_{ij} - X_{ip}) + \epsilon_i,$$

an unconstrained regression of $\mathbf{Y} - \mathbf{X}_p$ on $Z = XB$, where B is the $p \times (p-1)$ matrix with columns $\mathbf{e}_j - \mathbf{e}_p$. B has rank $p - 1$ and X full column rank, so Z has full column rank $p - 1$. The map $(\omega_1, \ldots, \omega_{p-1}) \mapsto \omega$ is a bijection onto the constraint set, and $\|\mathbf{Y} - X\omega\|^2$ equals the residual sum of squares of the reduced regression at $(\omega_1, \ldots, \omega_{p-1})$. Hence $\tilde{\omega}$ is this regression's estimator (completed by $\tilde{\omega}_p = 1 - \sum_{j<p} \tilde{\omega}_j$), and $SSE(\tilde{\omega}) = \|\mathbf{Y} - X\tilde{\omega}\|^2$ is its residual sum of squares, with $n - (p - 1)$ degrees of freedom.

Step 3. (σ^2.) Theorem 6.4, applied to the reduced regression (its proof uses only that the hat matrix of Z is the projection onto the column space of Z), gives $SSE(\tilde{\omega})/\sigma^2 \sim \chi^2(n - p + 1)$, independent of the reduced coefficients and hence of $\tilde{\omega}$. Thus

$$\tilde{\sigma}^2 = \frac{\|\mathbf{Y} - X\tilde{\omega}\|^2}{n - p + 1}$$

is unbiased, and $(n - p + 1)\tilde{\sigma}^2/\sigma^2 \sim \chi^2(n - p + 1)$.

Step 4. (Distribution of $\tilde{\omega}$.) $\tilde{\omega}$ is an affine function of $\hat{\omega} \sim N_p(\omega, \sigma^2 G)$, hence normal. Since $\mathbf{1}'\omega = 1$, $E\tilde{\omega} = \omega - \frac{1-1}{s}\mathbf{g} = \omega$. With $M = I - \mathbf{g}\mathbf{1}'/s$,

$$\operatorname{Var}\tilde{\omega} = \sigma^2 MGM' = \sigma^2\Big(G - \frac{\mathbf{g}\mathbf{g}'}{s} - \frac{\mathbf{g}\mathbf{g}'}{s} + \frac{\mathbf{g}(\mathbf{1}'G\mathbf{1})\mathbf{g}'}{s^2}\Big) = \sigma^2\Big(G - \frac{\mathbf{g}\mathbf{g}'}{s}\Big).$$

So $\tilde{\omega} \sim N_p(\omega, \sigma^2(G - \mathbf{g}\mathbf{g}'/s))$, a singular normal distribution concentrated on the hyperplane $\mathbf{1}'\omega = 1$: indeed $(G - \mathbf{g}\mathbf{g}'/s)\mathbf{1} = \mathbf{g} - \mathbf{g} = \mathbf{0}$. Replacing σ^2 by $\tilde{\sigma}^2$ gives t intervals for each ω_j with $n - p + 1$ degrees of freedom. ■

Exercise 7.3 (Bias and variance of restricted least squares)

Under assumption (B), let $\tilde{\beta}$ be the restricted estimator of Theorem 7.3, and $G = (X'X)^{-1}$, $W = \{CGC'\}^{-1}$.
(a) Show that $E\tilde{\beta} = \beta - GC'W(C\beta - \mathbf{m})$; in particular $\tilde{\beta}$ is unbiased when the restriction is true.
(b) Show that $\operatorname{Var}\tilde{\beta} = \sigma^2(G - GC'WCG)$, so that $\operatorname{Var}\hat{\beta} - \operatorname{Var}\tilde{\beta}$ is nonnegative definite whether or not the restriction is true.

Solution [Supplement].

(a) By Theorem 7.3, $\tilde{\beta} = M\hat{\beta} + GC'W\mathbf{m}$ with $M = I - GC'WC$. Since $E\hat{\beta} = \beta$,

$$E\tilde{\beta} = \beta - GC'WC\beta + GC'W\mathbf{m} = \beta - GC'W(C\beta - \mathbf{m}),$$

which equals β when $C\beta = \mathbf{m}$.

(b) $\operatorname{Var}\tilde{\beta} = M(\sigma^2 G)M'$. Using $G' = G$, $W' = W$ and $W(CGC')W = W$,

$$\begin{aligned} MGM' &= (G - GC'WCG)(I - C'WCG) \\ &= G - GC'WCG - GC'WCG + GC'W(CGC')WCG = G - GC'WCG. \end{aligned}$$

Hence $\operatorname{Var}\hat{\beta} - \operatorname{Var}\tilde{\beta} = \sigma^2(CG)'W(CG)$, and $\mathbf{a}'(CG)'W(CG)\mathbf{a} = \mathbf{b}'W\mathbf{b} \geq 0$ with $\mathbf{b} = CG\mathbf{a}$, since W is positive definite. So a correct restriction yields an unbiased estimator with smaller variance, while a false one buys the same variance reduction at the price of the bias in (a). ■

Chapter 8. Inference in Multiple Regression II

8.1 Extra sums of squares

When regressors are added to a model, SSR increases (and SSE decreases). The increase is the **extra sum of squares** due to the added regressors, and it measures their contribution *given* those already present.

For the full model $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \epsilon$, write

$$SS(\hat\beta_0, \ldots, \hat\beta_p) = \hat\beta' X'\mathbf{y} \quad (df = p+1), \qquad SS(\hat\beta_0) = n\bar y^2 \quad (df = 1),$$
$$SS(\hat\beta_1, \ldots, \hat\beta_p \mid \hat\beta_0) = SS(\hat\beta_0, \ldots, \hat\beta_p) - SS(\hat\beta_0) = SSR \quad (df = p).$$

If a submodel uses only $x_1, \ldots, x_q$ ($q < p$) with fit $\tilde\beta$, the extra sum of squares of $x_{q+1}, \ldots, x_p$ given $x_1, \ldots, x_q$ is

$$SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q) = SS(\hat\beta) - SS(\tilde\beta) = SSR(\text{full}) - SSR(\text{sub}) = SSE(\text{sub}) - SSE(\text{full}),$$

with $p - q$ degrees of freedom. In matrix form, partition $\beta = (\beta_1', \beta_2')'$ with $\beta_1 = (\beta_0, \ldots, \beta_q)'$, $X = (X_1, X_2)$, and let $H_1 = X_1(X_1'X_1)^{-1}X_1'$, $H = X(X'X)^{-1}X'$:

$$SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H - H_1)\mathbf{y}.$$

Theorem 8.1

(a) $H - H_1$ is symmetric idempotent of rank $p - q$, and $(H - H_1)(I - H) = 0$.

(b) Under normal errors, $SS(\beta_2 \mid \beta_1)/\sigma^2 \sim \chi^2(p-q, \lambda)$ with $\lambda = \beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$, independent of SSE.

(c) Hence, under $H_0 : \beta_{q+1} = \cdots = \beta_p = 0$,

$$F_0 = \frac{SS(\hat\beta_{q+1}, \ldots, \hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_q)/(p-q)}{MSE} \sim F(p-q, n-p-1).$$

Proof. (a) $HX = X$ gives $HX_1 = X_1$, hence $HH_1 = H_1$ and (transposing) $H_1H = H_1$. Then $(H-H_1)^2 = H - H_1 - H_1 + H_1 = H - H_1$, and $\operatorname{tr}(H - H_1) = (p+1) - (q+1)$. Also $(H - H_1)(I - H) = H - H - H_1 + H_1 = 0$.

(b) By Theorem 5.5(b) the noncentrality is $\frac{1}{2\sigma^2}(X\beta)'(H - H_1)X\beta$. Since $(H - H_1)X_1 = 0$ and $(H - H_1)X_2 = (I - H_1)X_2$, this equals $\beta_2' X_2'(I - H_1)X_2\beta_2/(2\sigma^2)$. Independence follows from (a) and Theorem 5.6.

(c) Under H_0, $\lambda = 0$. ■

This is the same test as the general linear hypothesis with $C = (0, I_{p-q})$ (Chapter 7), since $SSE(\text{sub}) - SSE(\text{full}) = Q$.

When the intercept is included in the regression sum of squares, one uses the uncorrected table: $SSR = \hat\beta' X'\mathbf{y}$ with $p + 1$ degrees of freedom and $SST = \mathbf{y}'\mathbf{y}$ with n; then $MSR/MSE \sim F(p + 1, n - p - 1)$ tests $\beta = \mathbf{0}$.

8.2 Partial and sequential F tests

Partial F test. The contribution of a single regressor x_j given all others is $SS(\hat\beta_j \mid \text{all others})$, with one degree of freedom. The test of $H_0 : \beta_j = 0$,

$$F_0 = \frac{SS(\hat\beta_j \mid \hat\beta_0, \ldots, \hat\beta_{j-1}, \hat\beta_{j+1}, \ldots, \hat\beta_p)}{MSE} \sim F(1, n-p-1),$$

is the **partial F test**. It is equivalent to the t test, since

$$SS(\hat\beta_j \mid \text{others}) = \frac{\hat\beta_j^2}{c_{jj}}, \qquad F_0 = \Big(\frac{\hat\beta_j}{\sqrt{c_{jj}MSE}}\Big)^2 = t_0^2.$$

(This follows from Theorem 7.3 with $C = \mathbf{e}_j'$: $Q = \hat\beta_j^2/c_{jj}$.) A non-significant partial F means that x_j adds little *once the other variables are in the model*; it does not mean that x_j is unrelated to y. With correlated regressors, every partial F can be small while the overall F is large.

Sequential F tests. Enter the regressors one at a time in a chosen order, and at each stage test the newly added variable given those already in:

$$SS(\hat\beta_1 \mid \hat\beta_0), \quad SS(\hat\beta_2 \mid \hat\beta_0, \hat\beta_1), \quad \ldots, \quad SS(\hat\beta_p \mid \hat\beta_0, \ldots, \hat\beta_{p-1}).$$

These sum to SSR, and each is compared with the MSE of the model at that stage (or of the full model). The decomposition depends on the order unless the centred regressors are mutually orthogonal. The tests are valid only if the MSE used is unbiased for σ^2, which requires that the omitted variables have no effect; otherwise the denominator is inflated.

8.3 Joint confidence regions and simultaneous intervals

Motivation. Several individual 95% intervals do not hold simultaneously with probability 95%. Joint regions and Bonferroni or Scheffé intervals control the error rate for all statements at once.

Theorem 8.2 (Joint confidence region)

Under normal errors,

$$\frac{(\hat\beta-\beta)'X'X(\hat\beta-\beta)}{(p+1)MSE} \sim F(p+1, n-p-1),$$

so $\{\beta : (\hat\beta-\beta)'X'X(\hat\beta-\beta) \le (p+1)MSE\, F_\alpha(p+1,n-p-1)\}$ is a $100(1-\alpha)\%$ confidence ellipsoid for β.

Proof. $\hat\beta - \beta = (X'X)^{-1}X'\epsilon$, so $(\hat\beta-\beta)'X'X(\hat\beta-\beta) = \epsilon' H \epsilon$, and $\epsilon' H\epsilon/\sigma^2 \sim \chi^2(p+1)$. It is independent of $SSE = \epsilon'(I-H)\epsilon$, since $H(I-H)=0$. ∎

Separate 95% intervals for $\beta_0, \ldots, \beta_p$ do not have joint coverage 95%. Two remedies:

Theorem 8.3 (Bonferroni and Scheffé intervals)

(a) (Bonferroni) The intervals $\hat\beta_j \pm t_{\alpha/(2(p+1))}(n-p-1)\sqrt{c_{jj}MSE}$, $j = 0, \ldots, p$, cover all β_j simultaneously with probability at least $1-\alpha$.

(b) (Scheffé) With probability $1-\alpha$, simultaneously for all $\mathbf{a} \in \mathbb{R}^{p+1}$,

$$\mathbf{a}'\beta \in \mathbf{a}'\hat\beta \pm \sqrt{(p+1)F_\alpha(p+1,n-p-1)}\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}.$$

Proof. (a) If A_j is the event that the j-th interval covers, $P(A_j^c) = \alpha/(p+1)$, and $P(\bigcap A_j) \ge 1 - \sum P(A_j^c) = 1-\alpha$.

(b) By the Cauchy–Schwarz inequality in the inner product defined by $(X'X)^{-1}$,

$$\max_{\mathbf{a}\neq\mathbf{0}} \frac{\{\mathbf{a}'(\hat\beta-\beta)\}^2}{\mathbf{a}'(X'X)^{-1}\mathbf{a}} = (\hat\beta-\beta)'X'X(\hat\beta-\beta),$$

so the event in (b) is exactly the joint confidence region of Theorem 8.2. ■

Taking $\mathbf{a} = \mathbf{x} = (1, x_1, \ldots, x_p)'$ in (b) gives a **confidence band** for the whole regression surface, valid for all $\mathbf{x}$ at once. For a few coefficients Bonferroni intervals are shorter; Scheffé intervals pay for covering infinitely many combinations.

8.4 Testing equality of regression models

Two populations follow

$$\mathbf{y}_1 = X_1\beta_1 + \epsilon_1, \qquad \mathbf{y}_2 = X_2\beta_2 + \epsilon_2,$$

with n_1 and n_2 observations, design matrices of size $n_i \times (p+1)$, and independent $N(\mathbf{0}, \sigma^2 I)$ errors with a *common* σ^2. To test $H_0 : \beta_1 = \beta_2$:

(1) Full model: fit each separately, $SSE(F) = SSE_1 + SSE_2$, with $n-2(p+1)$ degrees of freedom $(n = n_1 + n_2)$.

(2) Reduced model: stack $\mathbf{y} = (\mathbf{y}_1', \mathbf{y}_2')'$ and $X = (X_1', X_2')'$ and fit one β_0: $SSE(R)$, with $n-(p+1)$ degrees of freedom.

(3) Under H_0,

$$F_0 = \frac{\{SSE(R) - SSE(F)\}/(p+1)}{SSE(F)/(n-2p-2)} \sim F(p+1, n-2p-2).$$

The reduced model is the full model (with block-diagonal design $\mathrm{diag}(X_1, X_2)$ and parameter $(\beta_1', \beta_2')'$) under the $p+1$ linear restrictions $\beta_1 - \beta_2 = \mathbf{0}$, so this is again a general linear hypothesis. For k groups and $H_0 : \beta_1 = \cdots = \beta_k$, the numerator has $(k-1)(p+1)$ degrees of freedom and the denominator $n - k(p+1)$.

The test assumes equal error variances; with $k = 2$ this can be checked first by $F = MSE_1/MSE_2 \sim F(n_1 - p - 1, n_2 - p - 1)$.

8.5 Selected exercises

Motivation. The exercises derive the equality test of Section 8.4 from the difference of the two fitted coefficient vectors, rewrite partial F statistics through R^2, show when sequential sums of squares do not depend on the order, and compare Scheffé and t intervals.

Exercise 8.1 (Equality of two regressions via $\hat\beta_1 - \hat\beta_2$)

In the setting of Section 8.4, let $\hat\beta_1, \hat\beta_2$ be the separate least squares estimators, $G_i = (X_i'X_i)^{-1}$, $\hat\Delta = \hat\beta_1 - \hat\beta_2$ and $MSE = (SSE_1 + SSE_2)/(n-2p-2)$.
(a) Find the distribution of $\hat\Delta$, and show that $(n-2p-2)MSE/\sigma^2 \sim \chi^2(n-2p-2)$, independent of $\hat\Delta$.
(b) Show that under $H_0 : \beta_1 = \beta_2$,

$$F_1 = \frac{\hat\Delta'(G_1 + G_2)^{-1}\hat\Delta/(p+1)}{MSE} \sim F(p+1, n-2p-2).$$

(c) Show that F_1 equals the statistic F_0 of Section 8.4.

Solution.

(a) By Theorem 6.3, $\hat\beta_i \sim N_{p+1}(\beta_i, \sigma^2 G_i)$. The two samples are independent, so $\hat\beta_1$ and $\hat\beta_2$ are independent and $\hat\Delta \sim N_{p+1}(\beta_1-\beta_2, \sigma^2(G_1+G_2))$. By Theorem 6.4, $SSE_i/\sigma^2 \sim \chi^2(n_i-p-1)$ independently of $\hat\beta_i$; together with the independence of the samples, the four quantities $\hat\beta_1, SSE_1, \hat\beta_2, SSE_2$ are mutually independent. Hence $(SSE_1+SSE_2)/\sigma^2 \sim \chi^2(n-2p-2)$, independent of $\hat\Delta$.

(b) $\Sigma = \sigma^2(G_1+G_2)$ is positive definite. Under H_0, $\hat\Delta \sim N(\mathbf{0}, \Sigma)$, and with $A = \Sigma^{-1}$ we have $A\Sigma = I$, which is idempotent of rank $p+1$; Theorem 5.5(a) gives $\hat\Delta'\Sigma^{-1}\hat\Delta \sim \chi^2(p+1)$. Dividing by $(n-2p-2)MSE/\sigma^2$ and using (a),

$$F_1 = \frac{\hat\Delta'\Sigma^{-1}\hat\Delta/(p+1)}{\{(n-2p-2)MSE/\sigma^2\}/(n-2p-2)} \sim F(p+1, n-2p-2).$$

(c) Write the full model as one regression with design $X_F = \operatorname{diag}(X_1, X_2)$ and parameter $\mathbf{b} = (\beta_1', \beta_2')'$. Then $X_F'X_F = \operatorname{diag}(X_1'X_1, X_2'X_2)$, so the normal equations decouple, $\hat{\mathbf{b}} = (\hat\beta_1', \hat\beta_2')'$ and $SSE(F) = SSE_1 + SSE_2$. The hypothesis is $C\mathbf{b} = \mathbf{0}$ with $C = (I_{p+1}, -I_{p+1})$, and

$$C(X_F'X_F)^{-1}C' = G_1+G_2, \qquad C\hat{\mathbf{b}} = \hat\Delta, \qquad Q = \hat\Delta'(G_1+G_2)^{-1}\hat\Delta.$$

By Theorem 7.3 (Method II), $SSE(R) - SSE(F) = Q$, so $F_0 = \{Q/(p+1)\}/MSE = F_1$. ■

Exercise 8.2 (Extra sums of squares through R^2)

Let the full model (with intercept) have p regressors and coefficient of determination R_F^2, and the submodel the first q of them, with R_R^2.
(a) Show that the F statistic of Theorem 8.1(c) equals

$$F_0 = \frac{(R_F^2 - R_R^2)/(p-q)}{(1-R_F^2)/(n-p-1)}.$$

(b) Show that $R_F^2 \geq R_R^2$, with equality if and only if $\hat\beta_{q+1} = \cdots = \hat\beta_p = 0$ in the full fit.

Solution [Supplement].

(a) Both models contain the intercept and are fitted to the same $\mathbf{y}$, so they share SST, and $SSE = SST(1-R^2)$ for each. Hence $SS(\beta_2 \mid \beta_1) = SSE(\text{sub}) - SSE(\text{full}) = SST(R_F^2 - R_R^2)$ and $MSE = SST(1-R_F^2)/(n-p-1)$; SST cancels in the ratio.

(b) $R_F^2 - R_R^2 = \mathbf{y}'(H-H_1)\mathbf{y}/SST = \|(H-H_1)\mathbf{y}\|^2/SST \geq 0$, because $H - H_1$ is symmetric idempotent (Theorem 8.1(a)). By Theorem 6.5, the full-model residual is $(I-H)\mathbf{y} = (I-H_1)\mathbf{y} - X_{2\cdot1}\hat\beta_2$, so

$$(H-H_1)\mathbf{y} = (I-H_1)\mathbf{y} - (I-H)\mathbf{y} = X_{2\cdot1}\hat\beta_2.$$

$X_{2\cdot1} = (I-H_1)X_2$ has full column rank: if $X_{2\cdot1}\mathbf{a} = \mathbf{0}$, then $X_2\mathbf{a} = H_1X_2\mathbf{a}$ lies in the column space of X_1, contradicting the full rank of X unless $\mathbf{a} = \mathbf{0}$. Therefore $R_F^2 = R_R^2$ iff $X_{2\cdot1}\hat\beta_2 = \mathbf{0}$ iff $\hat\beta_2 = \mathbf{0}$. ■

Exercise 8.3 (When the order of entry does not matter)

Partition $X = (X_1, X_2)$ as in Section 8.1, X_1 containing the intercept, and suppose $X_1'X_2 = 0$ (for instance, the columns of X_2 are centred and uncorrelated with those of X_1). Let $H_2 = X_2(X_2'X_2)^{-1}X_2'$.
(a) Show that $H = H_1 + H_2$, and hence $SS(\beta_2 \mid \beta_1) = \mathbf{y}'H_2\mathbf{y}$, which does not involve X_1.
(b) Show that $\hat\beta_2 = (X_2'X_2)^{-1}X_2'\mathbf{y}$.
(c) Deduce that if the centred regressors are split into mutually orthogonal blocks, the sequential sums of squares of the blocks do not depend on the order of entry.

Solution [Supplement].

(a) $H_1H_2 = X_1(X_1'X_1)^{-1}(X_1'X_2)(X_2'X_2)^{-1}X_2' = 0$, and likewise $H_2H_1 = 0$. So $P = H_1 + H_2$ is symmetric with $P^2 = H_1 + H_2 = P$, an orthogonal projection. Its range contains the columns of X_1 and X_2 ($PX_1 = X_1$, $PX_2 = X_2$) and is contained in the column space of X, so it equals the column space of X, and $P = H$ by uniqueness of the orthogonal projection. Then $SS(\beta_2 \mid \beta_1) = \mathbf{y}'(H - H_1)\mathbf{y} = \mathbf{y}'H_2\mathbf{y}$.

(b) $(I - H_1)X_2 = X_2 - X_1(X_1'X_1)^{-1}X_1'X_2 = X_2$, so $X_{2\cdot 1} = X_2$, and by Theorem 6.5, $\hat\beta_2 = (X_2'X_2)^{-1}X_2'(I - H_1)\mathbf{y} = (X_2'X_2)^{-1}X_2'\mathbf{y}$, using $X_2'H_1 = 0$.

(c) Let the centred blocks $Z_1, \dots, Z_m$ satisfy $Z_j'Z_k = 0$ for $j \neq k$ and $\mathbf{1}'Z_j = \mathbf{0}$. Whatever blocks have entered before Z_j, together with the intercept they form an X_1 orthogonal to Z_j, so by (a) the sequential sum of squares of Z_j is $\mathbf{y}'H_{Z_j}\mathbf{y}$ in every order. These sum to SSR, by (a) applied repeatedly. ■

Exercise 8.4 (Scheffé intervals are wider than t intervals)

Let $\nu = n - p - 1$. Show that $(p+1)F_\alpha(p+1, \nu) \geq t_{\alpha/2}(\nu)^2$, so that for a single combination $\mathbf{a}'\beta$ the Scheffé interval of Theorem 8.3(b) contains the t interval of Theorem 7.2(b).

Solution [Supplement].

Step 1. Fix $\mathbf{a} \neq \mathbf{0}$ and let $T_{\mathbf{a}} = \mathbf{a}'(\hat\beta - \beta)/\sqrt{\mathbf{a}'(X'X)^{-1}\mathbf{a}\, MSE}$. By Theorem 7.2, $T_{\mathbf{a}} \sim t(\nu)$, so $T_{\mathbf{a}}^2 \sim F(1, \nu)$, whose distribution function is continuous and strictly increasing on $(0, \infty)$.

Step 2. By the proof of Theorem 8.3(b), $\sup_{\mathbf{b} \neq \mathbf{0}} T_{\mathbf{b}}^2 = (\hat\beta - \beta)'X'X(\hat\beta - \beta)/MSE$, and by Theorem 8.2 this is at most $c = (p+1)F_\alpha(p+1, \nu)$ with probability $1 - \alpha$.

Step 3. Since $T_{\mathbf{a}}^2 \leq \sup_{\mathbf{b}} T_{\mathbf{b}}^2$, $P(T_{\mathbf{a}}^2 \leq c) \geq 1 - \alpha = P(T_{\mathbf{a}}^2 \leq t_{\alpha/2}(\nu)^2)$. By the strict monotonicity in Step 1, $c \geq t_{\alpha/2}(\nu)^2$. The half-widths are $\sqrt{c}$ and $t_{\alpha/2}(\nu)$ times the same standard error, which proves the claim. ■

Chapter 10. Regression Diagnostics

10.1 What can go wrong

Fitting $\mathbf{y} = X\beta + \epsilon$, $\epsilon \sim N(\mathbf{0}, \sigma^2 I)$, by $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is justified only if the model is adequate. **Regression diagnostics** look for:

(1) violations of the error assumptions (normality, independence, equal variance), studied through residuals;

(2) a wrong functional form, remedied by transforming y or the x_j (Chapters 5 and 11);

(3) near-singularity of $X'X$ (multicollinearity, Chapter 15);

(4) **high-leverage points**: observations whose $\mathbf{x}_i$ is far from the bulk of the design;

(5) **outliers**: observations whose y_i is far from its fitted value;

(6) **influential observations**: observations whose deletion changes the fit substantially.

Points of types (4)–(6) need not coincide: a high-leverage point can lie on the line, and an outlier at the centre of the design may hardly move the fit.

10.2 The hat matrix and leverage

Motivation. A point far from the others in x-space can pull the fitted surface toward itself. The diagonal of the hat matrix measures this leverage and flags such points before we look at y.

The fitted values are $\hat{\mathbf{y}} = H\mathbf{y}$ with $H = X(X'X)^{-1}X' = (h_{ij})$, where $h_{ij} = \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_j$ and $\mathbf{x}_i' = (1, x_{i1}, \dots, x_{ip})$ is the i-th row of X.

Theorem 10.1 (Properties of the hat matrix)

(a) H is symmetric and idempotent with $\operatorname{rank} H = \operatorname{tr} H = \sum_i h_{ii} = p + 1$.

(b) $h_{ii} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$; hence $0 \le h_{ii} \le 1$ and $|h_{ij}| \le \tfrac{1}{2}$ for $i \neq j$.

(c) $HX = X$. If the model has an intercept, $H\mathbf{1} = \mathbf{1}$, i.e. every row of H sums to 1, and $h_{ii} \ge 1/n$.

(d) With an intercept, $h_{ii} = \frac{1}{n} + (\mathbf{x}_i^* - \bar{\mathbf{x}})'(\tilde X'\tilde X)^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$, where $\mathbf{x}_i^* = (x_{i1}, \dots, x_{ip})'$ and $\tilde X$ is the centred $n \times p$ matrix of regressors. For $p = 1$, $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar x)^2}{S_{xx}}$.

Proof. (a) $H' = H$, $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$, and $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = p + 1$.

(b) The (i,i) entry of $H = HH'$ is $\sum_j h_{ij}^2$. Since it is at least h_{ii}^2, $h_{ii} \ge h_{ii}^2$, so $0 \le h_{ii} \le 1$. For $j \neq i$, $h_{ij}^2 \le h_{ii} - h_{ii}^2 \le \tfrac{1}{4}$.

(c) $HX = X$ is direct; the intercept column gives $H\mathbf{1} = \mathbf{1}$. For $h_{ii} \ge 1/n$ see (d).

(d) Reparametrize with centred regressors: the column space of X is spanned by $\mathbf{1}$ and the columns of $\tilde X$, which are orthogonal to $\mathbf{1}$. Hence $H = \frac{1}{n}\mathbf{1}\mathbf{1}' + \tilde X(\tilde X'\tilde X)^{-1}\tilde X'$, and the diagonal gives the formula, the second term being nonnegative. For $p = 1$, $\tilde X'\tilde X = S_{xx}$. ∎

Since $\hat y_i = \sum_j h_{ij} y_j$, the weight of y_i in its own fitted value is $h_{ii} = \partial \hat y_i / \partial y_i$; h_{ii} is the **leverage** of observation i. By (d), it grows with the (Mahalanobis) distance of $\mathbf{x}_i^*$ from the centroid $\bar{\mathbf{x}}$: with $M(i) = (\mathbf{x}_i^* - \bar{\mathbf{x}})'\{\tilde X'\tilde X/(n-1)\}^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$,

$$M(i) = (n-1)\Big(h_{ii} - \frac{1}{n}\Big).$$

The average leverage is $\bar{h} = (p+1)/n$, and a common rule flags $h_{ii} > 2\bar{h}$ (or $3\bar{h}$) as high leverage.

10.3 Residuals

Motivation. Raw residuals have unequal variances, so their sizes are not comparable. Scaling them by their standard errors, and deleting the point in question, gives residuals that can be compared with t quantiles to detect outliers.

Theorem 10.2

$\mathbf{e} = (I-H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I-H))$. So $\operatorname{Var} e_i = \sigma^2(1-h_{ii})$, $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$, and $\operatorname{Corr}(e_i, e_j) = -h_{ij}/\sqrt{(1-h_{ii})(1-h_{jj})}$.

High-leverage points have small residual variance: the fit is pulled towards them. To compare residuals on a common scale:

Definition (Scaled residuals)

With $s^2 = MSE = \mathbf{e}'\mathbf{e}/(n-p-1)$ and $s^2_{(i)}$ the MSE computed without observation i:

- the **standardized** (internally studentized) residual is $r_i = \dfrac{e_i}{s\sqrt{1-h_{ii}}}$;
- the **studentized** (externally studentized) residual is $r_i^* = \dfrac{e_i}{s_{(i)}\sqrt{1-h_{ii}}}$.

r_i is not t-distributed, because e_i and s are dependent (a large $|e_i|$ inflates s). Deleting observation i removes the dependence. Let $X_{(i)}$, $\mathbf{y}_{(i)}$ denote X, $\mathbf{y}$ without row i, $\hat{\beta}_{(i)}$ the corresponding estimator, and $\tilde{y}_i = \mathbf{x}_i'\hat{\beta}_{(i)}$ the prediction of y_i from the other observations.

Theorem 10.3 (Deletion formulas)

(a) $\hat{\beta}_{(i)} = \hat{\beta} - \dfrac{(X'X)^{-1}\mathbf{x}_i e_i}{1-h_{ii}}$.

(b) $y_i - \tilde{y}_i = \dfrac{e_i}{1-h_{ii}}$ (the **deleted** or PRESS residual).

(c) $(n-p-2)s^2_{(i)} = (n-p-1)s^2 - \dfrac{e_i^2}{1-h_{ii}}$.

Proof [Supplement]. (a) $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. With $A = X'X$, the Sherman–Morrison formula gives

$$(A - \mathbf{x}_i\mathbf{x}_i')^{-1} = A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i'A^{-1}}{1-h_{ii}}.$$

Multiplying by $X'\mathbf{y} - \mathbf{x}_i y_i$ and using $A^{-1}X'\mathbf{y} = \hat{\beta}$, $\mathbf{x}_i'\hat{\beta} = \hat{y}_i$, $\mathbf{x}_i'A^{-1}\mathbf{x}_i = h_{ii}$:

$$\hat{\beta}_{(i)} = \hat{\beta} - A^{-1}\mathbf{x}_i y_i + \frac{A^{-1}\mathbf{x}_i(\hat{y}_i - h_{ii}y_i)}{1-h_{ii}} = \hat{\beta} + \frac{A^{-1}\mathbf{x}_i\{\hat{y}_i - h_{ii}y_i - (1-h_{ii})y_i\}}{1-h_{ii}} = \hat{\beta} - \frac{A^{-1}\mathbf{x}_i e_i}{1-h_{ii}}.$$

(b) $\tilde{y}_i = \mathbf{x}_i'\hat{\beta}_{(i)} = \hat{y}_i - h_{ii}e_i/(1-h_{ii})$, so $y_i - \tilde{y}_i = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$.

(c) For $j \neq i$, the deleted residual is $y_j - \mathbf{x}_j'\hat\beta_{(i)} = e_j + h_{ji}e_i/(1-h_{ii})$. Summing squares over $j \neq i$ and using $\sum_j e_j h_{ji} = 0$ (since $H\mathbf{e} = \mathbf{0}$) and $\sum_j h_{ji}^2 = h_{ii}$:

$$SSE_{(i)} = \sum_{j\neq i} e_j^2 + \frac{2e_i}{1-h_{ii}}\sum_{j\neq i} h_{ji}e_j + \frac{e_i^2}{(1-h_{ii})^2}\sum_{j\neq i} h_{ji}^2.$$

Here $\sum_{j\neq i} e_j^2 = SSE - e_i^2$, $\sum_{j\neq i} h_{ji}e_j = -h_{ii}e_i$, and $\sum_{j\neq i} h_{ji}^2 = h_{ii} - h_{ii}^2$. So

$$SSE_{(i)} = SSE - e_i^2 - \frac{2h_{ii}e_i^2}{1-h_{ii}} + \frac{h_{ii}e_i^2}{1-h_{ii}} = SSE - \frac{e_i^2}{1-h_{ii}}.$$

■

Theorem 10.4 (Distribution of the studentized residual)

(a) $r_i^* = \dfrac{y_i - \tilde y_i}{s_{(i)}\sqrt{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i}} \sim t(n-p-2)$.

(b) $r_i^* = r_i\sqrt{\dfrac{n-p-2}{n-p-1-r_i^2}}$, a monotone function of r_i.

Proof. (a) y_i is independent of $\hat\beta_{(i)}$ and $s_{(i)}^2$, which are computed from the other $n-1$ observations; so $y_i - \tilde y_i \sim N(0, \sigma^2\{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i\})$, independent of $(n-p-2)s_{(i)}^2/\sigma^2 \sim \chi^2(n-p-2)$. By Sherman–Morrison, $1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i = 1 + h_{ii} + h_{ii}^2/(1-h_{ii}) = 1/(1-h_{ii})$, and with Theorem 10.3(b) the ratio equals $e_i/\{s_{(i)}\sqrt{1-h_{ii}}\}$.

(b) Substitute Theorem 10.3(c), $s_{(i)}^2 = s^2(n-p-1-r_i^2)/(n-p-2)$. ■

So an observation is flagged as an outlier when $|r_i^*| > t_{\alpha/2}(n-p-2)$. Since n tests are performed, a Bonferroni level α/n is safer.

10.4 Influence measures

An observation is influential if deleting it changes the fitted model a lot. The measures below combine the residual and the leverage.

Theorem 10.5 (Influence measures)

(a) **DFFITS**: $\mathrm{DFFITS}_{(i)} = \dfrac{\hat y_i - \tilde y_{i(i)}}{s_{(i)}\sqrt{h_{ii}}} = \left(\dfrac{h_{ii}}{1-h_{ii}}\right)^{1/2} r_i^*$, where $\tilde y_{i(i)} = \mathbf{x}_i'\hat\beta_{(i)}$.

(b) **Cook's distance**: $D_{(i)} = \dfrac{(\hat\beta - \hat\beta_{(i)})'X'X(\hat\beta - \hat\beta_{(i)})}{(p+1)s^2} = \dfrac{r_i^2}{p+1}\cdot\dfrac{h_{ii}}{1-h_{ii}}$.

(c) **COVRATIO**: $\mathrm{COVRATIO}_{(i)} = \dfrac{\det\{s_{(i)}^2(X_{(i)}'X_{(i)})^{-1}\}}{\det\{s^2(X'X)^{-1}\}} = \dfrac{1}{(1-h_{ii})}\left(\dfrac{n-p-2+r_i^{*2}}{n-p-1}\right)^{-(p+1)}$.

(d) **Andrews–Pregibon**: with $Z = (X, \mathbf{y})$, $\mathrm{AP}_{(i)} = \dfrac{\det(Z_{(i)}'Z_{(i)})}{\det(Z'Z)} = 1 - h_{ii} - \dfrac{e_i^2}{SSE}$.

Proof [Supplement]. (a) By Theorem 10.3(a), $\hat y_i - \tilde y_{i(i)} = \mathbf{x}_i'(\hat\beta - \hat\beta_{(i)}) = h_{ii}e_i/(1-h_{ii})$; divide by $s_{(i)}\sqrt{h_{ii}}$.

(b) $(\hat\beta-\hat\beta_{(i)})'X'X(\hat\beta-\hat\beta_{(i)}) = \frac{e_i^2}{(1-h_{ii})^2}\mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i = \frac{h_{ii}e_i^2}{(1-h_{ii})^2}$; divide by $(p+1)s^2$.

(c) By the matrix determinant lemma,

$$\det(X_{(i)}'X_{(i)}) = \det(X'X)\,(1-h_{ii}),$$

so the ratio is $(s_{(i)}^2/s^2)^{p+1}/(1-h_{ii})$. By Theorem 10.3(c), $s_{(i)}^2/s^2 = (n-p-1-r_i^2)/(n-p-2)$, and by Theorem 10.4(b), $n-p-2+r_i^{*2} = (n-p-2)(n-p-1)/(n-p-1-r_i^2)$. Hence $s_{(i)}^2/s^2 = (n-p-1)/(n-p-2+r_i^{*2})$, which gives the formula.

(d) By the determinant lemma, $\det(Z_{(i)}'Z_{(i)})/\det(Z'Z) = 1-\mathbf{z}_i'(Z'Z)^{-1}\mathbf{z}_i$, where $\mathbf{z}_i' = (\mathbf{x}_i', y_i)$. The last quantity is the leverage of observation i in the regression on Z, which by the partitioned-inverse formula equals $h_{ii} + e_i^2/SSE$. ■

Rules of thumb (Belsley, Kuh and Welsch; Cook):

- $|\text{DFFITS}_{(i)}| > 2\sqrt{(p+1)/n}$;
- $D_{(i)} > F_{0.50}(p+1, n-p-1)$ (roughly 1): deleting i moves $\hat\beta$ outside the 50% confidence ellipsoid of Theorem 8.2;
- $|\text{COVRATIO}_{(i)} - 1| > 3(p+1)/n$;
- small $\text{AP}_{(i)}$.

Every measure grows with both $|r_i|$ and h_{ii}: influence is "outlyingness × leverage".

Example 10.1 (Two kinds of unusual point)

In a classical data set of $n = 21$ children (age at first word x, aptitude score y), the fit is $\hat y = 109.87 - 1.13x$. Observation 18 has $x = 42$, far from the others, with $h_{18,18} = 0.65 > 3\bar h = 0.29$: a high-leverage point. Its residual is small ($r^* = -0.85$), yet $D_{(18)} = 0.68$ and $\text{DFFITS}_{(18)} = -1.16$ are the largest: it is influential. Observation 19 has the largest residual, $r_{19}^* = 3.61 > t_{0.025}(18) = 2.10$: an outlier, but with ordinary leverage ($h = 0.053$), so its Cook's distance is only 0.22. The two points call for different actions: check the measurement of the outlier, and assess whether the conclusions depend on the single high-leverage point.

What to do. An unusual point is first checked for recording errors. If it is genuine, it may signal a wrong model (a missing variable or nonlinearity); robust regression or reporting fits with and without the point are better than silently deleting it. Influence diagnostics based on deleting one point at a time can miss groups of points that mask each other.

10.5 Selected exercises

Motivation. The deletion formulas answer concrete questions: what happens when a point lies exactly on the fitted surface, what a leverage of one means, and why the standardized residual cannot have a t distribution.

Exercise 10.1 (Deleting a point on the fitted surface)

Suppose $y_i = \hat y_i$ for some i, and that $X_{(i)}$ still has full column rank. Show directly from the normal equations that deleting observation i does not change the least squares estimator, and find $s_{(i)}^2$ in terms of s^2.

Solution.

Step 1. $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. Hence, for the full-data estimator $\hat\beta$,

$$X'_{(i)}(\mathbf{y}_{(i)} - X_{(i)}\hat\beta) = X'(\mathbf{y} - X\hat\beta) - \mathbf{x}_i(y_i - \mathbf{x}_i'\hat\beta) = \mathbf{0} - \mathbf{x}_i e_i = \mathbf{0},$$

since $X'\mathbf{e} = \mathbf{0}$ and $e_i = 0$.

Step 2. So $\hat\beta$ solves the normal equations of the reduced data. As $X'_{(i)}X_{(i)}$ is nonsingular, the solution is unique: $\hat\beta_{(i)} = \hat\beta$. (This is the case $e_i = 0$ of Theorem 10.3(a).)

Step 3. The remaining residuals are unchanged, so $SSE_{(i)} = \sum_{j\neq i} e_j^2 = SSE$, now with $n-p-2$ degrees of freedom: $s^2_{(i)} = \frac{n-p-1}{n-p-2}s^2$, which is larger than s^2 when $SSE > 0$.

Such a point has zero Cook's distance however large its leverage, illustrating that leverage alone does not make a point influential. ■

Exercise 10.2 (Leverage equal to one)

Suppose $h_{ii} = 1$.
(a) Show that $h_{ij} = 0$ for $j \neq i$, and that $\hat y_i = y_i$ and $e_i = 0$ for every response vector $\mathbf{y}$.
(b) Show that $X'_{(i)}X_{(i)}$ is singular, so that β cannot be estimated without observation i; conversely, if $X_{(i)}$ has full column rank, then $h_{ii} < 1$.
(c) In simple regression with $x = (0, \dots, 0, 1)$ (with $n \geq 2$), compute h_{nn}.

Solution [Supplement].

(a) By Theorem 10.1(b), $h_{ii} = h_{ii}^2 + \sum_{j\neq i} h_{ij}^2$, i.e. $1 = 1 + \sum_{j\neq i} h_{ij}^2$, so every $h_{ij} = 0$ $(j \neq i)$. Then $\hat y_i = \sum_j h_{ij}y_j = y_i$ and $e_i = 0$, whatever $\mathbf{y}$ is.

(b) By the matrix determinant lemma (as in the proof of Theorem 10.5(c)),

$$\det(X'_{(i)}X_{(i)}) = \det(X'X - \mathbf{x}_i\mathbf{x}_i') = \det(X'X)\,\{1 - \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i\} = \det(X'X)(1 - h_{ii}).$$

So $h_{ii} = 1$ makes $X'_{(i)}X_{(i)}$ singular, i.e. $X_{(i)}$ rank deficient. Conversely, if $X_{(i)}$ has full column rank, the determinant is positive, so $h_{ii} \neq 1$, and $h_{ii} < 1$ by Theorem 10.1(b).

(c) $\bar x = 1/n$ and $S_{xx} = (n-1)/n^2 + (1-1/n)^2 = (n-1)/n$. By Theorem 10.1(d),

$$h_{nn} = \frac{1}{n} + \frac{(1-1/n)^2}{(n-1)/n} = \frac{1}{n} + \frac{n-1}{n} = 1.$$

The other observations all have $x = 0$, so they carry no information about the slope: the single point at $x = 1$ determines the slope alone, and the line passes through it. ■

Exercise 10.3 (Standardized residuals are bounded)

Assume $h_{ii} < 1$.
(a) Show that $r_i^2 \leq n-p-1$. Conclude that r_i does not have a t distribution.
(b) Show that the map $r_i \mapsto r_i^*$ of Theorem 10.4(b) is a strictly increasing bijection from $(-\sqrt{n-p-1}, \sqrt{n-p-1})$ onto $\mathbb{R}$.

Solution [Supplement].

(a) By Theorem 10.3(c), $0 \leq (n-p-2)s^2_{(i)} = (n-p-1)s^2 - e_i^2/(1-h_{ii})$. Hence $e_i^2/\{s^2(1-h_{ii})\} \leq n-p-1$, i.e. $r_i^2 \leq n-p-1$, with equality iff $SSE_{(i)} = 0$. A t variable takes arbitrarily large values with positive probability, so r_i cannot be t-distributed.

(b) On $|r| < \sqrt{n-p-1}$ let $f(r) = r\sqrt{(n-p-2)/(n-p-1-r^2)}$. It is continuous and odd. For $0 \leq r_1 < r_2 < \sqrt{n-p-1}$, the factor r increases and the factor $(n-p-1-r^2)^{-1/2}$ increases, both

being positive for $r > 0$, so $f(r_1) < f(r_2)$; by oddness f is strictly increasing on the whole interval. As $r \uparrow \sqrt{n-p-1}$, $f(r) \to +\infty$, and $f(r) \to -\infty$ as $r \downarrow -\sqrt{n-p-1}$. By the intermediate value theorem f is onto $\mathbb{R}$. So r_i^* and r_i rank the observations identically, but only r_i^* has the unbounded $t(n-p-2)$ distribution. ■

Chapter 11. Variable Selection

With many candidate regressors we want a **parsimonious** model: enough variables to explain y well and predict accurately, but no more, since every extra variable adds variance to the estimates and predictions. This chapter quantifies the trade-off and describes search methods and criteria.

11.1 Consequences of omitting variables

Partition the **full model** as

$$\mathbf{y} = X\beta + \epsilon = X_k\beta_k + X_r\beta_r + \epsilon, \qquad \epsilon \sim N(\mathbf{0}, \sigma^2 I),$$

where X_k $(n \times (k+1))$ contains the intercept and k regressors and X_r $(n \times r,\ r = p-k)$ the others. The full-model estimator $\hat\beta = (\hat\beta_k', \hat\beta_r')'$ and $\hat\sigma^2 = SSE/(n-p-1)$ are compared with the **reduced model** estimators

$$\tilde\beta_k = (X_k'X_k)^{-1}X_k'\mathbf{y}, \qquad \tilde\sigma^2 = \frac{\mathbf{y}'\{I - X_k(X_k'X_k)^{-1}X_k'\}\mathbf{y}}{n-k-1}.$$

Theorem 11.1

Let $A = (X_k'X_k)^{-1}X_k'X_r$ (the **alias matrix**) and $H_k = X_k(X_k'X_k)^{-1}X_k'$. If the full model is true:

(a) $E\tilde\beta_k = \beta_k + A\beta_r$ and $\operatorname{Var}\tilde\beta_k = \sigma^2(X_k'X_k)^{-1}$; so $\operatorname{MSE}(\tilde\beta_k) = E(\tilde\beta_k - \beta_k)(\tilde\beta_k - \beta_k)' = \sigma^2(X_k'X_k)^{-1} + A\beta_r\beta_r'A'$.

(b) $E\tilde\sigma^2 = \sigma^2 + \dfrac{\beta_r'X_r'(I-H_k)X_r\beta_r}{n-k-1} \geq \sigma^2$.

(c) For predicting $E(y) = \mathbf{x}'\beta = \mathbf{x}_k'\beta_k + \mathbf{x}_r'\beta_r$ by $\tilde y = \mathbf{x}_k'\tilde\beta_k$:

$$\operatorname{MSE}(\tilde y) = \sigma^2\mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + (\mathbf{x}_k'A\beta_r - \mathbf{x}_r'\beta_r)^2.$$

Proof. (a) $E\tilde\beta_k = (X_k'X_k)^{-1}X_k'(X_k\beta_k + X_r\beta_r) = \beta_k + A\beta_r$; the variance is as usual; MSE = Var +bias $\cdot$ bias$'$.

(b) Theorem 5.4(a) with $I - H_k$, whose trace is $n-k-1$ and which annihilates $X_k\beta_k$.

(c) MSE = Var +bias^2 with bias $\mathbf{x}_k'(\beta_k + A\beta_r) - \mathbf{x}'\beta$. ■

So omitting relevant variables biases the remaining coefficients (unless $X_k'X_r = 0$) and inflates the residual variance estimate. In exchange the variance decreases:

Theorem 11.2

(a) $\operatorname{Var}(\hat\beta_k) - \operatorname{Var}(\tilde\beta_k)$ is nonnegative definite.

(b) $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{Var}(\mathbf{x}_k'\tilde\beta_k)$.

(c) If $\operatorname{Var}(\hat\beta_r) - \beta_r\beta_r'$ is nonnegative definite, then $\operatorname{Var}(\hat\beta_k) - \operatorname{MSE}(\tilde\beta_k)$ is nonnegative definite and $\operatorname{Var}(\mathbf{x}'\hat\beta) \geq \operatorname{MSE}(\mathbf{x}_k'\tilde\beta_k)$.

Proof. By the partitioned-inverse formula, with $B = X_r'(I-H_k)X_r$ (nonsingular),

$$\sigma^{-2}\operatorname{Var}(\hat\beta) = (X'X)^{-1} = \begin{pmatrix} (X_k'X_k)^{-1} + AB^{-1}A' & -AB^{-1} \\ -B^{-1}A' & B^{-1} \end{pmatrix}.$$

(a) $\operatorname{Var}\hat{\beta}_k - \operatorname{Var}\tilde{\beta}_k = \sigma^2 A B^{-1} A' \geq 0$.

(b) With $\mathbf{x} = (\mathbf{x}_k', \mathbf{x}_r')'$ and $\mathbf{u} = A'\mathbf{x}_k - \mathbf{x}_r$,

$$\sigma^{-2}\operatorname{Var}(\mathbf{x}'\hat{\beta}) = \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k + \mathbf{u}'B^{-1}\mathbf{u} \geq \mathbf{x}_k'(X_k'X_k)^{-1}\mathbf{x}_k.$$

(c) $\operatorname{Var}\hat{\beta}_r = \sigma^2 B^{-1}$, so the assumption reads $\sigma^2 B^{-1} \geq \beta_r\beta_r'$. Then $\operatorname{Var}\hat{\beta}_k - \operatorname{MSE}(\tilde{\beta}_k) = A(\sigma^2 B^{-1} - \beta_r\beta_r')A' \geq 0$, and similarly $\operatorname{Var}(\mathbf{x}'\hat{\beta}) - \operatorname{MSE}(\mathbf{x}_k'\tilde{\beta}_k) = \mathbf{u}'(\sigma^2 B^{-1} - \beta_r\beta_r')\mathbf{u} \geq 0$. ■

In words: dropping variables whose coefficients are small relative to their standard errors (so that $\beta_r\beta_r' \leq \operatorname{Var}\hat{\beta}_r$) reduces the mean squared error of estimation and prediction, even though it introduces bias. Conversely, if the reduced model is true ($\beta_r = \mathbf{0}$), both estimators are unbiased and the reduced one has smaller variance, so including irrelevant variables only costs precision.

11.2 Search methods

With p candidate regressors there are 2^p subsets (each variable in or out), so exhaustive search becomes infeasible for large p ($2^{50} \approx 10^{15}$).

All possible regressions (best subset). Fit all 2^p models. For each size k, keep the best model by R_k^2 (equivalently by SSE_k or MSE_k, since these are monotone in each other for fixed k), then compare the best models across k with a criterion that penalizes size (Section 11.3).

Backward elimination.

(1) Fit the full model.

(2) Compute the partial F statistic of each variable in the model.

(3) Let F_L be the smallest, and F_C the critical value at level α_{drop}.

(4) If $F_L < F_C$, delete that variable, refit and return to (2); otherwise stop.

This needs at most $p(p+1)/2$ fits.

Forward selection. Start with no regressors. At each stage add the variable that increases R^2 the most (equivalently, has the largest partial F given those already in), provided its partial F exceeds the critical value at level α_{add}; otherwise stop.

Stepwise selection. Forward selection, but after each addition the variables already in the model are re-examined by partial F tests and removed if they fall below the α_{drop} threshold (with $\alpha_{\text{drop}} \geq \alpha_{\text{add}}$ to avoid cycling). A variable that was useful early can become redundant once others enter.

The methods need not agree, and none guarantees the best subset. The levels used are not true significance levels, because of the repeated testing.

Example 11.1

In Hald's classical cement data ($n = 13$, four ingredients $x_1, \ldots, x_4$), x_2 and x_4 are highly correlated ($x_1 + x_2 + x_3 + x_4 \approx 100$). Backward elimination ($\alpha = 0.05$) removes x_3 (partial $F = 0.02$) and then x_4 (partial $F = 1.86 < F_{0.05}(1, 9) = 5.12$), ending with $\{x_1, x_2\}$ ($R^2 = 0.979$). Forward selection enters x_4 first (largest single $R^2 = 0.675$), then x_1, and stops because x_2 has partial $F = 5.03 < 5.12$, ending with $\{x_1, x_4\}$ ($R^2 = 0.972$). Stepwise selection with $\alpha_{\text{add}} = 0.10$ enters x_2 and then removes x_4 (partial $F = 1.86$), ending with $\{x_1, x_2\}$. Because

x_2 and x_4 carry nearly the same information, different routes end at different, almost equally good models.

11.3 Selection criteria

Motivation. Adding variables never reduces R^2, so it cannot choose a model size. Criteria such as adjusted R^2, Mallows' C_p, AIC and PRESS penalize complexity or estimate prediction error directly.

For a model with k regressors (plus intercept):

(1) residual mean square $MSE_k = SSE_k/(n-k-1)$;

(2) $R^2_k = 1 - SSE_k/SST$;

(3) adjusted R^2: $R^2_{a,k} = 1 - \dfrac{n-1}{n-k-1}(1-R^2_k) = 1 - \dfrac{MSE_k}{SST/(n-1)}$;

(4) Mallows' $C_k = \dfrac{SSE_k}{\hat\sigma^2} + 2(k+1) - n$, with $\hat\sigma^2$ the MSE of the full model.

R^2_k never decreases when a variable is added, so it can only compare models of the same size. MSE_k can increase when a variable is added (the loss of one degree of freedom may outweigh the decrease in SSE), and $R^2_{a,k}$ is a decreasing function of MSE_k: maximizing R^2_a is minimizing MSE_k.

Mallows' C_p. We judge the reduced model by the total standardized mean squared error of its fitted values at the n design points:

$$\Gamma_k = \frac{1}{\sigma^2}\sum_{i=1}^{n} \mathrm{MSE}(\tilde y_i), \qquad \tilde y_i = \mathbf{x}'_{ki}\tilde\beta_k.$$

Theorem 11.3

$$\Gamma_k = (k+1) + \frac{\beta'_r X'_r (I-H_k) X_r \beta_r}{\sigma^2} = \frac{E(SSE_k)}{\sigma^2} + 2(k+1) - n.$$

Hence C_k estimates Γ_k, and if the reduced model is correct ($\beta_r = \mathbf{0}$), $E(SSE_k) = (n-k-1)\sigma^2$ and $C_k \approx k+1$.

Proof [Supplement]. By Theorem 11.1(c), the variance part sums to

$$\sum_i \mathbf{x}'_{ki}(X'_kX_k)^{-1}\mathbf{x}_{ki} = \mathrm{tr}\Big\{(X'_kX_k)^{-1}\sum_i \mathbf{x}_{ki}\mathbf{x}'_{ki}\Big\} = \mathrm{tr}\{(X'_kX_k)^{-1}X'_kX_k\} = k+1.$$

The bias vector over the design points is $X_kA\beta_r - X_r\beta_r = -(I-H_k)X_r\beta_r$, with squared length $\beta'_rX'_r(I-H_k)X_r\beta_r$. By Theorem 11.1(b), this equals $E(SSE_k) - (n-k-1)\sigma^2$. Substituting gives the second expression. ∎

So we look for models with small C_k and C_k close to $k+1$; a C_k well above $k+1$ signals bias from omitted variables. For the full model, $C_p = p+1$ identically.

11.4 Selected exercises

Motivation. The first exercise makes the omitted-variable bias explicit for a straight line; the others link the selection criteria to the partial F statistic and derive AIC for the normal linear model.

Exercise 11.1 (Omitted-variable bias for a straight line)

The straight line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$ is fitted by least squares, but the true mean is $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ (errors uncorrelated with variance σ^2). Write $S_{11} = \sum (x_{i1} - \bar{x}_1)^2 > 0$, $S_{22} = \sum (x_{i2} - \bar{x}_2)^2$ and $S_{12} = \sum (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)$.
(a) Is $\hat{\beta}_1$ biased? Find $E\hat{\beta}_1$.
(b) Find the expectation of the residual mean square of the fitted line.

Solution.

(a) With $k_i = (x_{i1} - \bar{x}_1)/S_{11}$, $\hat{\beta}_1 = \sum k_i y_i$, $\sum k_i = 0$ and $\sum k_i x_{i1} = 1$ (Section 4.1). Hence

$$E\hat{\beta}_1 = \sum k_i(\beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2}) = \beta_1 + \beta_2 \sum k_i x_{i2} = \beta_1 + \beta_2 \frac{S_{12}}{S_{11}}.$$

So $\hat{\beta}_1$ is biased unless $\beta_2 = 0$ or $S_{12} = 0$ (the omitted regressor is uncorrelated with x_1 in the sample). The factor S_{12}/S_{11} is the slope of the regression of x_2 on x_1, the second entry of the alias matrix A of Theorem 11.1: $\hat{\beta}_1$ picks up the part of the effect of x_2 that travels with x_1.

(b) By Theorem 11.1(b) with $k = 1$ and $X_r = \mathbf{x}_2$, the excess is $\beta_2^2 \, \mathbf{x}_2'(I - H_k)\mathbf{x}_2/(n-2)$, where $\mathbf{x}_2'(I - H_k)\mathbf{x}_2$ is the residual sum of squares of the regression of x_2 on $(1, x_1)$, namely $S_{22} - S_{12}^2/S_{11}$ (Theorem 3.3). So

$$E(MSE) = \sigma^2 + \frac{\beta_2^2 (S_{22} - S_{12}^2/S_{11})}{n-2} \geq \sigma^2.$$

■

Exercise 11.2 (Adjusted R^2 and the partial F statistic)

A model with k regressors (and intercept) is enlarged by r further regressors, and F is the partial F statistic for the added regressors, computed with the residual mean square of the larger model. Assuming $SSE_{k+r} > 0$, show that the adjusted R^2 increases if and only if $F > 1$.

Solution [Supplement].

Step 1. Since $R_a^2 = 1 - MSE/\{SST/(n-1)\}$ with SST common to both models, R_a^2 increases iff $MSE_{k+r} < MSE_k$.

Step 2. By definition $F = \{(SSE_k - SSE_{k+r})/r\}/MSE_{k+r}$, so $SSE_k = SSE_{k+r} + rF\,MSE_{k+r} = (n - k - r - 1 + rF)MSE_{k+r}$, and

$$MSE_k = \frac{SSE_k}{n-k-1} = \frac{n-k-r-1+rF}{n-k-1} \, MSE_{k+r}.$$

Step 3. With $MSE_{k+r} > 0$, $MSE_k > MSE_{k+r}$ iff $n - k - r - 1 + rF > n - k - 1$ iff $F > 1$. So adjusted R^2 admits a variable far more easily than a significance test, which would require $F > F_\alpha$ (about 4 for one variable at $\alpha = 0.05$). ■

Exercise 11.3 (Mallows' C_k and the partial F statistic)

Let F_0 be the partial F statistic for dropping the $r = p - k$ regressors of X_r from the full model. Show that

$$C_k = (k+1) + (p-k)(F_0 - 1),$$

so that $C_k \leq k + 1$ if and only if $F_0 \leq 1$.

Solution [Supplement]. Let $\hat{\sigma}^2 = SSE/(n-p-1)$ be the full-model residual mean square. By definition of F_0, $SSE_k = SSE + (p-k)F_0\hat{\sigma}^2 = \{(n-p-1) + (p-k)F_0\}\hat{\sigma}^2$. Substituting into $C_k = SSE_k/\hat{\sigma}^2 + 2(k+1) - n$,

$$C_k = (n-p-1) + (p-k)F_0 + 2(k+1) - n = (k+1) + (p-k)F_0 - (p-k).$$

Since $F_0 \approx 1$ on average when $\beta_r = \mathbf{0}$ (its mean is $\nu/(\nu-2)$ with $\nu = n-p-1$), this matches $C_k \approx k+1$ in Theorem 11.3. ■

Exercise 11.4 (AIC for the normal linear model)

For a model with k regressors and intercept and $\epsilon \sim N(\mathbf{0}, \sigma^2 I)$, define $\mathrm{AIC}_k = -2\,l_{\max} + 2(\text{number of parameters})$, where $l_{\max}$ is the maximized log-likelihood and the parameters are β_k and σ^2. Show that

$$\mathrm{AIC}_k = n \log \frac{SSE_k}{n} + 2(k+1) + n(\log 2\pi + 1) + 2,$$

so that, up to a constant common to all models, $\mathrm{AIC}_k = n \log(SSE_k/n) + 2(k+1)$.

Solution [Supplement].

Step 1. The log-likelihood is $l(\beta, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) - \frac{1}{2\sigma^2}\|\mathbf{y} - X_k\beta\|^2$. For each σ^2 it is maximized over β by the least squares estimator (Theorem 6.1), giving $-\frac{n}{2}\log(2\pi\sigma^2) - SSE_k/(2\sigma^2)$.

Step 2. Assume $SSE_k > 0$ (true with probability one). The derivative in σ^2 is $-\frac{n}{2\sigma^2} + \frac{SSE_k}{2\sigma^4}$, positive for $\sigma^2 < SSE_k/n$ and negative beyond, so the maximum is at $\hat{\sigma}^2 = SSE_k/n$, and

$$l_{\max} = -\frac{n}{2}\log\frac{2\pi SSE_k}{n} - \frac{n}{2}.$$

Step 3. There are $k+2$ parameters, so $\mathrm{AIC}_k = n\log(SSE_k/n) + n\log 2\pi + n + 2(k+2)$, which is the claimed expression. Like C_k, it trades the decrease of SSE_k against a penalty of 2 per parameter. ■

Chapter 12. Generalized Least Squares

12.1 The general linear model

So far the errors were uncorrelated with equal variance. In time series, clustered or heteroscedastic data this fails. The **general linear regression model** is

$$\mathbf{y} = X\beta + \epsilon, \qquad \epsilon \sim N_n(\mathbf{0}, \sigma^2 V),$$

with V a known positive definite matrix, X of size $n \times (p+1)$ and full column rank. Ordinary least squares (OLS) is no longer the BLUE.

Since V is symmetric positive definite, $V = P\Lambda P'$ with P orthogonal, and $T = P\Lambda^{1/2}$ satisfies $V = TT'$. Multiply the model by T^{-1}:

$$\mathbf{z} = W\beta + \eta, \qquad \mathbf{z} = T^{-1}\mathbf{y}, \quad W = T^{-1}X, \quad \eta = T^{-1}\epsilon,$$

and $\operatorname{Var}\eta = \sigma^2 T^{-1}V(T^{-1})' = \sigma^2 T^{-1}TT'(T')^{-1} = \sigma^2 I_n$. The transformed model satisfies the usual assumptions.

Theorem 12.1 (Generalized least squares)

The BLUE of β is the **generalized least squares** (GLS) estimator

$$\hat\beta_G = (W'W)^{-1}W'\mathbf{z} = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y},$$

which minimizes $(\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. It satisfies $E\hat\beta_G = \beta$ and $\operatorname{Var}\hat\beta_G = \sigma^2(X'V^{-1}X)^{-1}$.

Proof. Apply the Gauss–Markov theorem (Theorem 7.1) to the transformed model; linear unbiased estimators in $\mathbf{z}$ are the same as those in $\mathbf{y}$, since T is invertible. Using $(T^{-1})'T^{-1} = (TT')^{-1} = V^{-1}$, $W'W = X'V^{-1}X$ and $W'\mathbf{z} = X'V^{-1}\mathbf{y}$. Also $\|\mathbf{z}-W\beta\|^2 = (\mathbf{y}-X\beta)'V^{-1}(\mathbf{y}-X\beta)$. Finally $E\hat\beta_G = (X'V^{-1}X)^{-1}X'V^{-1}X\beta = \beta$ and $\operatorname{Var}\hat\beta_G = (X'V^{-1}X)^{-1}X'V^{-1}(\sigma^2V)V^{-1}X(X'V^{-1}X)^{-1} = \sigma^2(X'V^{-1}X)^{-1}$. ∎

When V is diagonal, $V = \operatorname{diag}(v_i)$, GLS is **weighted least squares** with weights $w_i = 1/v_i$: minimize $\sum w_i(y_i - \mathbf{x}_i'\beta)^2$, giving less weight to noisier observations.

12.2 Properties

OLS under correlated errors. The OLS estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is still unbiased, but

$$\operatorname{Var}\hat\beta = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1},$$

which is at least $\sigma^2(X'V^{-1}X)^{-1}$ (by Gauss–Markov), with equality when $V = I$ (more generally, when the column space of X is invariant under V).

Residuals and σ^2. The GLS residual vector is $\mathbf{e}_G = \mathbf{y} - X\hat\beta_G = \{I - X(X'V^{-1}X)^{-1}X'V^{-1}\}\mathbf{y}$, with

$$E\mathbf{e}_G = \mathbf{0}, \qquad \operatorname{Var}\mathbf{e}_G = \sigma^2\{V - X(X'V^{-1}X)^{-1}X'\}.$$

Theorem 12.2

$SSE(\hat\beta_G) = \mathbf{e}_G' V^{-1}\mathbf{e}_G$ satisfies $E\,SSE(\hat\beta_G) = (n-p-1)\sigma^2$, so

$$\tilde\sigma^2 = \frac{(\mathbf{y} - X\hat\beta_G)' V^{-1}(\mathbf{y} - X\hat\beta_G)}{n-p-1}$$

is unbiased for σ^2, and $\widehat{\mathrm{Var}}(\hat\beta_G) = \tilde\sigma^2 (X'V^{-1}X)^{-1}$.

Proof. $SSE(\hat\beta_G)$ is the residual sum of squares of the transformed model, $\|\mathbf{z} - W\hat\beta_G\|^2$, whose expectation is $(n-p-1)\sigma^2$ (Theorem 6.2). Directly: by Theorem 5.4(a),

$$\begin{aligned} E(\mathbf{e}_G' V^{-1}\mathbf{e}_G) &= \sigma^2 \mathrm{tr}[V^{-1}\{V - X(X'V^{-1}X)^{-1}X'\}] \\ &= \sigma^2\{n - \mathrm{tr}((X'V^{-1}X)^{-1}X'V^{-1}X)\} = \sigma^2(n-p-1). \end{aligned}$$

■

By contrast, the OLS estimate $\hat\sigma^2 = \mathbf{y}'(I-H)\mathbf{y}/(n-p-1)$ has

$$E\hat\sigma^2 = \frac{\sigma^2}{n-p-1}[\mathrm{tr}V - \mathrm{tr}\{(X'X)^{-1}X'VX\}],$$

which in general differs from σ^2, so the usual OLS standard errors $\hat\sigma^2(X'X)^{-1}$ are biased in two ways under correlated errors.

Under normality, all the inference of Chapters 6–8 carries over to the transformed model: for instance $\hat\beta_G \sim N(\beta, \sigma^2(X'V^{-1}X)^{-1})$, $(n-p-1)\tilde\sigma^2/\sigma^2 \sim \chi^2(n-p-1)$, and t and F tests use $\tilde\sigma^2$.

Example 12.1 (Averages with unequal precision)

Suppose $y_i = \mu + \epsilon_i$ where y_i is the average of m_i measurements, so $\mathrm{Var}\,\epsilon_i = \sigma^2/m_i$ and $V = \mathrm{diag}(1/m_i)$. Then $X = \mathbf{1}$ and

$$\hat\mu_G = \frac{\sum m_i y_i}{\sum m_i}, \qquad \mathrm{Var}\,\hat\mu_G = \frac{\sigma^2}{\sum m_i},$$

the grand average of all measurements. The OLS estimate $\bar y$ has variance $\frac{\sigma^2}{n^2}\sum \frac{1}{m_i} \geq \frac{\sigma^2}{\sum m_i}$ by the harmonic–arithmetic mean inequality.

Example 12.2 (AR(1) errors)

If $\epsilon_t = \rho\epsilon_{t-1} + u_t$ with $|\rho| < 1$ and u_t i.i.d. $N(0, \sigma_u^2)$, then $\mathrm{Var}\,\epsilon = \frac{\sigma_u^2}{1-\rho^2}V$ with $V_{st} = \rho^{|s-t|}$. The transformation T^{-1} amounts to the familiar quasi-differencing $y_t - \rho y_{t-1}$, $\mathbf{x}_t - \rho\mathbf{x}_{t-1}$ for $t \geq 2$, with the first observation scaled by $\sqrt{1-\rho^2}$. In practice ρ is unknown and is estimated from OLS residuals (feasible GLS).

Unknown V. V is rarely known. It is modelled with few parameters (equal correlation within clusters, AR(1) in time, variance proportional to a power of a regressor) and estimated from the data, typically from OLS residuals; GLS with the estimate plugged in is **feasible GLS**.

12.3 Selected exercises

Motivation. The exercises prove the facts about OLS under correlated errors that Section 12.2 states, work out weighted least squares for a regression through the origin, and verify the AR(1)

transformation of Example 12.2.

Exercise 12.1 (OLS under a general covariance)

Let $\mathbf{y} = X\beta + \epsilon$ with $E\epsilon = \mathbf{0}$, $\operatorname{Var}\epsilon = \sigma^2 V$, V known positive definite and X of full column rank $p+1$. Let $\tilde\beta = (X'X)^{-1}X'\mathbf{y}$ and $\hat\sigma^2 = \mathbf{y}'(I-H)\mathbf{y}/(n-p-1)$.
(a) Derive $E\tilde\beta$ and $\operatorname{Var}\tilde\beta$.
(b) Derive $E\hat\sigma^2$.
(c) Show that if $VX = XQ$ for some matrix Q, then $\tilde\beta = \hat\beta_G$.
(d) Let $V = (1-\rho)I_n + \rho J_n$ with $-1/(n-1) < \rho < 1$ (equicorrelated errors), and let X contain the intercept column. Show that OLS and GLS coincide.

Solution.

(a) $E\tilde\beta = (X'X)^{-1}X'X\beta = \beta$ and

$$\operatorname{Var}\tilde\beta = (X'X)^{-1}X'(\sigma^2 V)X(X'X)^{-1} = \sigma^2(X'X)^{-1}X'VX(X'X)^{-1}.$$

(b) By Theorem 5.4(a) with mean $X\beta$, covariance $\sigma^2 V$ and $(I-H)X = 0$,

$$\begin{aligned} E\{\mathbf{y}'(I-H)\mathbf{y}\} &= \sigma^2 \operatorname{tr}\{(I-H)V\} = \sigma^2\big[\operatorname{tr}V - \operatorname{tr}\{X(X'X)^{-1}X'V\}\big] \\ &= \sigma^2\big[\operatorname{tr}V - \operatorname{tr}\{(X'X)^{-1}X'VX\}\big], \end{aligned}$$

using $\operatorname{tr}(AB) = \operatorname{tr}(BA)$. Dividing by $n-p-1$ gives the formula of Section 12.2.

(c) VX has full column rank (V is nonsingular), so XQ does, and the square matrix Q is nonsingular. Then $V^{-1}X = XQ^{-1}$ and $X'V^{-1} = (Q^{-1})'X'$, so

$$\hat\beta_G = \{(Q^{-1})'X'X\}^{-1}(Q^{-1})'X'\mathbf{y} = (X'X)^{-1}Q'(Q^{-1})'X'\mathbf{y} = (X'X)^{-1}X'\mathbf{y} = \tilde\beta.$$

(d) V has eigenvalue $1+(n-1)\rho$ on $\mathbf{1}$ and $1-\rho$ on $\mathbf{1}^\perp$, both positive, so V is positive definite. If the intercept is the first column, $\mathbf{1} = X\mathbf{e}_1$, so $J_nX = \mathbf{1}\mathbf{1}'X = X\mathbf{e}_1\mathbf{1}'X$ and

$$VX = (1-\rho)X + \rho X\mathbf{e}_1\mathbf{1}'X = X\{(1-\rho)I + \rho\,\mathbf{e}_1\mathbf{1}'X\} = XQ.$$

By (c), $\tilde\beta = \hat\beta_G$: with equicorrelated errors and an intercept, OLS is BLUE, although its usual standard errors and $\hat\sigma^2$ are still biased by (a)–(b). ■

Exercise 12.2 (Weighted least squares through the origin)

Let $y_i = \beta x_i + \epsilon_i$, $i = 1,\dots,n$, with $x_i > 0$, uncorrelated errors, $E\epsilon_i = 0$ and $\operatorname{Var}\epsilon_i = \sigma^2 x_i$.
(a) Show that the GLS estimator is $\hat\beta_G = \bar y/\bar x$, with $\operatorname{Var}\hat\beta_G = \sigma^2/(n\bar x)$.
(b) Find the variance of the OLS estimator $\tilde\beta = \sum x_i y_i / \sum x_i^2$.
(c) Verify directly that $\operatorname{Var}\tilde\beta \ge \operatorname{Var}\hat\beta_G$, with equality iff all x_i are equal.

Solution [Supplement].

(a) $V = \operatorname{diag}(x_i)$ and $X = (x_1,\dots,x_n)'$, so $X'V^{-1}X = \sum x_i^2/x_i = \sum x_i$ and $X'V^{-1}\mathbf{y} = \sum y_i$. By Theorem 12.1, $\hat\beta_G = \sum y_i/\sum x_i = \bar y/\bar x$ and $\operatorname{Var}\hat\beta_G = \sigma^2/\sum x_i = \sigma^2/(n\bar x)$.

(b) $\operatorname{Var}\tilde\beta = \sum x_i^2 \operatorname{Var} y_i/(\sum x_i^2)^2 = \sigma^2 \sum x_i^3/(\sum x_i^2)^2$.

(c) We need $(\sum x_i^2)^2 \le \sum x_i \sum x_i^3$. With $a_i = x_i^{1/2}$ and $b_i = x_i^{3/2}$, this is the Cauchy–Schwarz inequality $(\sum a_i b_i)^2 \le \sum a_i^2 \sum b_i^2$, with equality iff $\mathbf{b}$ is proportional to $\mathbf{a}$, i.e. $x_i = b_i/a_i$ is constant. This agrees with the Gauss–Markov argument of Section 12.2. The GLS estimator is the classical *ratio estimator*, natural when the spread of y grows with x. ■

Exercise 12.3 (The AR(1) transformation)

Let $\epsilon_t = \rho\epsilon_{t-1} + u_t$ $(t = 2, \dots, n)$ with $|\rho| < 1$, $E\epsilon_1 = 0$, $\operatorname{Var}\epsilon_1 = \sigma_u^2/(1-\rho^2)$, and let $u_2, \dots, u_n$ have mean 0, variance σ_u^2, and be uncorrelated with each other and with ϵ_1.
(a) Show that $\operatorname{Cov}(\epsilon_s, \epsilon_t) = \sigma_u^2\rho^{|s-t|}/(1-\rho^2)$.
(b) Let $\eta_1 = \sqrt{1-\rho^2}\,\epsilon_1$ and $\eta_t = \epsilon_t - \rho\epsilon_{t-1}$ for $t \geq 2$, i.e. $\eta = L\epsilon$ for a lower bidiagonal matrix L. Show that $\operatorname{Var}\eta = \sigma_u^2 I_n$.
(c) Deduce that OLS applied to $L\mathbf{y}$ and LX is exactly GLS.

Solution [Supplement].

(a) Each ϵ_s is a linear combination of $\epsilon_1, u_2, \dots, u_s$, so u_t is uncorrelated with ϵ_s for $s < t$.

Step 1. (Variances.) By induction, if $\operatorname{Var}\epsilon_{t-1} = \sigma_u^2/(1-\rho^2)$ then $\operatorname{Var}\epsilon_t = \rho^2\operatorname{Var}\epsilon_{t-1} + \sigma_u^2 = \sigma_u^2/(1-\rho^2)$.

Step 2. (Covariances.) For $h \geq 1$,

$$\operatorname{Cov}(\epsilon_{t+h}, \epsilon_t) = \rho\operatorname{Cov}(\epsilon_{t+h-1}, \epsilon_t) + \operatorname{Cov}(u_{t+h}, \epsilon_t) = \rho\operatorname{Cov}(\epsilon_{t+h-1}, \epsilon_t).$$

Iterating down to $h = 0$ gives $\rho^h\operatorname{Var}\epsilon_t = \sigma_u^2\rho^h/(1-\rho^2)$.

(b) For $t \geq 2$, $\eta_t = u_t$, so these have variance σ_u^2 and are mutually uncorrelated. Also $\operatorname{Var}\eta_1 = (1-\rho^2)\operatorname{Var}\epsilon_1 = \sigma_u^2$ and $\operatorname{Cov}(\eta_1, u_t) = \sqrt{1-\rho^2}\operatorname{Cov}(\epsilon_1, u_t) = 0$. Hence $\operatorname{Var}\eta = \sigma_u^2 I_n$.

(c) With $V_{st} = \rho^{|s-t|}$, (a) says $\operatorname{Var}\epsilon = \frac{\sigma_u^2}{1-\rho^2}V$, and (b) says $L\{\frac{\sigma_u^2}{1-\rho^2}V\}L' = \sigma_u^2 I$, i.e. $LVL' = (1-\rho^2)I$. L is nonsingular (triangular with nonzero diagonal), so $V^{-1} = L'L/(1-\rho^2)$. Then

$$\hat{\beta}_G = (X'V^{-1}X)^{-1}X'V^{-1}\mathbf{y} = (X'L'LX)^{-1}X'L'L\mathbf{y},$$

the OLS estimator for the transformed data $L\mathbf{y}$, LX; the constant $1-\rho^2$ cancels. These are the quasi-differences of Example 12.2. ■

Chapter 13. Indicator Variables

13.1 One indicator variable

Qualitative explanatory variables (sex, treatment, region) enter a regression through **indicator** (dummy) variables taking the values 0 and 1.

Parallel lines. Suppose y is the score on a test, x_1 the hours of preparation, and $x_2 = 1$ for students who attended a review session and 0 otherwise. The model

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

gives two response functions with a common slope:

$$E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + \beta_1 x_1, & x_2 = 1. \end{cases}$$

β_1 is the effect of x_1 in both groups, and β_2 is the vertical distance between the parallel lines: the group effect at any fixed x_1. Fitting one model, rather than one line per group, uses all n observations to estimate the common slope and σ^2.

Interaction. If the effect of x_1 may differ between groups, add the product term:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \epsilon_i, \qquad E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_2 = 0, \\ (\beta_0 + \beta_2) + (\beta_1 + \beta_3) x_1, & x_2 = 1. \end{cases}$$

Now β_2 is the difference in intercepts and β_3 the difference in slopes. The hypotheses of Chapter 5 become ordinary tests in one model:

- parallel lines: $H_0 : \beta_3 = 0$ (a t test);
- identical lines: $H_0 : \beta_2 = \beta_3 = 0$ (a partial F test with 2 degrees of freedom).

These agree with the separate-fit tests of Section 5.3 because the interaction model with $n_1 + n_2$ observations has exactly the four parameters of the two separate lines and the same residual sum of squares.

Example 13.1

In a study of 20 firms, y is the time to adopt an innovation, x_1 the firm size and $x_2 = 1$ for stock companies. The fit $\hat{y} = 33.87 - 0.102 x_1 + 8.06 x_2$ says that adoption is faster for larger firms (about 1 month per 10 size units) and that stock companies take about 8 months longer at every size. Adding the interaction gives $\hat{\beta}_3 = -0.0004$ with standard error 0.018, so $t_0 = -0.02$: the parallel-lines model is adequate.

13.2 Several categories

A qualitative variable with q categories needs $q - 1$ indicators. With three categories E_1, E_2, E_3, define

$$x_2 = \mathbf{1}\{E_1\}, \qquad x_3 = \mathbf{1}\{E_2\},$$

with E_3 the **baseline**:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \epsilon_i, \qquad E(y) = \begin{cases} (\beta_0 + \beta_2) + \beta_1 x_1, & E_1, \\ (\beta_0 + \beta_3) + \beta_1 x_1, & E_2, \\ \beta_0 + \beta_1 x_1, & E_3. \end{cases}$$

So β_2 and β_3 compare E_1 and E_2 with the baseline, and $\beta_2 - \beta_3$ compares E_1 with E_2.

The dummy variable trap. Using q indicators together with an intercept makes the columns of X linearly dependent: the indicators sum to the intercept column $\mathbf{1}$. Then $\operatorname{rank} X <$ number of columns, $X'X$ is singular, and β is not identifiable. Either drop one indicator (baseline coding, as above), drop the intercept, or impose a constraint such as $\sum(\text{effects}) = 0$.

Interactions and several factors. Separate slopes are obtained with the products $x_1 x_2$ and $x_1 x_3$:

$$E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_2 + \beta_5 x_1 x_3,$$

so that in E_1 the line is $(\beta_0 + \beta_2) + (\beta_1 + \beta_4) x_1$, and so on. Several qualitative factors each get their own set of indicators, and interactions between factors are products of their indicators. A model containing only indicators (no quantitative regressors) is an analysis of variance model written as a regression.

13.3 Piecewise linear regression

Sometimes the slope changes at a known point x_w (a threshold or a policy change), while the response stays continuous. With $x_2 = \mathbf{1}\{x_1 > x_w\}$,

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 (x_{i1} - x_w) x_{i2} + \epsilon_i,$$

so that

$$E(y) = \begin{cases} \beta_0 + \beta_1 x_1, & x_1 \le x_w, \\ (\beta_0 - \beta_2 x_w) + (\beta_1 + \beta_2) x_1, & x_1 > x_w. \end{cases}$$

The two lines meet at x_w, β_2 is the change in slope, and $H_0 : \beta_2 = 0$ tests for a break. The regressor $(x_1 - x_w)_+ = (x_1 - x_w) x_2$ is a **truncated power basis** function; several knots $x_{w_1} < x_{w_2} < \cdots$ add one such term each, giving a linear spline. If a jump at x_w is also allowed, add x_2 itself.

Example 13.2

Unit cost y against lot size x_1 for eight lots, with a suspected change at $x_w = 500$:

x_1	300	340	400	480	570	650	720	800
y	475	440	452	377	355	257	249	139

The fit is $\hat{y} = 589.5 - 0.395 x_1 - 0.389 (x_1 - 500)_+$: the cost falls by 0.40 per unit up to 500 and by $0.40 + 0.39 = 0.78$ per unit beyond.

13.4 Selected exercises

Motivation. Indicator models are ordinary regressions, so the general results of Chapters 6–8 identify their estimates: a jump test for a broken line, the two-sample t test, the pooled within-group slope, and the equivalence of the interaction model with two separate fits.

Exercise 13.1 (Testing for a jump)

Consider

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 (x_{i1} - 40) x_{i2} + \beta_3 x_{i3} + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

where x_1 is quantitative and $x_{i2} = x_{i3} = 1$ if $x_{i1} > 40$ and 0 otherwise. The mean response varies over the range of x_1. Write down hypotheses for testing whether the mean response is discontinuous at $x_1 = 40$, and explain how to carry out the test.

Solution.

Step 1. (The mean function.) For $x_1 \le 40$, $E(y) = \beta_0 + \beta_1 x_1$; for $x_1 > 40$,

$$E(y) = (\beta_0 - 40\beta_2 + \beta_3) + (\beta_1 + \beta_2) x_1.$$

So the model consists of two lines with arbitrary intercepts and slopes on either side of 40; β_2 is the change in slope.

Step 2. (The jump.) The left line at $x_1 = 40$ gives $\beta_0 + 40\beta_1$, and the right line tends to $\beta_0 - 40\beta_2 + \beta_3 + 40(\beta_1 + \beta_2) = \beta_0 + 40\beta_1 + \beta_3$ as $x_1 \downarrow 40$. The jump is β_3, so the hypotheses are

$$H_0 : \beta_3 = 0 \ \text{ (continuous at 40)} \qquad \text{vs} \qquad H_1 : \beta_3 \neq 0.$$

Step 3. (The test.) Assume the design matrix with columns $\mathbf{1}$, $\mathbf{x}_1$, $(\mathbf{x}_1 - 40)\mathbf{x}_2$, $\mathbf{x}_3$ has full rank, which holds for instance when there are at least two distinct values of x_1 on each side of 40 (the model is a reparametrization of two separate lines). Fit the model by least squares and compute

$$t_0 = \frac{\hat\beta_3}{\sqrt{c_{33} MSE}}, \qquad MSE = \frac{SSE}{n-4},$$

with c_{33} the diagonal entry of $(X'X)^{-1}$ for β_3. Under H_0, $t_0 \sim t(n-4)$ (Section 6.5); reject when $|t_0| > t_{\alpha/2}(n-4)$. Equivalently (Section 8.2), compare the full model with the continuous broken line of Section 13.3 by the partial F statistic $F_0 = t_0^2$ with $(1, n-4)$ degrees of freedom. ■

Exercise 13.2 (An indicator alone is the two-sample t test)

Let $y_i = \beta_0 + \beta_1 d_i + \epsilon_i$ with $d_i \in \{0, 1\}$, $n_0 \ge 1$ observations with $d = 0$, $n_1 \ge 1$ with $d = 1$, $n = n_0 + n_1 \ge 3$, and group means $\bar y_0, \bar y_1$. Show that $\hat\beta_0 = \bar y_0$, $\hat\beta_1 = \bar y_1 - \bar y_0$, that MSE is the pooled variance s_p^2, and that the t statistic for $\beta_1 = 0$ is the pooled two-sample t statistic.

Solution [Supplement].

Step 1. This is a simple regression on $x = d$ with the two design points $a = 0$, $b = 1$. By Exercise 3.2, $\hat\beta_1 = \bar y_1 - \bar y_0$ and the fitted line passes through $(0, \bar y_0)$, so $\hat\beta_0 = \bar y_0$; also $S_{xx} = n_0 n_1 / n$.

Step 2. The fitted value of each observation is its group mean, so

$$SSE = \sum_{d_i = 0} (y_i - \bar y_0)^2 + \sum_{d_i = 1} (y_i - \bar y_1)^2, \qquad MSE = \frac{SSE}{n-2} = s_p^2.$$

Step 3. By Section 4.1,

$$t_0 = \frac{\hat\beta_1}{\sqrt{MSE / S_{xx}}} = \frac{\bar y_1 - \bar y_0}{s_p \sqrt{n/(n_0 n_1)}} = \frac{\bar y_1 - \bar y_0}{s_p \sqrt{1/n_0 + 1/n_1}},$$

which is the pooled two-sample t statistic, with the same $n - 2$ degrees of freedom. ■

Exercise 13.3 (The common slope of parallel lines)

In the parallel-lines model $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \epsilon_i$ of Section 13.1, let group $g \in \{0, 1\}$ (with $x_2 = g$) have mean $\bar{x}_g$ of x_1, mean $\bar{y}_g$, and within-group sums $S_{xx,g} = \sum_{i \in g}(x_{i1} - \bar{x}_g)^2$, $S_{xy,g} = \sum_{i \in g}(x_{i1} - \bar{x}_g)(y_i - \bar{y}_g)$, with $S_{xx,0} + S_{xx,1} > 0$. Show that

$$\hat{\beta}_1 = \frac{S_{xy,0} + S_{xy,1}}{S_{xx,0} + S_{xx,1}}, \qquad \hat{\beta}_2 = (\bar{y}_1 - \bar{y}_0) - \hat{\beta}_1(\bar{x}_1 - \bar{x}_0).$$

Solution [Supplement].

Step 1. Apply Theorem 6.5 with $X_1 = (\mathbf{1}, \mathbf{x}_2)$ and $X_2 = \mathbf{x}_1$. The column space of X_1 is spanned by the group indicators $\mathbf{1}_{G_0} = \mathbf{1} - \mathbf{x}_2$ and $\mathbf{1}_{G_1} = \mathbf{x}_2$, which are orthogonal. So $H_1\mathbf{v}$ replaces each entry of $\mathbf{v}$ by the mean of its group, and $(I - H_1)\mathbf{v}$ subtracts the group means.

Step 2. Hence $\mathbf{e} = (I - H_1)\mathbf{y}$ has entries $y_i - \bar{y}_{g(i)}$ and $X_{2\cdot1} = (I - H_1)\mathbf{x}_1$ has entries $x_{i1} - \bar{x}_{g(i)}$. Since $\|X_{2\cdot1}\|^2 = S_{xx,0} + S_{xx,1} > 0$, the design has full rank, and

$$\hat{\beta}_1 = \frac{X_{2\cdot1}'\mathbf{e}}{X_{2\cdot1}'X_{2\cdot1}} = \frac{S_{xy,0} + S_{xy,1}}{S_{xx,0} + S_{xx,1}}.$$

Step 3. The residuals of the full fit are orthogonal to $\mathbf{1}_{G_0}$ and $\mathbf{1}_{G_1}$, so they sum to zero within each group. Averaging the fitted equation over each group gives $\bar{y}_0 = \hat{\beta}_0 + \hat{\beta}_1\bar{x}_0$ and $\bar{y}_1 = \hat{\beta}_0 + \hat{\beta}_2 + \hat{\beta}_1\bar{x}_1$. Subtracting yields $\hat{\beta}_2$.

So the common slope pools the within-group information only, and $\hat{\beta}_2$ compares the group means *adjusted* to a common value of x_1 (the analysis of covariance). ■

Exercise 13.4 (The interaction model is two separate fits)

Fit the interaction model $E(y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$ ($x_2 \in \{0, 1\}$) to the pooled data, and separately fit lines $b_{0g} + b_{1g}x_1$ in each group g (each group having at least two distinct x_1 values). Show that both give the same fitted values and $SSE = SSE_0 + SSE_1$, and that $\hat{\beta}_0 = b_{00}$, $\hat{\beta}_1 = b_{10}$, $\hat{\beta}_2 = b_{01} - b_{00}$, $\hat{\beta}_3 = b_{11} - b_{10}$.

Solution [Supplement].

Step 1. Let Z have columns $\mathbf{1}_{G_0}$, $\mathbf{x}_1\mathbf{1}_{G_0}$, $\mathbf{1}_{G_1}$, $\mathbf{x}_1\mathbf{1}_{G_1}$ (products taken entrywise), the design of the separate fits. The interaction design has columns

$$\mathbf{1} = \mathbf{1}_{G_0} + \mathbf{1}_{G_1}, \quad \mathbf{x}_1 = \mathbf{x}_1\mathbf{1}_{G_0} + \mathbf{x}_1\mathbf{1}_{G_1}, \quad \mathbf{x}_2 = \mathbf{1}_{G_1}, \quad \mathbf{x}_1\mathbf{x}_2 = \mathbf{x}_1\mathbf{1}_{G_1},$$

so $X = ZA$ for a 4×4 matrix A; it is nonsingular, since conversely $\mathbf{1}_{G_0} = \mathbf{1} - \mathbf{x}_2$ and $\mathbf{x}_1\mathbf{1}_{G_0} = \mathbf{x}_1 - \mathbf{x}_1\mathbf{x}_2$ express Z through X.

Step 2. Ordering the observations by group, Z is block diagonal, so $\|\mathbf{y} - Z\mathbf{b}\|^2 = \sum_g \|\mathbf{y}_g - b_{0g}\mathbf{1} - b_{1g}\mathbf{x}_{1,g}\|^2$ is minimized by minimizing each term separately: the least squares fit on Z is the pair of separate lines, with $SSE = SSE_0 + SSE_1$. Each block has full rank by the assumption on the x_1 values, so Z and hence X have full rank.

Step 3. By Exercise 6.3, X and Z give the same fitted values and SSE, so $X\hat{\beta} = Z\hat{\mathbf{b}}$. By Step 1,

$$X\hat{\beta} = \hat{\beta}_0\mathbf{1}_{G_0} + \hat{\beta}_1\mathbf{x}_1\mathbf{1}_{G_0} + (\hat{\beta}_0 + \hat{\beta}_2)\mathbf{1}_{G_1} + (\hat{\beta}_1 + \hat{\beta}_3)\mathbf{x}_1\mathbf{1}_{G_1},$$

and since the columns of Z are linearly independent, $b_{00} = \hat{\beta}_0$, $b_{10} = \hat{\beta}_1$, $b_{01} = \hat{\beta}_0 + \hat{\beta}_2$ and $b_{11} = \hat{\beta}_1 + \hat{\beta}_3$. This justifies the remark after the interaction model in Section 13.1. ■

Chapter 15. Multicollinearity and Biased Estimation

The least squares estimator $\hat\beta = (X'X)^{-1}X'\mathbf{y}$ is the BLUE with covariance $\sigma^2(X'X)^{-1}$. When $X'X$ is nearly singular this covariance is huge, and a slightly biased estimator with much smaller variance can be far more accurate. This chapter studies the problem and two remedies.

Throughout we standardize: y is centred and each regressor is centred and scaled so that $\sum_i x_{ij} = 0$ and $\sum_i x_{ij}^2 = 1$. Then X is $n \times p$ (no intercept column), $X'X$ is the correlation matrix of the regressors, and $X'\mathbf{y}$ is proportional to their correlations with y. Coefficients on the original scale are recovered from $\tilde\beta_j = \hat\beta_j\, s_y/s_j$ and $\tilde\beta_0 = \bar y - \sum_j \tilde\beta_j \bar x_j$, where s_j is the scale used for x_j.

15.1 Multicollinearity

Multicollinearity means that the regressors are (nearly) linearly dependent. Exact dependence makes X rank-deficient and $\hat\beta$ undefined; near dependence makes $(X'X)^{-1}$ huge and $\hat\beta$ unstable.

Example 15.1 (Two regressors)

With two standardized regressors with correlation r_{12},

$$X'X = \begin{pmatrix} 1 & r_{12} \\ r_{12} & 1 \end{pmatrix}, \qquad \operatorname{Var}\hat\beta = \frac{\sigma^2}{1-r_{12}^2}\begin{pmatrix} 1 & -r_{12} \\ -r_{12} & 1 \end{pmatrix},$$

so $\operatorname{Var}\hat\beta_j = \sigma^2/(1-r_{12}^2) \to \infty$ as $|r_{12}| \to 1$, and $\operatorname{Corr}(\hat\beta_1, \hat\beta_2) = -r_{12}$. With $r_{12} = 0.99$ the variances are 50 times those under orthogonal regressors, and confidence intervals $\hat\beta_j \pm t_{\alpha/2}(n-3)\sqrt{MSE/(1-r_{12}^2)}$ are 7 times as wide.

Eigenvalues. If $\lambda_1, \dots, \lambda_p$ are the eigenvalues of $X'X$,

$$\sum_j \operatorname{Var}\hat\beta_j = \sigma^2 \operatorname{tr}(X'X)^{-1} = \sigma^2 \sum_j \frac{1}{\lambda_j},$$

so a single small eigenvalue blows up the total variance. The eigenvector of a small eigenvalue shows which combination of regressors is nearly constant.

Theorem 15.1 (Variance inflation factor)

Let R_j^2 be the coefficient of determination of the regression of x_j on the other regressors. Then c_{jj}, the j-th diagonal entry of $(X'X)^{-1}$, equals $(1-R_j^2)^{-1}$, so

$$\operatorname{Var}\hat\beta_j = \frac{\sigma^2}{1-R_j^2}.$$

Proof. Write $X = (\mathbf{x}_j, X_{-j})$. By the partitioned-inverse formula,

$$c_{jj} = \{\mathbf{x}_j'\mathbf{x}_j - \mathbf{x}_j' X_{-j}(X_{-j}'X_{-j})^{-1}X_{-j}'\mathbf{x}_j\}^{-1} = \{\mathbf{x}_j'(I - H_{-j})\mathbf{x}_j\}^{-1},$$

which is the inverse of the residual sum of squares of $\mathbf{x}_j$ on X_{-j}. Since $\mathbf{x}_j$ is centred with $\mathbf{x}_j'\mathbf{x}_j = 1$, its total sum of squares is 1, so the residual sum of squares is $1 - R_j^2$. ∎

$\mathrm{VIF}_j = (1-R_j^2)^{-1}$ is the **variance inflation factor**: the factor by which $\operatorname{Var}\hat\beta_j$ exceeds its value under orthogonal regressors. $\mathrm{VIF}_j > 10$ (i.e. $R_j^2 > 0.9$) is a common warning sign.

15.2 Ridge regression

Motivation. Under multicollinearity least squares remains unbiased but its variance explodes. Ridge regression adds kI to $X^\top X$; this biases the estimates but, for suitable k, always reduces mean squared error.

Measure the accuracy of an estimator b by the total mean squared error $E\|b-\beta\|^2$. For least squares,

$$E\|\hat\beta-\beta\|^2 = \operatorname{tr}\operatorname{Var}\hat\beta = \sigma^2\sum_j \frac{1}{\lambda_j}.$$

Definition (Ridge estimator)

For $k>0$, $\hat\beta(k)=(X'X+kI_p)^{-1}X'\mathbf{y}$, the minimizer of $\|\mathbf{y}-X\beta\|^2+k\|\beta\|^2$.

Adding k to the diagonal lifts all eigenvalues by k, so $X'X+kI$ is well conditioned. If $X'X=I$, then $\hat\beta(k)=\hat\beta/(1+k)$: ridge **shrinks** least squares towards $\mathbf{0}$. In general $\hat\beta(k)=(X'X+kI)^{-1}X'X\hat\beta$, which is biased: $E\hat\beta(k)=(X'X+kI)^{-1}X'X\beta\neq\beta$.

Canonical form. Let $X'X=P\Lambda P'$ with P orthogonal and $\Lambda=\operatorname{diag}(\lambda_1,\dots,\lambda_p)$. With $Z=XP$ and $\alpha=P'\beta$, the model is $\mathbf{y}=Z\alpha+\epsilon$ with $Z'Z=\Lambda$. Then $\hat\alpha=\Lambda^{-1}Z'\mathbf{y}$, $\operatorname{Var}\hat\alpha=\sigma^2\Lambda^{-1}$, and

$$\hat\alpha(k)=P'\hat\beta(k)=(\Lambda+kI)^{-1}Z'\mathbf{y}, \qquad \hat\alpha_j(k)=\frac{\lambda_j}{\lambda_j+k}\hat\alpha_j :$$

each canonical coefficient is shrunk, most strongly in directions with small λ_j. Since P is orthogonal, $\|\hat\beta(k)-\beta\|=\|\hat\alpha(k)-\alpha\|$.

Theorem 15.2 (Mean squared error of ridge)

$$E\|\hat\beta(k)-\beta\|^2=\sigma^2\sum_{j=1}^{p}\frac{\lambda_j}{(\lambda_j+k)^2}+k^2\sum_{j=1}^{p}\frac{\alpha_j^2}{(\lambda_j+k)^2}.$$

The first term (total variance) decreases in k and the second (squared bias) increases. There is always a $k>0$ with $E\|\hat\beta(k)-\beta\|^2<E\|\hat\beta-\beta\|^2$.

Proof [Supplement]. $\hat\alpha_j(k)$ has mean $\frac{\lambda_j}{\lambda_j+k}\alpha_j$ and variance $\frac{\lambda_j^2}{(\lambda_j+k)^2}\cdot\frac{\sigma^2}{\lambda_j}$; its bias is $-\frac{k}{\lambda_j+k}\alpha_j$. Adding variance and squared bias over j gives the formula. Call it $g(k)$. Then $g(0)=\sigma^2\sum 1/\lambda_j=E\|\hat\beta-\beta\|^2$, and

$$g'(k)=-2\sigma^2\sum_j\frac{\lambda_j}{(\lambda_j+k)^3}+2k\sum_j\frac{\lambda_j\alpha_j^2}{(\lambda_j+k)^3}.$$

At $k=0$, $g'(0)=-2\sigma^2\sum_j\lambda_j^{-2}<0$, so $g(k)<g(0)$ for small $k>0$. ■

In the orthonormal case $X'X=I$, $g(k)=\frac{p\sigma^2+k^2\|\alpha\|^2}{(1+k)^2}$, minimized at $k=p\sigma^2/\|\beta\|^2$ (set $g'(k)=0$). More generally, a separate constant k_j for each canonical direction, $\hat\alpha_j=\frac{\lambda_j}{\lambda_j+k_j}\hat\alpha_j$ (generalized ridge), has mean squared error $\sum_j\frac{\sigma^2\lambda_j+k_j^2\alpha_j^2}{(\lambda_j+k_j)^2}$, minimized at $k_j=\sigma^2/\alpha_j^2$.

Choosing k. The optimal k depends on unknown σ^2 and β. Practical choices:

- the **ridge trace**: plot $\hat\beta_j(k)$ against k and take the smallest k at which the coefficients stabilize;

- the plug-in rule $\hat{k} = p\hat{\sigma}^2/\hat{\beta}'\hat{\beta}$ (Hoerl, Kennard and Baldwin), possibly iterated with $\hat{\beta}(k)$ in place of $\hat{\beta}$;
- cross-validation, e.g. minimizing $\text{PRESS}(k) = \sum_i \{y_i - \mathbf{x}_i'\hat{\beta}_{(i)}(k)\}^2$. As in Theorem 10.3, with $A(k) = X(X'X + kI)^{-1}X' = (a_{ij})$ and $e_i = y_i - \mathbf{x}_i'\hat{\beta}(k)$,

$$\text{PRESS}(k) = \sum_{i=1}^{n} \Big(\frac{e_i}{1 - a_{ii}}\Big)^2,$$

so no refitting is needed (leave-one-out cross-validation).

15.3 Principal component regression

Let $\mathbf{e}_1, \dots, \mathbf{e}_p$ be orthonormal eigenvectors of $X'X$ with eigenvalues $l_1 \geq \cdots \geq l_p > 0$, and $E_r = (\mathbf{e}_1, \dots, \mathbf{e}_r)$. The **principal components** are the columns of $Z_r = XE_r$, and they are orthogonal: $Z_r'Z_r = \text{diag}(l_1, \dots, l_r)$.

All components ($r = p$). $X\beta = XE_pE_p'\beta = Z\alpha$ with $\alpha = E_p'\beta$, and

$$\hat{\beta} = E_p\hat{\alpha} = E_p(Z'Z)^{-1}Z'\mathbf{y} = \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}, \qquad \text{Var}\,\hat{\beta} = \sigma^2 \sum_{k=1}^{p} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

which is just least squares.

Leading components ($r < p$). Regress $\mathbf{y}$ on Z_r only and transform back:

$$\tilde{\beta}_r = E_r(Z_r'Z_r)^{-1}Z_r'\mathbf{y} = \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'\mathbf{y}.$$

Theorem 15.3 (Principal component regression)

$$E\tilde{\beta}_r = \beta - \sum_{k=r+1}^{p} \mathbf{e}_k\mathbf{e}_k'\beta, \qquad \text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k',$$

and

$$E\|\tilde{\beta}_r - \beta\|^2 = \sigma^2 \sum_{k=1}^{r} \frac{1}{l_k} + \sum_{k=r+1}^{p} (\mathbf{e}_k'\beta)^2.$$

Proof. Using $X'X = \sum_h l_h\mathbf{e}_h\mathbf{e}_h'$ and orthonormality, $E\tilde{\beta}_r = \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'X'X\beta = \sum_{k\leq r} \mathbf{e}_k\mathbf{e}_k'\beta$, and $\beta = \sum_{k\leq p} \mathbf{e}_k\mathbf{e}_k'\beta$. Next,

$$\text{Var}\,\tilde{\beta}_r = \sigma^2 \sum_{k,k'\leq r} \frac{1}{l_k l_{k'}}\mathbf{e}_k\mathbf{e}_k'X'X\mathbf{e}_{k'}\mathbf{e}_{k'}' = \sigma^2 \sum_{k\leq r} \frac{1}{l_k}\mathbf{e}_k\mathbf{e}_k'.$$

The trace of the variance plus the squared length of the bias gives the mean squared error. ■

Dropping the components with the smallest l_k removes the largest variance terms σ^2/l_k at the cost of the bias $\sum_{k>r}(\mathbf{e}_k'\beta)^2$. This pays off when the small-eigenvalue directions carry little of β. PCR is a "hard" version of ridge: ridge shrinks every canonical coefficient by $l_k/(l_k + k)$, PCR keeps the first r and sets the rest to 0.

Choosing r. Common rules: keep components until the cumulative proportion $\sum_{k\le r} l_k / \sum_k l_k$ (the share of the total variance of the standardized regressors) is large enough; look for an elbow in the scree plot of l_k; or choose r by cross-validation. Note that the components are chosen without looking at y; partial least squares instead builds components that are correlated with y.

15.4 Selected exercises

Motivation. The exercises restate the variance inflation factor on the original scale of the data, compute ridge regression as least squares on augmented data, and show that ridge solves a norm-constrained least squares problem.

Exercise 15.1 (The VIF on the original scale)

In $Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i$ with uncorrelated errors of variance σ^2 and unstandardized regressors, let $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2 > 0$ and let R_j^2 be the coefficient of determination of the regression (with intercept) of x_j on the other regressors. Show that the least squares estimator satisfies

$$\operatorname{Var} \hat{\beta}_j = \frac{\sigma^2}{S_{jj}} \operatorname{VIF}_j, \qquad \operatorname{VIF}_j = (1 - R_j^2)^{-1}.$$

Solution.

Step 1. (Centring.) Let $C = I - J_n/n$, the projection onto $\mathbf{1}^\perp$, and $\tilde{X} = CX_s$, where X_s holds the p regressor columns. By Theorem 6.5 with $X_1 = \mathbf{1}$, the slope vector is $\hat{\beta}_s = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'C\mathbf{y} = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'\mathbf{y}$, so $\operatorname{Var}\hat{\beta}_s = (\tilde{X}'\tilde{X})^{-1}\tilde{X}'(\sigma^2 I)\tilde{X}(\tilde{X}'\tilde{X})^{-1} = \sigma^2(\tilde{X}'\tilde{X})^{-1}$.

Step 2. (Partitioned inverse.) Write $\tilde{X} = (\tilde{\mathbf{x}}_j, \tilde{X}_{-j})$ and $\tilde{H}_{-j} = \tilde{X}_{-j}(\tilde{X}'_{-j}\tilde{X}_{-j})^{-1}\tilde{X}'_{-j}$. As in the proof of Theorem 15.1, the j-th diagonal entry of $(\tilde{X}'\tilde{X})^{-1}$ is $\{\tilde{\mathbf{x}}_j'(I - \tilde{H}_{-j})\tilde{\mathbf{x}}_j\}^{-1}$.

Step 3. (A residual sum of squares.) The column space of $(\mathbf{1}, X_{-j})$ is spanned by $\mathbf{1}$ and the columns of $\tilde{X}_{-j}$, which are orthogonal to $\mathbf{1}$; so its projection is $J_n/n + \tilde{H}_{-j}$, as in the proof of Theorem 10.1(d). Since $\tilde{H}_{-j}\mathbf{1} = \mathbf{0}$, the residual of $\mathbf{x}_j$ on $(\mathbf{1}, X_{-j})$ is

$$(I - J_n/n - \tilde{H}_{-j})\mathbf{x}_j = \tilde{\mathbf{x}}_j - \tilde{H}_{-j}\tilde{\mathbf{x}}_j = (I - \tilde{H}_{-j})\tilde{\mathbf{x}}_j,$$

whose squared length is $\tilde{\mathbf{x}}_j'(I - \tilde{H}_{-j})\tilde{\mathbf{x}}_j$. The total sum of squares of x_j is S_{jj}, so this residual sum of squares equals $S_{jj}(1 - R_j^2)$.

Step 4. Combining, $\operatorname{Var}\hat{\beta}_j = \sigma^2/\{S_{jj}(1 - R_j^2)\}$. The factor σ^2/S_{jj} is the variance in the simple regression on x_j alone (Theorem 4.1(a)), which is also the value when x_j is uncorrelated with the other regressors, and VIF_j is the inflation due to collinearity. Theorem 15.1 is the case $S_{jj} = 1$. ■

Exercise 15.2 (Ridge regression as augmented least squares)

Let $E(\mathbf{y} \mid X) = X\beta$ and $\operatorname{Var}(\mathbf{y} \mid X) = \sigma^2 I_n$, X of size $n \times p$ with singular value decomposition $X = UDV'$ (U of size $n \times p$ with orthonormal columns, $D = \operatorname{diag}(d_1, \ldots, d_p)$ with $d_j \ge 0$, V orthogonal). For $\lambda > 0$ let

$$X_* = \begin{pmatrix} X \\ \sqrt{\lambda} I_p \end{pmatrix}, \qquad \mathbf{y}_* = \begin{pmatrix} \mathbf{y} \\ \mathbf{0}_p \end{pmatrix}, \qquad \hat{\beta} = (X_*'X_*)^{-1}X_*'\mathbf{y}_*.$$

Express $E(\hat{\beta} \mid X)$ and $\operatorname{Var}(\hat{\beta} \mid X)$ through U, D, V, β and σ.

Solution.

Step 1. $X_*'X_* = X'X + \lambda I$ and $X_*'\mathbf{y}_* = X'\mathbf{y}$, so $\hat\beta = (X'X+\lambda I)^{-1}X'\mathbf{y}$ is the ridge estimator with $k=\lambda$; it is defined even if X is rank deficient, because $X'X+\lambda I$ is positive definite.

Step 2. $X'X = VDU'UDV' = VD^2V'$ and $X'X+\lambda I = V(D^2+\lambda I)V'$, so $(X'X+\lambda I)^{-1} = V(D^2+\lambda I)^{-1}V'$ and

$$\hat\beta = V(D^2+\lambda I)^{-1}V'VDU'\mathbf{y} = V(D^2+\lambda I)^{-1}DU'\mathbf{y}.$$

Step 3. (Mean.) $E(\hat\beta \mid X) = V(D^2+\lambda I)^{-1}DU'UDV'\beta = V\,\mathrm{diag}\Big(\dfrac{d_j^2}{d_j^2+\lambda}\Big)V'\beta.$

Step 4. (Variance.) With $M = V(D^2+\lambda I)^{-1}DU'$, $\mathrm{Var}(\hat\beta \mid X) = \sigma^2 MM'$ and $U'U = I$ give

$$\mathrm{Var}(\hat\beta \mid X) = \sigma^2 V(D^2+\lambda I)^{-1}D^2(D^2+\lambda I)^{-1}V' = \sigma^2 V\,\mathrm{diag}\Big(\frac{d_j^2}{(d_j^2+\lambda)^2}\Big)V'.$$

In the coordinates $\alpha = V'\beta$ these are the canonical formulas of Section 15.2 with $\lambda_j = d_j^2$: each component is shrunk by $d_j^2/(d_j^2+\lambda)$. The p extra rows act as pseudo-observations $0 = \sqrt{\lambda}\beta_j + \text{error}$ that pull β towards $\mathbf{0}$. ■

Exercise 15.3 (Ridge as constrained least squares)

Let X $(n\times p)$ have full column rank, $\hat\beta$ be the least squares estimator, $Q(\beta) = (\beta-\hat\beta)'X'X(\beta-\hat\beta)$, and $c>0$.
(a) Find the minimizer of $Q(\beta)$ subject to $\beta'\beta \le c$.
(b) For fixed $k\ge 0$, find the mean and variance of $\hat\beta(k) = (X'X+kI)^{-1}X'\mathbf{y}$.
(c) Using the singular value decomposition $X = UDV'$, show that $\hat\beta(k)$ shrinks $\hat\beta$ towards $\mathbf{0}$.

Solution.

(a) If $\hat\beta'\hat\beta \le c$, then $\hat\beta$ is feasible with $Q(\hat\beta)=0$, and it is the unique minimizer since $X'X$ is positive definite. Assume now $\hat\beta'\hat\beta > c$.

Step 1. For $k\ge 0$, $Q(\beta)+k\|\beta\|^2$ is strictly convex with gradient $2X'X(\beta-\hat\beta)+2k\beta$, which vanishes at $(X'X+kI)^{-1}X'X\hat\beta = (X'X+kI)^{-1}X'\mathbf{y} = \hat\beta(k)$. So $\hat\beta(k)$ is its unique minimizer.

Step 2. With $X'X = P\Lambda P'$ and $\hat\alpha = P'\hat\beta$, $\|\hat\beta(k)\|^2 = \sum_j \lambda_j^2\hat\alpha_j^2/(\lambda_j+k)^2$. This is continuous and strictly decreasing in k (since $\hat\alpha \ne \mathbf{0}$), equals $\|\hat\beta\|^2 > c$ at $k=0$, and tends to 0. Hence there is a unique $k^*>0$ with $\|\hat\beta(k^*)\|^2 = c$.

Step 3. For any feasible $\beta \ne \hat\beta(k^*)$, by Step 1,

$$Q(\beta) > Q(\hat\beta(k^*)) + k^*(\|\hat\beta(k^*)\|^2 - \|\beta\|^2) = Q(\hat\beta(k^*)) + k^*(c-\|\beta\|^2) \ge Q(\hat\beta(k^*)).$$

So the constrained minimizer is the ridge estimator $\hat\beta(k^*)$, the ridge constant being the Lagrange multiplier of the constraint.

(b) $\hat\beta(k) = M\mathbf{y}$ with $M = (X'X+kI)^{-1}X'$, so

$$E\hat\beta(k) = (X'X+kI)^{-1}X'X\beta, \qquad \mathrm{Var}\,\hat\beta(k) = \sigma^2(X'X+kI)^{-1}X'X(X'X+kI)^{-1}.$$

These formulas are for a fixed k; the k^* of (a) depends on $\mathbf{y}$, so they describe the constrained estimator only conditionally on its multiplier.

(c) As in Exercise 15.2, $\hat\beta(k) = V\,\mathrm{diag}\{d_j^2/(d_j^2+k)\}V'\hat\beta$ with all $d_j>0$ by full rank. In the orthonormal coordinates $V'\beta$, each component of $\hat\beta$ is multiplied by $d_j^2/(d_j^2+k) \in (0,1]$, strictly less than 1 for $k>0$; hence $\|\hat\beta(k)\| < \|\hat\beta\|$ whenever $k>0$ and $\hat\beta \ne \mathbf{0}$, and the smallest singular values are shrunk most. ■

Chapter 16. Logistic Regression

16.1 A model for a binary response

When y takes only the values 0 and 1, $E(y \mid x) = P(y = 1 \mid x) =: \pi(x)$ is a probability. The linear model $y = \beta_0 + \beta_1 x + \epsilon$ is unsuitable: fitted values can leave $[0, 1]$, the errors take only two values for each x (so they are not normal), and $\mathrm{Var}(y \mid x) = \pi(x)\{1 - \pi(x)\}$ varies with x. Empirically, $\pi(x)$ often follows an S-shaped curve. We therefore model

$$\pi(x) = G(\beta_0 + \beta_1 x)$$

with a cdf G, the inverse **link function**. Common choices: the logistic $G(t) = e^t/(1 + e^t)$, the normal cdf Φ (probit), and the Gumbel $G(t) = \exp(-e^{-t})$. The logistic choice gives the **logistic regression model**

$$\pi(x) = \frac{\exp(\beta_0 + \beta_1 x)}{1 + \exp(\beta_0 + \beta_1 x)}, \qquad \text{equivalently} \qquad \operatorname{logit} \pi(x) = \log \frac{\pi(x)}{1 - \pi(x)} = \beta_0 + \beta_1 x.$$

The curve increases in x if $\beta_1 > 0$ and decreases if $\beta_1 < 0$.

Odds and odds ratios. The **odds** of $y = 1$ are $\pi/(1 - \pi) = P(y = 1)/P(y = 0) \in [0, \infty)$, with $\pi = \text{odds}/(1 + \text{odds})$. Under the logistic model

$$\frac{\pi(x)}{1 - \pi(x)} = e^{\beta_0}(e^{\beta_1})^x, \qquad \frac{\pi(x+1)/\{1 - \pi(x+1)\}}{\pi(x)/\{1 - \pi(x)\}} = e^{\beta_1}.$$

So e^{β_1} is the **odds ratio** for a unit increase in x, the same at every x. For a binary x (exposed $= 1$), e^{β_1} is the odds ratio between the two groups; for instance $\hat\beta_1 = 2.0$ means the odds of $y = 1$ are $e^2 \approx 7.4$ times higher in the exposed group.

16.2 Estimation

Motivation. For a binary response the linear model fails (fitted probabilities outside $[0, 1]$, nonconstant variance). Logistic regression models the log-odds linearly, and its coefficients are estimated by maximum likelihood with iterative reweighted least squares.

With $\mathbf{x}_i = (1, x_{i1}, \ldots, x_{ip})'$ and $\pi_i = \pi(\mathbf{x}_i) = e^{\mathbf{x}_i'\beta}/(1 + e^{\mathbf{x}_i'\beta})$, independent observations $y_i \sim \text{Bernoulli}(\pi_i)$ give the likelihood $L(\beta) = \prod \pi_i^{y_i}(1 - \pi_i)^{1-y_i}$ and

$$l(\beta) = \sum_{i=1}^{n} \{y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)\} = \sum_{i=1}^{n} \{y_i \mathbf{x}_i'\beta - \log(1 + e^{\mathbf{x}_i'\beta})\}.$$

Theorem 16.1

With $\pi(\beta) = (\pi_1, \ldots, \pi_n)'$ and $W(\beta) = \mathrm{diag}\{\pi_i(1 - \pi_i)\}$,

$$\frac{\partial l}{\partial \beta} = X'\{\mathbf{y} - \pi(\beta)\}, \qquad \frac{\partial^2 l}{\partial \beta \partial \beta'} = -X'W(\beta)X.$$

The Hessian is negative definite (if X has full column rank), so l is strictly concave and the MLE, when it exists, is the unique root of $X'\{\mathbf{y} - \pi(\hat\beta)\} = \mathbf{0}$.

Proof. $\frac{d}{dt}\log(1+e^t) = \frac{e^t}{1+e^t}$ and $\frac{d^2}{dt^2}\log(1+e^t) = \frac{e^t}{(1+e^t)^2} = G(t)\{1-G(t)\}$; apply the chain rule with $t_i = \mathbf{x}_i'\beta$. Since $0 < \pi_i < 1$, $\mathbf{a}'X'WX\mathbf{a} = \sum_i \pi_i(1-\pi_i)(\mathbf{x}_i'\mathbf{a})^2 > 0$ for $\mathbf{a} \neq \mathbf{0}$. ■

The likelihood equations are nonlinear and are solved by Newton–Raphson:

$$\beta^{(t+1)} = \beta^{(t)} + \{X'W^{(t)}X\}^{-1}X'\{\mathbf{y} - \pi^{(t)}\} = \{X'W^{(t)}X\}^{-1}X'W^{(t)}\mathbf{z}^{(t)},$$

with the **working response**

$$\mathbf{z}^{(t)} = X\beta^{(t)} + \{W^{(t)}\}^{-1}\{\mathbf{y} - \pi^{(t)}\}.$$

Each step is a weighted least squares regression of $\mathbf{z}^{(t)}$ on X with weights $\pi_i^{(t)}(1-\pi_i^{(t)})$:

$$\beta^{(t+1)} = \arg\min_{\beta}\{\mathbf{z}^{(t)} - X\beta\}'W^{(t)}\{\mathbf{z}^{(t)} - X\beta\},$$

hence the name **iteratively reweighted least squares** (IRLS). A common start is $\beta^{(0)} = \mathbf{0}$. Since the Hessian does not depend on $\mathbf{y}$, Newton–Raphson coincides with Fisher scoring, and $\{X'W(\hat\beta)X\}^{-1}$ estimates the asymptotic covariance of $\hat\beta$. Wald tests and intervals use $\hat\beta_j/\mathrm{se}(\hat\beta_j) \approx N(0,1)$; likelihood ratio tests compare $2\{l(\hat\beta_F) - l(\hat\beta_R)\}$ with a chi-square distribution.

If the two classes can be perfectly separated by a hyperplane in x, l increases without bound along a direction and the MLE does not exist.

16.3 Two-by-two tables

Motivation. With a single binary predictor, logistic regression reduces to a 2×2 table, and its slope is the log odds ratio. This connects the model with classical contingency-table analysis.

Binary exposure (x) and outcome (y) data arise from three designs:

- **cohort** (prospective) studies follow exposed and unexposed groups and record outcomes;
- **case–control** (retrospective) studies sample cases ($y = 1$) and controls ($y = 0$) and record past exposure;
- **cross-sectional** studies sample individuals and record both at one time.

With counts n_{xy}:

	$y=0$	$y=1$	total
$x=0$	n_{00}	n_{01}	n_{0+}
$x=1$	n_{10}	n_{11}	n_{1+}

The **odds ratio** is $OR = \dfrac{\pi_1/(1-\pi_1)}{\pi_0/(1-\pi_0)}$ with $\pi_x = P(y=1 \mid x)$, estimated by $\widehat{OR} = \dfrac{n_{11}n_{00}}{n_{10}n_{01}}$.

Theorem 16.2

In the logistic model $\mathrm{logit}\, P(y=1 \mid x) = \beta_0 + \beta_1 x$ with binary x, the MLEs satisfy

$$\hat\pi_1 = \frac{e^{\hat\beta_0+\hat\beta_1}}{1+e^{\hat\beta_0+\hat\beta_1}} = \frac{n_{11}}{n_{1+}}, \qquad \hat\pi_0 = \frac{e^{\hat\beta_0}}{1+e^{\hat\beta_0}} = \frac{n_{01}}{n_{0+}}, \qquad e^{\hat\beta_1} = \frac{n_{11}n_{00}}{n_{10}n_{01}}.$$

Proof. The log-likelihood is

$$l = n_{11}(\beta_0+\beta_1) + n_{01}\beta_0 - n_{1+}\log(1+e^{\beta_0+\beta_1}) - n_{0+}\log(1+e^{\beta_0}).$$

The model is saturated (two parameters for two probabilities), so by invariance the MLEs of π_1, π_0 are the sample proportions n_{11}/n_{1+} and n_{01}/n_{0+}. Then $e^{\hat\beta_1} = \frac{\hat\pi_1/(1-\hat\pi_1)}{\hat\pi_0/(1-\hat\pi_0)} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ■

Why the odds ratio. In a case–control study the row proportions do not estimate π_x, because the numbers of cases and controls are fixed by design. But the odds ratio is symmetric in x and y:

$$\frac{P(y=1 \mid x=1)P(y=0 \mid x=0)}{P(y=0 \mid x=1)P(y=1 \mid x=0)} = \frac{P(x=1 \mid y=1)P(x=0 \mid y=0)}{P(x=0 \mid y=1)P(x=1 \mid y=0)},$$

since both equal $\frac{p_{11}p_{00}}{p_{10}p_{01}}$ in terms of the joint cell probabilities. So the odds ratio, and hence the slope β_1 of a logistic regression, can be estimated from case–control data; only the intercept is distorted by the sampling.

Example 16.1

In a trial, 140 subjects receive a supplement and 140 a placebo; 17 and 31 catch a cold, respectively. With $x = 1$ for the supplement, $\widehat{OR} = \frac{17 \cdot 109}{123 \cdot 31} = 0.49$: the odds of a cold are about halved, and $\hat\beta_1 = \log 0.49 = -0.72$. The standard error is $\sqrt{\frac{1}{17} + \frac{1}{123} + \frac{1}{31} + \frac{1}{109}} = 0.33$, so the Wald statistic is -2.19, significant at the 5% level.

16.4 Selected exercises

Motivation. Three consequences of the likelihood equations: fitted and observed counts agree, the standard error used in Example 16.1 comes from the information matrix, and complete separation destroys the MLE.

Exercise 16.1 (Fitted counts equal observed counts)

(a) Show that the MLE of a logistic regression satisfies $X'\hat\pi = X'\mathbf{y}$; in particular, with an intercept, $\sum_i \hat\pi_i = \sum_i y_i$.
(b) For a binary x with all four cell counts n_{xy} positive, derive Theorem 16.2 directly from (a).

Solution [Supplement].

(a) By Theorem 16.1 the MLE solves $X'\{\mathbf{y} - \pi(\hat\beta)\} = \mathbf{0}$. The intercept row of X' is $\mathbf{1}'$, which gives $\sum \hat\pi_i = \sum y_i$: the expected number of successes under the fit equals the observed number. Each other regressor gives $\sum_i x_{ij}\hat\pi_i = \sum_i x_{ij} y_i$.

(b) With $\mathbf{x}_i = (1, x_i)'$ and $x_i \in \{0, 1\}$, $\hat\pi_i$ takes the value $\hat\pi_1$ when $x_i = 1$ and $\hat\pi_0$ when $x_i = 0$. The two equations of (a) read

$$n_{0+}\hat\pi_0 + n_{1+}\hat\pi_1 = n_{01} + n_{11}, \qquad n_{1+}\hat\pi_1 = n_{11}.$$

Hence $\hat\pi_1 = n_{11}/n_{1+}$ and $\hat\pi_0 = n_{01}/n_{0+}$, both in $(0, 1)$ because all cells are positive. Then $\hat\beta_0 = \operatorname{logit}\hat\pi_0$ and $\hat\beta_1 = \operatorname{logit}\hat\pi_1 - \operatorname{logit}\hat\pi_0$ solve the likelihood equations, and by the strict concavity of Theorem 16.1 they are the unique MLE. Finally $e^{\hat\beta_1} = \frac{n_{11}/n_{10}}{n_{01}/n_{00}}$. ■

Exercise 16.2 (Standard error of the log odds ratio)

In the setting of Exercise 16.1(b), show that the entry of $\{X'W(\hat\beta)X\}^{-1}$ for β_1 equals

$$\frac{1}{n_{11}} + \frac{1}{n_{10}} + \frac{1}{n_{01}} + \frac{1}{n_{00}},$$

the squared standard error used in Example 16.1.

Solution [Supplement].

Step 1. $X'WX = \sum_i w_i \mathbf{x}_i \mathbf{x}_i'$ with $w_i = \hat{\pi}_i(1-\hat{\pi}_i)$. Grouping by x, with $a = n_{0+}\hat{\pi}_0(1-\hat{\pi}_0)$ and $b = n_{1+}\hat{\pi}_1(1-\hat{\pi}_1)$,

$$X'WX = a\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix} + b\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix} = \begin{pmatrix}a+b & b\\ b & b\end{pmatrix}.$$

Step 2. Its determinant is $(a+b)b - b^2 = ab > 0$, and the $(2,2)$ entry of the inverse is $(a+b)/(ab) = 1/a + 1/b$.

Step 3. By Exercise 16.1(b), $a = n_{0+} \cdot \frac{n_{01}}{n_{0+}} \cdot \frac{n_{00}}{n_{0+}} = \frac{n_{01}n_{00}}{n_{0+}}$, so $\frac{1}{a} = \frac{n_{00}+n_{01}}{n_{01}n_{00}} = \frac{1}{n_{01}} + \frac{1}{n_{00}}$. In the same way $\frac{1}{b} = \frac{1}{n_{11}} + \frac{1}{n_{10}}$. Adding gives the claim. ■

Exercise 16.3 (Complete separation)

Suppose there is a vector $\mathbf{b}$ with $\mathbf{x}_i'\mathbf{b} > 0$ whenever $y_i = 1$ and $\mathbf{x}_i'\mathbf{b} < 0$ whenever $y_i = 0$. Show that $l(\beta) < 0$ for all β but $l(t\mathbf{b}) \to 0$ as $t \to \infty$, so that the MLE does not exist.

Solution [Supplement].

Step 1. For every β, $0 < \pi_i < 1$, so each term $y_i \log \pi_i + (1-y_i)\log(1-\pi_i)$ is negative and $l(\beta) < 0$.

Step 2. Along $\beta = t\mathbf{b}$: if $y_i = 1$, the term is $\log \pi_i = -\log(1 + e^{-t\mathbf{x}_i'\mathbf{b}}) \to 0$ since $\mathbf{x}_i'\mathbf{b} > 0$; if $y_i = 0$, it is $\log(1-\pi_i) = -\log(1 + e^{t\mathbf{x}_i'\mathbf{b}}) \to 0$ since $\mathbf{x}_i'\mathbf{b} < 0$. Hence $l(t\mathbf{b}) \to 0$.

Step 3. So $\sup_\beta l(\beta) = 0$, but no β attains it by Step 1: the MLE does not exist. Along the direction $\mathbf{b}$ the fitted probabilities tend to 0 or 1; in practice separation shows up as huge coefficients and fitted probabilities numerically equal to 0 or 1. ■